

Additional Practice

A1.5.01

Problems 1–2: Here is a shape puzzle. The sum of each row and column is shown.

1. Select *all* the true statements.

☒ A.  +  = 22

☒ B.  +  = 14

☐ C.  +  = 15

☐ D.  = 12

☒ E.  +  = 19

		= 17
		= 24
		= 19 = 22

2. Show or explain why this statement is *false*:  = 8

Explanations vary. Since two houses equal 24, we know that one house is 12. According to the puzzle, one house and one star add up to 19. This makes the star equal to 19 minus 12, which is 7, not 8.

Problems 3–4: Here is a shape puzzle.

3. Determine the solution for this puzzle.

Shape	Value
Heart	5
Flower	-12
House	4

			= -3
			= -3
= -7	= -8	= 9	

4. Explain or show your thinking.

Explanations vary. Based on columns 1 and 2, the heart must be 1 more than the house. Based on column 3, the house and the heart must add up to 9. Therefore, the house must be 5 and the heart must be 4. I used those values to determine that the flower must be -12.

Problems 5–6: Use these two equations:

$$x + y + y = 25$$

$$x + x + y = 20$$



5. Draw a shape puzzle to represent these equations.

Responses vary.

6. Determine the values of x and y .

$$x = \underline{\quad 5 \quad} \qquad y = \underline{\quad 10 \quad}$$

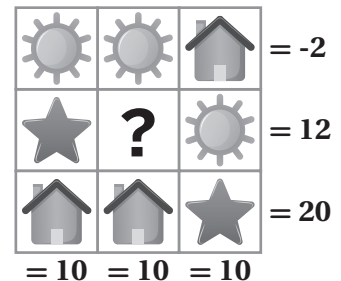
Problems 7–9: Here is a shape puzzle.

7. Determine the missing shape in the center of this puzzle.
Circle your choice.

Sun Star House

8. Show or explain your thinking.

Explanations vary. Based on the three columns all adding up to the same value, I know that all three different shapes add up to 10. The middle column must be all three different shapes. Since the star is the shape not represented in this column, the shape must be a star.



9. If the missing shape has a value of 8, what are the values of the other two shapes?

The sun has a value of -4 and the house has a value of 6.

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Additional Practice

A1.5.01

Problems 1–2: Here is a shape puzzle. The sum of each row and column is shown.

1. Select all the true statements.

☒ A. $\star + \star = 22$

☒ B. $\star + \star = 14$

☐ C. $\star + \star = 15$

☐ D. $\star = 12$

☒ E. $\star + \star = 19$

$$\begin{array}{|c|c|} \hline \star & \star \\ \hline \star & \star \\ \hline \end{array} \begin{array}{l} = 17 \\ = 24 \\ = 19 \end{array}$$

2. Show or explain why this statement is false: $\star = 8$

Explanations vary. Since two houses equal 24, we know that one house is 12. According to the puzzle, one house and one star add up to 19. This makes the star equal to 19 minus 12, which is 7, not 8.

Problems 3–4: Here is a shape puzzle.

3. Determine the solution for this puzzle.

Shape	Value
Heart	5
Flower	-12
House	4

$$\begin{array}{|c|c|c|} \hline \heartsuit & \star & \star \\ \hline \heartsuit & \star & \star \\ \hline \end{array} \begin{array}{l} = -3 \\ = -3 \\ = -7 \end{array}$$

4. Explain or show your thinking.

Explanations vary. Based on columns 1 and 2, the heart must be 1 more than the house. Based on column 3, the house and the heart must add up to 9. Therefore, the house must be 5 and the heart must be 4. I used those values to determine that the flower must be -12.

Unit 5 Lesson 1

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Name: _____ Date: _____ Period: _____

Problems 5–6: Use these two equations:

$$x + y + y = 25$$

$$x + x + y = 20$$

$$\begin{array}{|c|c|c|} \hline \star & \star & \star \\ \hline \star & \star & \star \\ \hline \end{array} \begin{array}{l} = 25 \\ = 20 \end{array}$$

5. Draw a shape puzzle to represent these equations.

Responses vary.

6. Determine the values of x and y .

$$x = 5 \quad y = 10$$

Problems 7–9: Here is a shape puzzle.

7. Determine the missing shape in the center of this puzzle. Circle your choice.

Sun ☒ Star ☐ House

$$\begin{array}{|c|c|c|} \hline \star & \star & \star \\ \hline \star & ? & \star \\ \hline \star & \star & \star \\ \hline \end{array} \begin{array}{l} = -2 \\ = 12 \\ = 20 \end{array}$$

8. Show or explain your thinking.

Explanations vary. Based on the three columns all adding up to the same value, I know that all three different shapes add up to 10. The middle column must be all three different shapes. Since the star is the shape not represented in this column, the shape must be a star.

9. If the missing shape has a value of 8, what are the values of the other two shapes?

The sun has a value of -4 and the house has a value of 6.

Unit 5 Lesson 1

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.C.6
2	2	HSA.REI.C.6
3	1	HSA.REI.C.6
4	2	HSA.REI.C.6
5	1	HSA.REI.C.6
6	1	HSA.REI.C.6
7	1	HSA.REI.C.6
8	2	HSA.REI.C.6
9	1	HSA.REI.C.6

Notes:

Additional Practice

A1.5.02

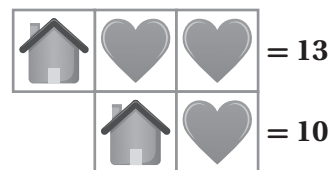
1. Solve this system of equations. Use the shape puzzle if it helps with your thinking.

$$2x + y = 13$$

$$x + y = 10$$

$$x = \underline{\quad 3 \quad}$$

$$y = \underline{\quad 7 \quad}$$



2. Which equation is the result of adding these two equations?

$$-3x + 2y = 1$$

$$3x + 5y = -15$$

☒ A. $7y = -14$

B. $7y = 16$

C. $-6x + 7y = 16$

D. $6x + 7y = -14$

3. Which system of equations is solved in the fewest steps using elimination by addition?

A. $-x + 4y = 5$
 $4x - y = 12$

B. $-4x + 3y = -16$
 $-2x + 3y = 12$

C. $5x - y = 7$
 $5x - 2y = 14$

☒ D. $-6x + 2y = 12$
 $6x + 7y = 3$

4. Which ordered pair is the solution to this system of equations?

$$-2x + y = 11$$

$$2x - 3y = -25$$

A. $(-3, 8)$

☒ B. $(-2, 7)$

C. $(2, -7)$

D. $(4, 7)$

Problems 5–8: Determine the solution for each system of equations. Show your thinking.

5. $4x - 7y = -3$

$4x + 7y = 67$

Responses vary.

$8x = 64$

$x = 8$

$4(8) - 7y = -3$

$32 - 7y = -3$

$-7y = -35$

$y = 5$

$x = \underline{\quad 8 \quad}$

$y = \underline{\quad 5 \quad}$

6. $5x - y = 4$

$-5x + 3y = -3$

Responses vary.

$2y = 1$

$y = 0.5, \text{ or equivalent}$

$5x - (0.5) = 4$

$5x = 4.5$

$x = 0.9$

$x = \underline{\quad 0.5 \quad}$

$y = \underline{\quad 0.9 \quad}$

7. $-x + 8y = 18$

$-3x + 8y = 6$

Responses vary.

$-x + 8y = 18$

$-(-3x + 8y = 6)$

$2x = 12$

$x = 6$

$-(6) + 8y = 18$

$8y = 24$

$y = 3$

$x = \underline{\quad 6 \quad}$

$y = \underline{\quad 3 \quad}$

8. $5x - 3y = -26$

$5x + 4y = -12$

Responses vary.

$5x - 3y = -26$

$-(5x + 4y = -12)$

$-7y = -14$

$y = 2$

$5x - 6 = -26$

$5x = -20$

$x = -4$

$x = \underline{\quad -4 \quad}$

$y = \underline{\quad 2 \quad}$

Name: _____ Date: _____ Period: _____

Additional Practice

A1.5.02

1. Solve this system of equations. Use the shape puzzle if it helps with your thinking.



$$2x + y = 13$$

$$x + y = 10$$

$$x = \underline{3}$$

$$y = \underline{7}$$

2. Which equation is the result of adding these two equations?

$$-3x + 2y = 1$$

$$3x + 5y = -15$$

☒ A. $7y = -14$

B. $7y = 16$

C. $-6x + 7y = 16$

D. $6x + 7y = -14$

3. Which system of equations is solved in the fewest steps using elimination by addition?

A. $-x + 4y = 5$
 $4x - y = 12$

B. $-4x + 3y = -16$
 $-2x + 3y = 12$

C. $5x - y = 7$
 $5x - 2y = 14$

☒ D. $-6x + 2y = 12$
 $6x + 7y = 3$

4. Which ordered pair is the solution to this system of equations?

$$-2x + y = 11$$

$$2x - 3y = -25$$

A. $(-3, 8)$

☒ B. $(-2, 7)$

C. $(2, -7)$

D. $(4, 7)$

Unit 5 Lesson 2

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Name: _____ Date: _____ Period: _____

Problems 5–8: Determine the solution for each system of equations. Show your thinking.

5. $4x - 7y = -3$

$$4x + 7y = 67$$

Responses vary.

$$8x = 64$$

$$x = 8$$

$$4(8) - 7y = -3$$

$$32 - 7y = -3$$

$$-7y = -35$$

$$y = 5$$

$$x = \underline{8} \quad y = \underline{5}$$

6. $5x - y = 4$

$$-5x + 3y = -3$$

Responses vary.

$$2y = 1$$

$$y = 0.5, \text{ or equivalent}$$

$$5x - (0.5) = 4$$

$$5x = 4.5$$

$$x = 0.9$$

$$x = \underline{0.5} \quad y = \underline{0.9}$$

7. $-x + 8y = 18$

$$-3x + 8y = 6$$

Responses vary.

$$-x + 8y = 18$$

$$-(-3x + 8y) = 6$$

$$2x = 12$$

$$x = 6$$

$$-(6) + 8y = 18$$

$$8y = 24$$

$$y = 3$$

$$x = \underline{6} \quad y = \underline{3}$$

8. $5x - 3y = -26$

$$5x + 4y = -12$$

Responses vary.

$$5x - 3y = -26$$

$$-(5x + 4y) = -12$$

$$-7y = -14$$

$$y = 2$$

$$5x - 6 = -26$$

$$5x = -20$$

$$x = -4$$

$$x = \underline{-4} \quad y = \underline{2}$$

Unit 5 Lesson 2

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.C.6
2	1	HSA.REI.C.6
3	1	HSA.REI.C.6
4	1	HSA.REI.C.6
5	1	HSA.REI.C.6
6	1	HSA.REI.C.6
7	1	HSA.REI.C.6
8	1	HSA.REI.C.6

Notes:

Additional Practice

A1.5.03

1. Select *all* equations that are equivalent to $2x - 9y = -5$.

☒ A. $-2x + 9y = 5$

☐ B. $4x + 18y = 10$

☒ C. $-6x + 27y = 15$

☐ D. $6x + 27y = -15$

☒ E. $4x - 18y = -10$

2. Diondre and Gia are solving this system of equations. They disagree about what the first step should be to eliminate a variable.

$$5x - 3y = -13$$

$$-10x + y = -21$$

Diondre's strategy: Multiply $5x - 3y = -13$ by -2 and then add the equations.

Gia's strategy: Multiply $-10x + y = -21$ by 3 and then add the equations.

Whose strategy will eliminate a variable once the equations are added?
Circle your choice.

A. Diondre's

☒ B. Gia's

C. Both

D. Neither

Explain your thinking.

Explanations vary. Gia's strategy will create a term that can be eliminated by adding the equations together because the $-3y$ in the first equation and the $3y$ in the new second equation will eliminate the y when added. If Diondre multiplied the first equation by -2 , this will result in both x -values having the coefficient of -10 . The equations will need to be **subtracted** to eliminate the x -value, not added.

3. Select *all* of the following equations that are equivalent to $-2x + 12y = 8$.

☐ A. $4x + 8y = 48$

☐ B. $20x + 120y = 80$

☒ C. $x - 6y = -4$

☒ D. $-10x + 60y = 40$

☐ E. $-2x - 12y = -8$

Problems 4–5: Determine the solution for each system of equations. Show your thinking.

4. $-2x + 9y = 24$
 $8x - 3y = -30$

Responses vary.

$4(-2x + 9y = 24)$
 $8x - 3y = -30$

$-8x + 36y = 96$
 $\underline{8x - 3y = -30}$
 $33y = 66$
 $y = 2$

$8x - 3(2) = -30$
 $8x - 6 = -30$
 $8x = -24$
 $x = -3$

$x = \underline{-3}$ $y = \underline{2}$

5. $-x + 4y = -30$
 $3x - 2y = 20$

Responses vary.

$3(-x + 4y = -30)$
 $3x - 2y = 20$

$-3x + 12y = -90$
 $\underline{3x - 2y = 20}$
 $10y = -70$
 $y = -7$

$3x - 2(-7) = 20$
 $3x + 14 = 20$
 $3x = 6$
 $x = 2$

$x = \underline{2}$ $y = \underline{-7}$

6. Solve the system of equations using the methods below.

$-x + 6y = 21$
 $4x - y = 31$

Eliminate the x -term first.

$4(-x + 6y = 21)$
 $4x - y = 31$

$-4x + 24y = 84$
 $\underline{4x - y = 31}$
 $23y = 115$
 $y = 5$

$4x - (5) = 31$
 $4x = 36$
 $x = 9$

The solution is ($\underline{9}$, $\underline{5}$).

Eliminate the y -term first.

$-x + 6y = 21$
 $6(4x - y = 31)$

$-x + 6y = 21$
 $\underline{24x - 6y = 186}$
 $23x = 207$
 $x = 9$

$-(9) + 6y = 31$
 $-9 + 6y = 31$
 $6y = 40$
 $y = 5$

The solution is ($\underline{9}$, $\underline{5}$).

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Additional Practice

A1.5.03

1. Select *all* equations that are equivalent to $2x - 9y = -5$.

- ☒ A. $-2x + 9y = 5$ ☐ B. $4x + 18y = 10$
☒ C. $-6x + 27y = 15$ ☐ D. $6x + 27y = -15$
☒ E. $4x - 18y = -10$

2. Diondre and Gia are solving this system of equations. They disagree about what the first step should be to eliminate a variable.

$$\begin{aligned} 5x - 3y &= -13 \\ -10x + y &= -21 \end{aligned}$$

Diondre's strategy: Multiply $5x - 3y = -13$ by -2 and then add the equations.

Gia's strategy: Multiply $-10x + y = -21$ by 3 and then add the equations.

Whose strategy will eliminate a variable once the equations are added? Circle your choice.

- A. Diondre's ☐ B. Gia's ☒ C. Both ☐ D. Neither

Explain your thinking.

Explanations vary. Gia's strategy will create a term that can be eliminated by adding the equations together because the $-3y$ in the first equation and the $3y$ in the new second equation will eliminate the y when added. If Diondre multiplied the first equation by -2 , this will result in both x -values having the coefficient of -10 . The equations will need to be subtracted to eliminate the x -value, not added.

3. Select *all* of the following equations that are equivalent to $-2x + 12y = 8$.

- ☐ A. $4x + 8y = 48$ ☐ B. $20x + 120y = 80$
☒ C. $x - 6y = -4$ ☒ D. $-10x + 60y = 40$
☐ E. $-2x - 12y = -8$

Unit 5 Lesson 3

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Problems 4–5: Determine the solution for each system of equations. Show your thinking.

4. $-2x + 9y = 24$
 $8x - 3y = -30$

Responses vary.

$$\begin{aligned} 4(-2x + 9y) &= 4(24) \\ 8x - 3y &= -30 \\ -8x + 36y &= 96 \\ 8x - 3y &= -30 \\ 33y &= 66 \\ y &= 2 \end{aligned}$$

$$\begin{aligned} 8x - 3(2) &= -30 \\ 8x - 6 &= -30 \\ 8x &= -24 \\ x &= -3 \end{aligned}$$

$$x = -3 \quad y = 2$$

5. $-x + 4y = -30$
 $3x - 2y = 20$

Responses vary.

$$\begin{aligned} 3(-x + 4y) &= 3(-30) \\ 3x - 2y &= 20 \\ -3x + 12y &= -90 \\ 3x - 2y &= 20 \\ 10y &= -70 \\ y &= -7 \end{aligned}$$

$$\begin{aligned} 3x - 2(-7) &= 20 \\ 3x + 14 &= 20 \\ 3x &= 6 \\ x &= 2 \end{aligned}$$

$$x = 2 \quad y = -7$$

6. Solve the system of equations using the methods below.

$$\begin{aligned} -x + 6y &= 21 \\ 4x - y &= 31 \end{aligned}$$

Eliminate the x -term first.

$$\begin{aligned} 4(-x + 6y) &= 4(21) \\ 4x - y &= 31 \\ -4x + 24y &= 84 \\ 4x - y &= 31 \\ 23y &= 115 \\ y &= 5 \end{aligned}$$

$$\begin{aligned} 4x - (5) &= 31 \\ 4x &= 36 \\ x &= 9 \end{aligned}$$

The solution is $(9, 5)$.

Eliminate the y -term first.

$$\begin{aligned} -x + 6y &= 21 \\ 6(4x - y) &= 6(31) \\ -x + 6y &= 21 \\ 24x - 6y &= 186 \\ 23x &= 207 \\ x &= 9 \end{aligned}$$

$$\begin{aligned} -(9) + 6y &= 21 \\ -9 + 6y &= 21 \\ 6y &= 30 \\ y &= 5 \end{aligned}$$

The solution is $(9, 5)$.

Unit 5 Lesson 3

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.C.5, HSA.REI.C.6
2	2	HSA.REI.C.5, HSA.REI.C.6
3	1	HSA.REI.C.5, HSA.REI.C.6
4	1	HSA.REI.C.5, HSA.REI.C.6
5	1	HSA.REI.C.5, HSA.REI.C.6
6	1	HSA.REI.C.5, HSA.REI.C.6

Notes:

Additional Practice

A1.5.04

1. Identify the solution to this system of equations.

$$\begin{aligned} y &= x + 5 \\ y &= -x + 9 \end{aligned}$$

- A. $(-2, 3)$ B. $(0, 9)$
C. $(2, 7)$ D. $(4, 5)$

2. Identify the solution to this system of equations.

$$\begin{aligned} x - y &= -6 \\ -3x - 2y &= 8 \end{aligned}$$

- A. $(-4, 2)$** B. $(-3, 3)$
 C. $(-1, 5)$ D. $(4, -2)$

Problems 3–4: Show or explain what your first step would be for solving each system of equations.

3. $3x - 2y = -13$
 $y = 5$

Responses vary. I would substitute 5 for y in the first equation. I would have a new equation with just one variable: $3x - 2(5) = -13$.

4. $y = -4x + 12$
 $6x + 5y = 11$

Responses vary. I would substitute the expression $-4x + 12$ for y in the second equation. The new equation would be $6x + 5(-4x + 12) = 11$.

5. Show or explain how you would solve one of the systems of equations using the substitution method.

System A

$$\begin{aligned} 2x - 3y &= 9 \\ 5x - 4y &= 11 \end{aligned}$$

System B

$$\begin{aligned} y &= 4x - 3 \\ -3x + y &= 7 \end{aligned}$$

Responses vary. I would solve System B using the substitution method. The first equation has the variable y already isolated and can be substituted into the second equation without any additional steps. In System A, both equations are in standard form, so the x - and y -intercepts can be determined quickly, which makes the system most efficiently solved by elimination.

Problems 6–7: Solve each system of equations.

6. $3x - 5y = 18$

$3x = 4y + 15$

(1, -3)

7. $4x + 2y = 21$

$2y = 25 - 5x$

(4, 2.5) (or equivalent)

8. Match each system of equations with its solution.

System of equations

Solution

a. $3f - 2y = 11$

$y = 3f - 13$

a (5, 2)

b. $x + 8y = -1$

$2x - 6y = -13$

c $\left(2, -\frac{2}{3}\right)$

c. $4x = -12y$

$10 + 6y = 3x$

b $\left(-5, \frac{1}{2}\right)$

9. Bard is solving this system of equations: $-3x + 2y = 16$
 $5x - 3y = -18$

Bard begins by rearranging the first equation to isolate the y variable: $y = 8 + 1.5x$.

Then Bard substitutes the expression $8 + 1.5x$ for y in the second equation, as shown:

$5x - 3(8 + 1.5x) = -18$

$y = 8 + 1.5x$

$5x - 24 - 4.5x = -18$

$y = 8 + 1.5(-84)$

$0.5x - 24 = -18$

$y = -118$

$0.5x = -42$

$x = -84$

Does Bard's solution of $(-84, -118)$ make both equations in the system true?

Explain your thinking.

No, Explanations vary. When I substitute $(-84, -118)$ into $5x - 3y$, the result is not -18 .

Name: _____ Date: _____ Period: _____

Additional Practice

A1.5.04

1. Identify the solution to this system of equations.

$$\begin{aligned} y &= x + 5 \\ y &= -x + 9 \end{aligned}$$

- A. $(-2, 3)$ B. $(0, 9)$
C. $(2, 7)$ D. $(4, 5)$

2. Identify the solution to this system of equations.

$$\begin{aligned} x - y &= -6 \\ -3x - 2y &= 8 \end{aligned}$$

- A. $(-4, 2)$** B. $(-3, 3)$
 C. $(-1, 5)$ D. $(4, -2)$

Problems 3–4: Show or explain what your first step would be for solving each system of equations.

3. $3x - 2y = -13$
 $y = 5$

Responses vary. I would substitute 5 for y in the first equation. I would have a new equation with just one variable: $3x - 2(5) = -13$.

4. $y = -4x + 12$
 $6x + 5y = 11$

Responses vary. I would substitute the expression $-4x + 12$ for y in the second equation. The new equation would be $6x + 5(-4x + 12) = 11$.

5. Show or explain how you would solve one of the systems of equations using the substitution method.

System A

$$\begin{aligned} 2x - 3y &= 9 \\ 5x - 4y &= 11 \end{aligned}$$

System B

$$\begin{aligned} y &= 4x - 3 \\ -3x + y &= 7 \end{aligned}$$

Responses vary. I would solve System B using the substitution method. The first equation has the variable y already isolated and can be substituted into the second equation without any additional steps. In System A, both equations are in standard form, so the x - and y -intercepts can be determined quickly, which makes the system most efficiently solved by elimination.

Unit 5 Lesson 4

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Problems 6–7: Solve each system of equations.

6. $3x - 5y = 18$
 $3x = 4y + 15$
 $(1, -3)$

7. $4x + 2y = 21$
 $2y = 25 - 5x$
 $(4, 2.5)$ (or equivalent)

8. Match each system of equations with its solution.

System of equations

Solution

a. $3f - 2g = 11$
 $y = 3f - 13$

a $(5, 2)$

b. $x + 8y = -1$
 $2x - 6y = -13$

c $(2, -\frac{2}{3})$

c. $4x = -12y$
 $10 + 6y = 3x$

b $(-5, \frac{1}{2})$

9. Bard is solving this system of equations: $-3x + 2y = 16$
 $5x - 3y = -18$

Bard begins by rearranging the first equation to isolate the y variable: $y = 8 + 1.5x$. Then Bard substitutes the expression $8 + 1.5x$ for y in the second equation, as shown:

$$5x - 3(8 + 1.5x) = -18$$

$$y = 8 + 1.5x$$

$$5x - 24 - 4.5x = -18$$

$$y = 8 + 1.5(-84)$$

$$0.5x - 24 = -18$$

$$y = -118$$

$$0.5x = -42$$

$$x = -84$$

Does Bard's solution of $(-84, -118)$ make both equations in the system true? Explain your thinking.

No, Explanations vary. When I substitute $(-84, -118)$ into $5x - 3y$, the result is not -18 .

Unit 5 Lesson 4

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.C.6
2	1	HSA.REI.C.6
3	1	HSA.REI.C.6
4	1	HSA.REI.C.6
5	2	HSA.REI.C.6
6	1	HSA.REI.C.6
7	1	HSA.REI.C.6
8	1	HSA.REI.C.6
9	2	HSA.REI.C.6

Notes:

Additional Practice

A1.5.05

1. The point $(2, 1)$ is on the line $y = 3x - 5$. Explain how you can determine if this point is the solution to this system of equations:

$$y = 3x - 5$$

$$y = x - 1$$

Explanations vary. You can graph the equation $y = x - 1$ and see if the point $(2, 1)$ lies on the line.

2. Here is a system of equations.

$$y = -\frac{1}{2}x + 1$$

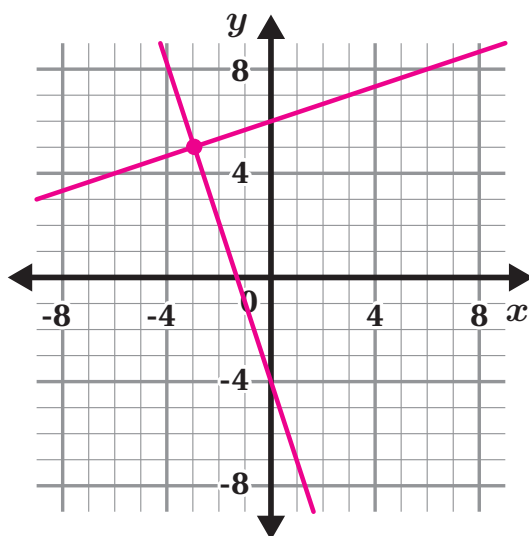
$$y = \frac{1}{2}x - 3$$

How many solutions does this system of equations have? Explain your thinking.

One. Explanations vary. The equations that make up this system of equations have different slopes so that means there is one solution.

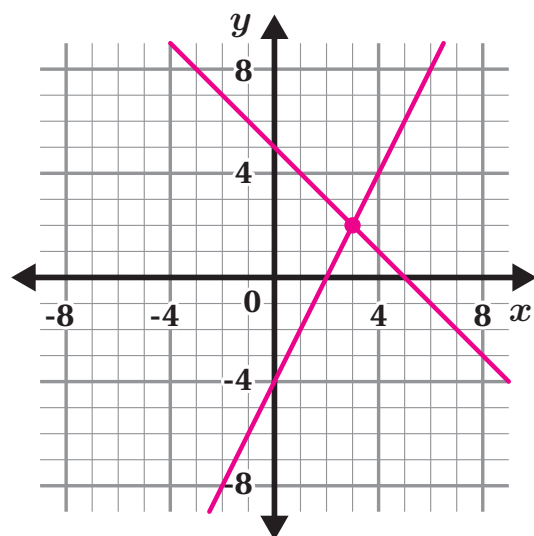
Problems 3–4: Solve this system of equations. Write the solution as a coordinate pair.

3. $y = -3x - 4$
 $y = \frac{1}{3}x + 6$



Solution: $(-3, 5)$

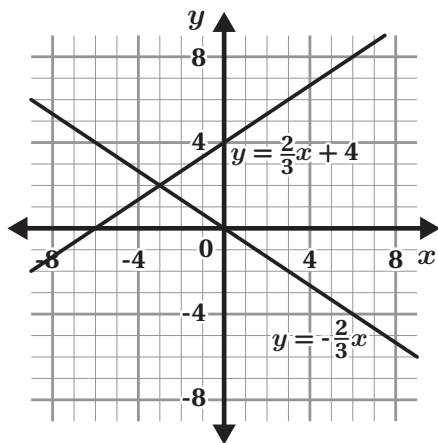
4. $y = 2x - 4$
 $y = -x + 5$



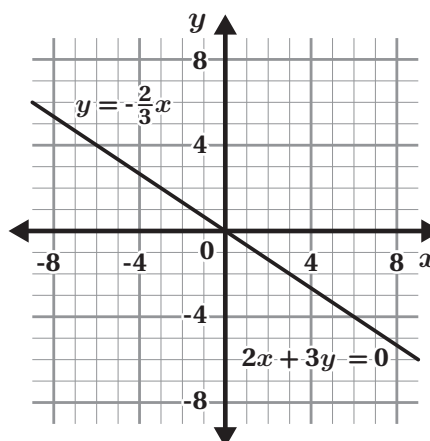
Solution: $(3, 2)$

5. Which graph shows a system of equations that has no solutions?

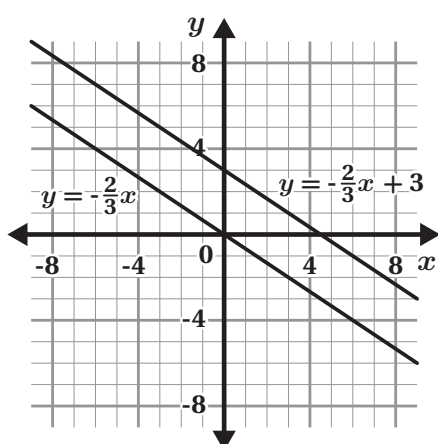
A.



B.



C.



6. Match each system of equations to the number of solutions it has.

Equation

Solution

a. $y = -3x + 4$
 $y = 3x + 4$

b no solutions

b. $3x + y = 4$
 $y = -3x - 4$

a one solution

c. $3x + y = 4$
 $y = -3x + 4$

c infinitely many solutions

7. A system of equations has infinitely many solutions. Select *all* of the statements that must be true about the equations in this system.

☐ A. The equations have different slopes.

☒ B. The equations have the same slope.

☐ C. The equations have different y -intercepts.

☒ D. The equations have the same y -intercept.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.5.05

1. The point (2, 1) is on the line $y = 3x - 5$. Explain how you can determine if this point is the solution to this system of equations:

$$y = 3x - 5$$

$$y = x - 1$$

Explanations vary. You can graph the equation $y = x - 1$ and see if the point (2, 1) lies on the line.

2. Here is a system of equations.

$$y = -\frac{1}{2}x + 1$$

$$y = \frac{1}{2}x - 3$$

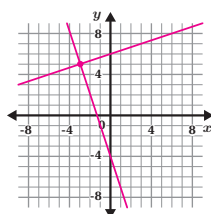
How many solutions does this system of equations have? Explain your thinking.

One. Explanations vary. The equations that make up this system of equations have different slopes so that means there is one solution.

Problems 3–4: Solve this system of equations. Write the solution as a coordinate pair.

3. $y = -3x - 4$

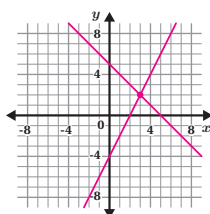
$$y = \frac{1}{3}x + 6$$



Solution: $(-3, 5)$

4. $y = 2x - 4$

$$y = -x + 5$$



Solution: $(3, 2)$

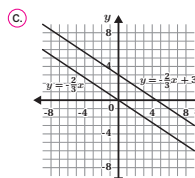
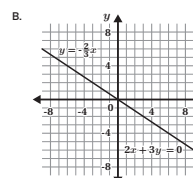
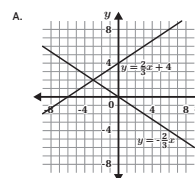
Unit 5 Lesson 5

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Name: _____ Date: _____ Period: _____

5. Which graph shows a system of equations that has no solutions?



6. Match each system of equations to the number of solutions it has.

Equation

a. $y = -3x + 4$
 $y = 3x + 4$

b. $3x + y = 4$
 $y = -3x - 4$

c. $3x + y = 4$
 $y = -3x + 4$

Solution

b no solutions

a one solution

c infinitely many solutions

7. A system of equations has infinitely many solutions. Select all of the statements that must be true about the equations in this system.

- ☐ A. The equations have different slopes. ☒ B. The equations have the same slope.
☐ C. The equations have different y-intercepts. ☒ D. The equations have the same y-intercept.

Unit 5 Lesson 5

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSA.REI.D.11
2	2	HSA.REI.C.6
3	1	HSA.REI.C.6, HSA.REI.D.11
4	1	HSA.REI.C.6, HSA.REI.D.11
5	1	HSA.REI.C.6
6	1	HSA.REI.C.6
7	1	HSA.REI.C.6

Notes:

Additional Practice

A1.5.08

Problems 1–3: Show or explain what your *first step* would be to solve each system of equations.

1. $7x - 12y = 20$

$15x + 12y = -9$

Responses vary.
Add the equations together to eliminate y .

2. $y = \frac{1}{2}x - 8$
 $y = -\frac{2}{3}x + 6$

Responses vary.
Both y 's are isolated.
Substitute $\frac{1}{2}x - 8$ in for y in the second equation.

3. $5x - 9y = 21$

$x = -6$

Responses vary.
Substitute -6 for x in the first equation.

Problems 4–5: Determine how many solutions each equation has. If there is one solution, what is the solution?

4. $5x - 10y = 40$

$3x + 10y = 8$

No Solution

One Solution

Infinitely many solutions

$(6, -1)$

5. $8x - 16y = 18$

$x = 2y + 9$

No Solution

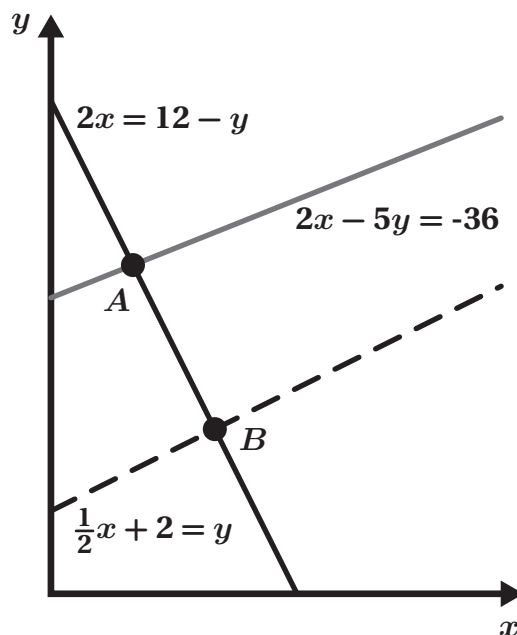
One Solution

Infinitely many solutions

6. Determine the coordinates of points A and B , the intersections of the lines on the graph.

$A = (2, 8)$

$B = (4, 4)$



7. Consider the equation $4x - 3y = 12$. Match each description with an equation that would make a system of equations with the given number of solutions. Show or explain your thinking.

Description	Equation
a. One solution	<u> b </u> $4x - 3y = 6$
b. No solution	<u> c </u> $y = \frac{4}{3}x - 4$
c. Infinitely many solutions	<u> a </u> $4x + y = 20$

Responses vary.

When I subtracted $4x - 3y = 6$ from $4x - 3y = 12$, I got $0 = 6$ which is false. So, this system has no solution.

When I substituted $\frac{4}{3}x - 4$ in for y in $4x - 3y = 12$, I got $0 = 0$ which is true. So, this system has infinitely many solutions.

When I subtracted $4x + y = 20$ from $4x - 3y = 12$, I got $-4y = -8$ which means that $y = 2$. This system will have one solution.

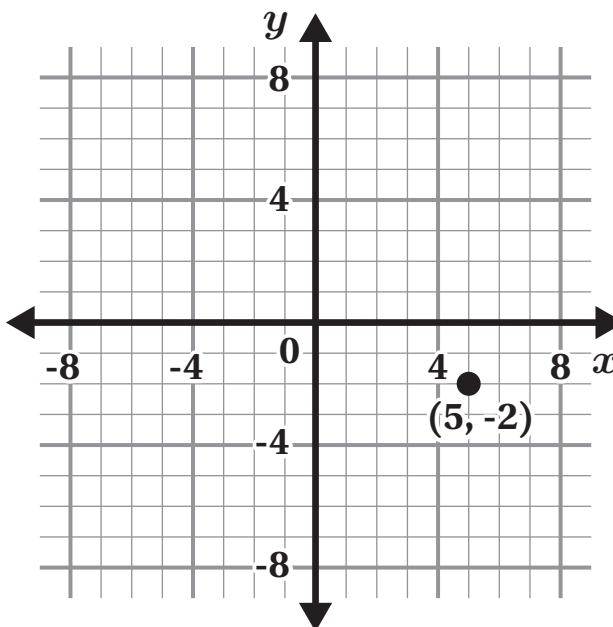
8. Write a system of equations where $(5, -2)$ is the solution.

Use the graph if it helps your thinking.

Responses vary.

$y = -x + 3$

$y = x - 7$



Name: _____ Date: _____ Period: _____

Additional Practice

A1.5.08

Problems 1–3: Show or explain what your *first* step would be to solve each system of equations.

1. $7x - 12y = 20$

$15x + 12y = -9$

Responses vary. Add the equations together to eliminate y .

2. $y = \frac{1}{2}x - 8$

$y = -\frac{2}{3}x + 6$

Responses vary. Both y 's are isolated. Substitute $\frac{1}{2}x - 8$ in for y in the second equation.

3. $5x - 9y = 21$

$x = -6$

Responses vary. Substitute -6 for x in the first equation.

Problems 4–5: Determine how many solutions each equation has. If there is one solution, what is the solution?

4. $5x - 10y = 40$

$3x + 10y = 8$

No Solution

One Solution

Infinitely many solutions

(6, -1)

5. $8x - 16y = 18$

$x = 2y + 9$

No Solution

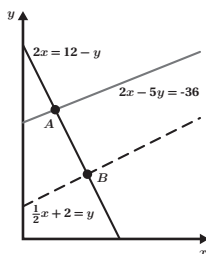
One Solution

Infinitely many solutions

6. Determine the coordinates of points A and B , the intersections of the lines on the graph.

$A = (2, 8)$

$B = (4, 4)$



Unit 5 Lesson 8

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Name: _____ Date: _____ Period: _____

7. Consider the equation $4x - 3y = 12$. Match each description with an equation that would make a system of equations with the given number of solutions. Show or explain your thinking.

Description

Equation

a. One solution

b $4x - 3y = 6$

b. No solution

c $y = \frac{4}{3}x - 4$

c. Infinitely many solutions

a $4x + y = 20$

Responses vary.

When I subtracted $4x - 3y = 6$ from $4x - 3y = 12$, I got $0 = 6$ which is false. So, this system has no solution.

When I substituted $\frac{4}{3}x - 4$ in for y in $4x - 3y = 12$, I got $0 = 0$ which is true. So, this system has infinitely many solutions.

When I subtracted $4x + y = 20$ from $4x - 3y = 12$, I got $-4y = -8$ which means that $y = 2$. This system will have one solution.

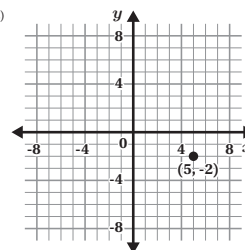
8. Write a system of equations where $(5, -2)$ is the solution.

Use the graph if it helps your thinking.

Responses vary.

$y = -x + 3$

$y = x - 7$



Unit 5 Lesson 8

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.C.6
2	1	HSA.REI.C.6
3	1	HSA.REI.C.6
4	1	HSA.REI.C.6
5	1	HSA.REI.C.6
6	1	HSA.REI.C.6
7	2	HSA.REI.C.6
8	2	HSA.REI.C.6

Notes:

Additional Practice

A1.5.09

1. Here is a system of inequalities.

$$y \leq -x + 4$$

$$2x + y > 8$$

Select *all* points that are solutions to the system of inequalities.

☒ A. (5, -1)

☒ B. (6, -3)

☐ C. (0, 0)

☒ D. (7, -5)

☐ E. (-5, 7)

☐ F. (6, 3)

Problems 2–3: The student council has \$300 to spend on burgers for a barbecue. They want to serve at least 240 burgers. A package of veggie burgers, x , costs \$10. A package of turkey burgers, y , costs \$20. The student council wrote this system of inequalities:

$$10x + 12y \geq 240$$

$$10x + 20y \leq 300$$

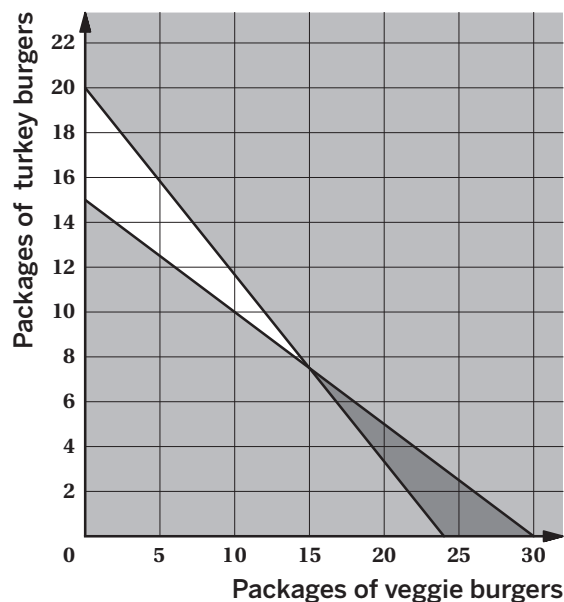
2. What does the inequality $10x + 20y \leq 300$ represent?

Responses vary. The inequality represents the combinations of packages of veggie burgers and turkey burgers that will cost no more than \$300.

3. Here is the graph of the system of inequalities.

Which of the following combinations of packages of veggie burgers and turkey burgers could the organizers buy?

- A. 0 veggie burger packages, 15 turkey burger packages
- B. 10 veggie burger packages, 12 turkey burger packages
- C. 15 veggie burger packages, 8 turkey burger packages
- ☒ D. 25 veggie burger packages, 2 turkey burger packages



Problems 4–5: Andre needs to buy batteries for his business.

AA batteries come in packs of 24 and cost \$15 per pack. D batteries come in packs of 12 and cost \$15 per pack. Andre wants to buy at least 180 batteries and spend no more than \$150.

Here is a systems of inequalities to represent Andre's constraints.

$$24x + 12y \geq 180$$

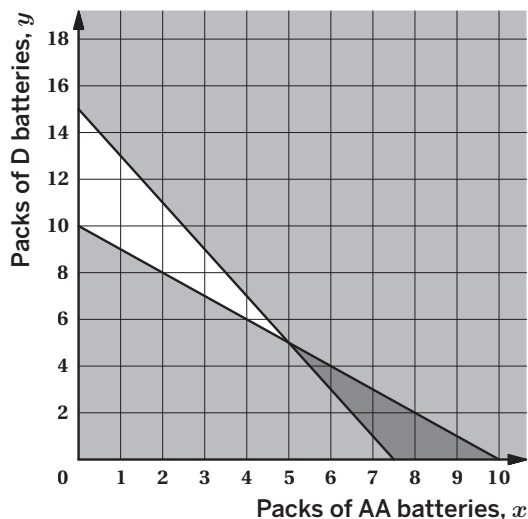
$$15x + 15y \leq 150$$

4. What is a combination of AA batteries and D batteries Andre that meets both constraints?

Responses vary. 8 packs of AA batteries and 2 packs of D batteries

5. What is a combination of AA batteries and D batteries that meets no constraints?

Responses vary. 2 packs of AA batteries and 10 packs of D batteries



Problems 6–7: A dog groomer charges \$24 to give a small dog a bath and \$30 to give a large dog a bath. A bath takes 30 minutes for a small dog and 1.25 hours for a large dog. The groomer works up to 6 hours each day, and she needs to earn at least \$180 a day.

$$24x + 30y \geq 180$$

$$0.5x + 1.25y \leq 6$$

6. Identify which of the following points meets the groomer's requirements. Explain your thinking.

A. (4, 4)

B. (6, 2)

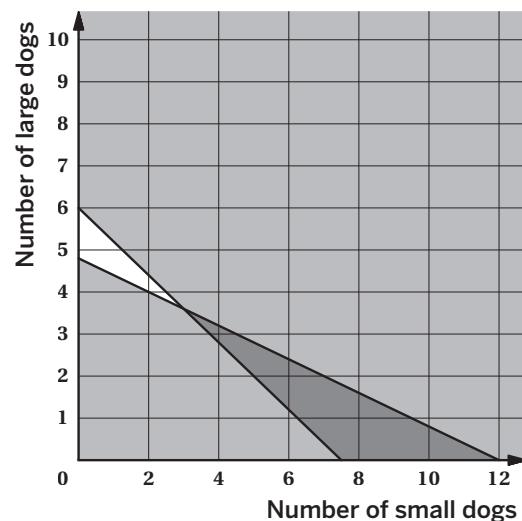
C. (7, 3)

D. (9, 2)

Explanations vary. The point (6, 2) is shaded by both inequalities, so I know it is a solution to the system of inequalities.

7. Explain why (8, 1.5) may be a solution to the systems of inequalities but does not represent a combination that meets both constraints.

Responses vary. The point (8, 1.5) is in the solution region of the system, but it is not possible to give a fractional number of dog baths.



Name: _____ Date: _____ Period: _____

Additional Practice

A1.5.09

1. Here is a system of inequalities.

$$y \leq -x + 4$$

$$2x + y > 8$$

Select all points that are solutions to the system of inequalities.

- ☒ A. (5, -1) ☒ B. (6, -3) ☐ C. (0, 0)
☒ D. (7, -5) ☐ E. (-5, 7) ☐ F. (6, 3)

Problems 2–3: The student council has \$300 to spend on burgers for a barbecue. They want to serve at least 240 burgers. A package of veggie burgers, x , costs \$10. A package of turkey burgers, y , costs \$20. The student council wrote this system of inequalities:

$$10x + 12y \geq 240$$

$$10x + 20y \leq 300$$

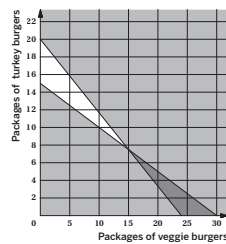
2. What does the inequality $10x + 20y \leq 300$ represent?

Responses vary. The inequality represents the combinations of packages of veggie burgers and turkey burgers that will cost no more than \$300.

3. Here is the graph of the system of inequalities.

Which of the following combinations of packages of veggie burgers and turkey burgers could the organizers buy?

- A. 0 veggie burger packages, 15 turkey burger packages
 B. 10 veggie burger packages, 12 turkey burger packages
 C. 15 veggie burger packages, 8 turkey burger packages
☒ D. 25 veggie burger packages, 2 turkey burger packages



Unit 5 Lesson 9

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Name: _____ Date: _____ Period: _____

Problems 4–5: Andre needs to buy batteries for his business.

AA batteries come in packs of 24 and cost \$15 per pack. D batteries come in packs of 12 and cost \$15 per pack. Andre wants to buy at least 180 batteries and spend no more than \$150.

Here is a systems of inequalities to represent Andre's constraints.

$$24x + 12y \geq 180$$

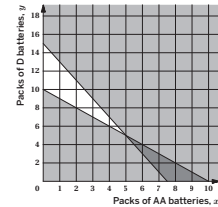
$$15x + 15y \leq 150$$

4. What is a combination of AA batteries and D batteries Andre that meets both constraints?

Responses vary. 8 packs of AA batteries and 2 packs of D batteries

5. What is a combination of AA batteries and D batteries that meets no constraints?

Responses vary. 2 packs of AA batteries and 10 packs of D batteries



Problems 6–7: A dog groomer charges \$24 to give a small dog a bath and \$30 to give a large dog a bath. A bath takes 30 minutes for a small dog and 1.25 hours for a large dog. The groomer works up to 6 hours each day, and she needs to earn at least \$180 a day.

$$24x + 30y \geq 180$$

$$0.5x + 1.25y \leq 6$$

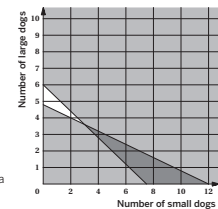
6. Identify which of the following points meets the groomer's requirements. Explain your thinking.

- A. (4, 4) ☒ B. (6, 2)
 C. (7, 3) D. (9, 2)

Explanations vary. The point (6, 2) is shaded by both inequalities, so I know it is a solution to the system of inequalities.

7. Explain why (8, 1.5) may be a solution to the systems of inequalities but does not represent a combination that meets both constraints.

Responses vary. The point (8, 1.5) is in the solution region of the system, but it is not possible to give a fractional number of dog baths.



Unit 5 Lesson 9

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.D.12
2	1	HSA.CED.A.3
3	1	HSA.CED.A.3
4	1	HSA.CED.A.3
5	3	HSA.CED.A.3
6	1	HSA.CED.A.3
7	2	HSA.CED.A.3

Notes:

Additional Practice

A1.5.10

Problems 1–2: Emika graphed the boundary lines for this system of inequalities.

$$2x + 5y \leq 10$$

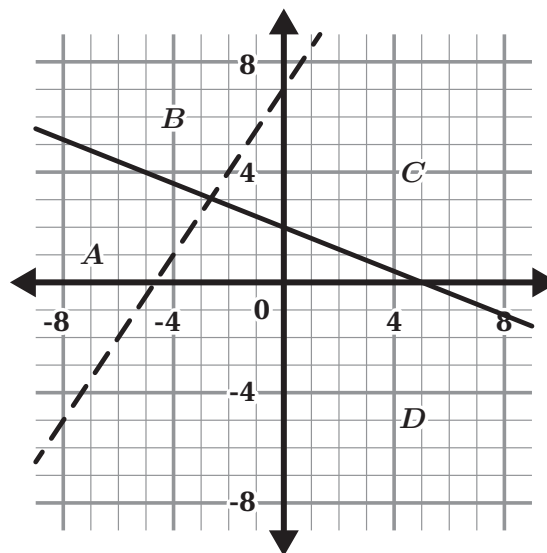
$$2y > 3x + 14$$

1. Which letter represents the solution region to the system of inequalities? Circle one:

☒ A B C D

2. Is the point $(4, -5)$ a solution to this system? Circle one:

Yes ☒ No



Problems 3–5: Taryn graphed the first inequality and the boundary line of the second inequality.

$$4x + y \leq 5$$

$$-x + 2y > -6$$

3. Complete the graph of the second inequality.
4. Explain how you knew where to shade the second inequality.

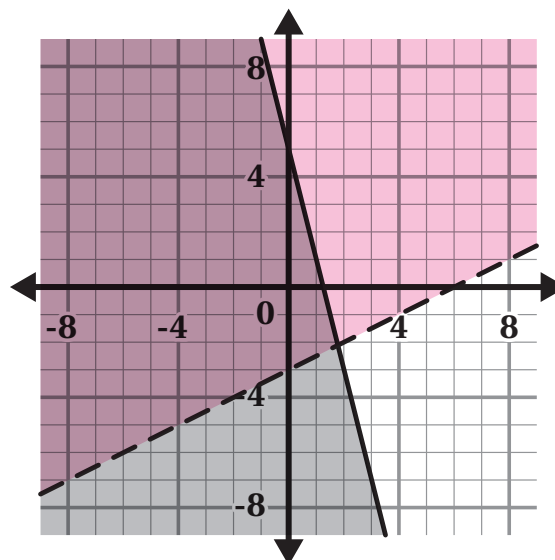
Explanations vary. I plugged $(0, 0)$ into $-x + 2y > -6$. Since $(0, 0)$ makes the inequality true, that side of the boundary line should be shaded.

5. Is $(0, -3)$ a solution to this system? Explain your thinking.

No. Explanations vary. The point $(0, -3)$ does not make the second inequality true:

$$-(0) + 2(-3) > -6$$

$$-6 > -6 \quad \text{False}$$



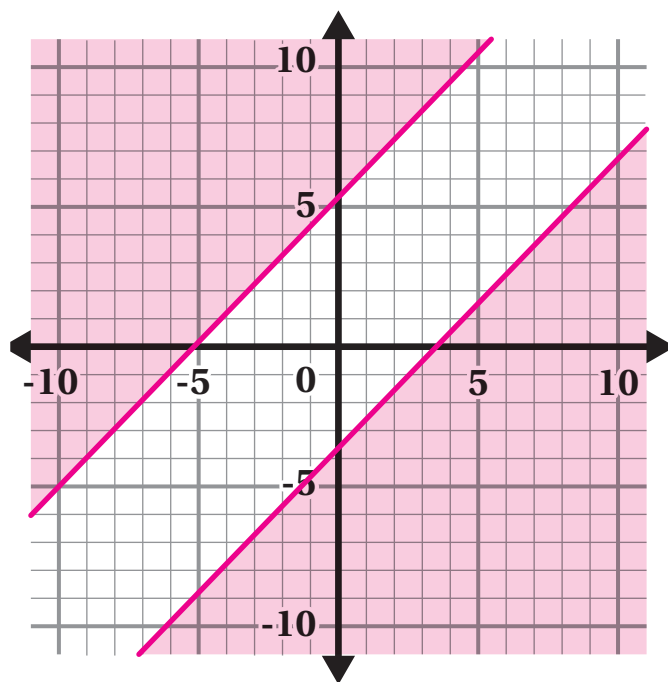
Problems 6–7: A coordinate plane is provided.

6. Create a graph of a system of inequalities that has no solutions.

Responses vary.

7. Explain how you know it has no solution.

Explanations vary. The boundary lines are parallel and the shading is in opposite directions, so the shaded regions will never overlap.



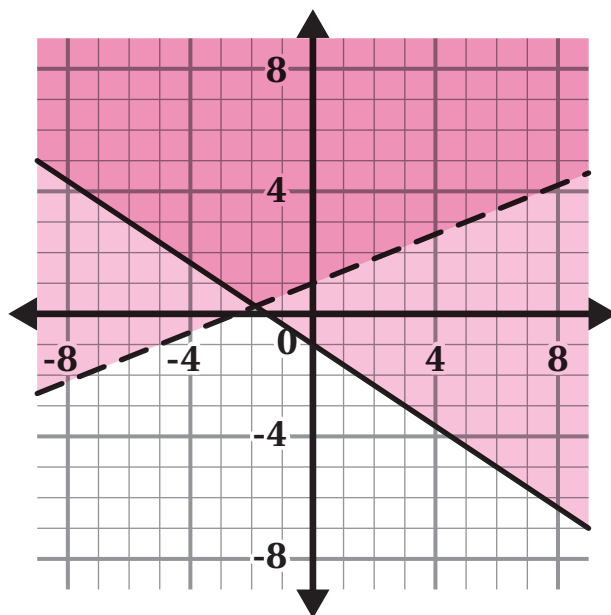
Problems 8–9: Randall graphed the boundary lines of this system of inequalities:

$$-2x + 5y > 5$$

$$2x + 3y \geq -3$$

8. Complete the graph of the system of inequalities.
9. Identify a coordinate pair that is in the solution region.

Responses vary. (2, 6).



Name: _____ Date: _____ Period: _____

Additional Practice

A1.5.10

Problems 1–2: Emika graphed the boundary lines for this system of inequalities.

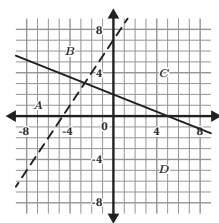
$$\begin{aligned} 2x + 5y &\leq 10 \\ 2y &> 3x + 14 \end{aligned}$$

1. Which letter represents the solution region to the system of inequalities? Circle one:

☒ A ☐ B ☐ C ☐ D

2. Is the point $(4, -5)$ a solution to this system? Circle one:

Yes ☒ No



Problems 3–5: Taryn graphed the first inequality and the boundary line of the second inequality.

$$\begin{aligned} 4x + y &\leq 5 \\ -x + 2y &> -6 \end{aligned}$$

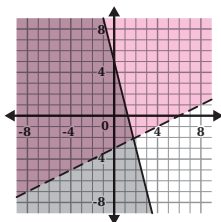
3. Complete the graph of the second inequality.

4. Explain how you knew where to shade the second inequality.

Explanations vary. I plugged $(0, 0)$ into $-x + 2y > -6$. Since $(0, 0)$ makes the inequality true, that side of the boundary line should be shaded.

5. Is $(0, -3)$ a solution to this system? Explain your thinking.

*No. Explanations vary. The point $(0, -3)$ does not make the second inequality true: $-(0) + 2(-3) > -6$
 $-6 > -6$ False*



Unit 5 Lesson 10

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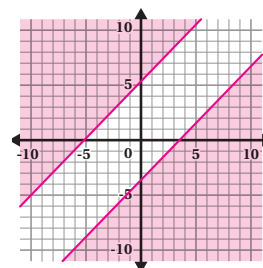
Name: _____ Date: _____ Period: _____

Problems 6–7: A coordinate plane is provided.

6. Create a graph of a system of inequalities that has no solutions.
Responses vary.

7. Explain how you know it has no solution.

Explanations vary. The boundary lines are parallel and the shading is in opposite directions, so the shaded regions will never overlap.

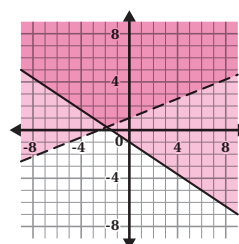


Problems 8–9: Randall graphed the boundary lines of this system of inequalities:

$$\begin{aligned} -2x + 5y &> 5 \\ 2x + 3y &\geq -3 \end{aligned}$$

8. Complete the graph of the system of inequalities.

9. Identify a coordinate pair that is in the solution region.
Responses vary. $(2, 6)$.



Unit 5 Lesson 10

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.D.12
2	1	HSA.REI.D.12
3	1	HSA.REI.D.12
4	2	HSA.REI.D.12
5	2	HSA.REI.D.12
6	2	HSA.REI.D.12
7	2	HSA.REI.D.12
8	2	HSA.REI.D.12
9	1	HSA.REI.D.12

Notes:

Additional Practice

A1.6.01

Problems 1–3: Wendy is observing a colony of mice in which its population doubles every week. The table shows the population of mice over time, in weeks.

1. What was the population of the mice at the beginning of the experiment?

6 mice

2. What will the population of mice be after 6 weeks?

384 mice

Time (weeks)	Population
0	?
1	12
2	24
3	48
4	96
...	...
6	?

3. Ricardo observed a different colony of mice that he believes populates at a rate slower than Wendy's mice. He wrote this equation: $p = 12 \cdot 1.5^t$. He used p to represent the population of mice in his colony and t for time, in weeks. Explain what the 12 and 1.5 mean in this situation.

The 12 represents the number of mice in Ricardo's colony at the start of his observation, and the 1.5 is the rate at which they are growing each week.

4. In the table, x represents the number of minutes and y represents the number of bacteria cells in a sample.

x	0	8	16	24	32	40	48	56
y	4	8	16	32	64	128	256	512

Select *all* the statements that describe the growth.

- ☒ **A.** Every 8 minutes, the number of bacteria cells doubles.
- ☐ **B.** Every 4 minutes, the number of bacteria cells increases by 8.
- ☐ **C.** At the end of the experiment, there are 56 bacteria cells in the sample.
- ☒ **D.** At the beginning of the experiment, there are 4 bacteria cells in the sample.
- ☐ **E.** At the beginning of the experiment, there are 8 bacteria cells in the sample.

Problems 5–7: Caroline has a new toy fish whose mass quadruples every hour when you add water. The fish’s weight was initially 2 grams.

5. Select *all* the expressions that represent the weight of Caroline’s toy fish, in grams, after 3 hours?

☒ A. $2 \cdot 4 \cdot 4 \cdot 4$

☐ B. $2 \cdot 2 \cdot 2$

☒ C. $2 \cdot 4^3$

☐ D. $2 + 4 \cdot 4 \cdot 4$

☐ E. $2 \cdot 3^4$

☐ F. $2 + 4^3$

6. Complete the table.

Time (hrs)	0	1	2	3	4
Weight (grams)	2	8	32	128	512

Problems 7–8: Dion is reading a book about a zombie apocalypse. The story began when 10 zombies entered a convention center where people were attending a concert. The total number of infected people increased by 2.5 times each day.

7. Write an equation that represents this situation where n represents the total number of people infected by the zombies and t represents the number of days that have passed since the apocalypse began.

$$n = 10(2.5)^t \text{ (or equivalent)}$$

8. Approximately how many people will be infected after 4 days have passed?

About 391 people

Name: _____ Date: _____ Period: _____

Additional Practice

A1.6.01

Problems 1–3: Wendy is observing a colony of mice in which its population doubles every week. The table shows the population of mice over time, in weeks.

1. What was the population of the mice at the beginning of the experiment?
6 mice

2. What will the population of mice be after 6 weeks?
384 mice

Time (weeks)	Population
0	?
1	12
2	24
3	48
4	96
...	...
6	?

3. Ricardo observed a different colony of mice that he believes populates at a rate slower than Wendy's mice. He wrote this equation: $p = 12 \cdot 1.5^t$. He used p to represent the population of mice in his colony and t for time, in weeks. Explain what the 12 and 1.5 mean in this situation.

The 12 represents the number of mice in Ricardo's colony at the start of his observation, and the 1.5 is the rate at which they are growing each week.

4. In the table, x represents the number of minutes and y represents the number of bacteria cells in a sample.

x	0	8	16	24	32	40	48	56
y	4	8	16	32	64	128	256	512

Select all the statements that describe the growth.

- ☒ **A.** Every 8 minutes, the number of bacteria cells doubles.
☐ **B.** Every 4 minutes, the number of bacteria cells increases by 8.
☐ **C.** At the end of the experiment, there are 56 bacteria cells in the sample.
☒ **D.** At the beginning of the experiment, there are 4 bacteria cells in the sample.
☐ **E.** At the beginning of the experiment, there are 8 bacteria cells in the sample.

Unit A1.6 Lesson 1

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Name: _____ Date: _____ Period: _____

Problems 5–7: Caroline has a new toy fish whose mass quadruples every hour when you add water. The fish's weight was initially 2 grams.

5. Select *all* the expressions that represent the weight of Caroline's toy fish, in grams, after 3 hours?

- ☒ **A.** $2 \cdot 4 \cdot 4 \cdot 4$ ☐ **B.** $2 \cdot 2 \cdot 2$ ☒ **C.** $2 \cdot 4^3$
☐ **D.** $2 + 4 \cdot 4 \cdot 4$ ☐ **E.** $2 \cdot 3^3$ ☐ **F.** $2 + 4^3$

6. Complete the table.

Time (hrs)	0	1	2	3	4
Weight (grams)	2	8	32	128	512

Problems 7–8: Dion is reading a book about a zombie apocalypse. The story began when 10 zombies entered a convention center where people were attending a concert. The total number of infected people increased by 2.5 times each day.

7. Write an equation that represents this situation where n represents the total number of people infected by the zombies and t represents the number of days that have passed since the apocalypse began.

$n = 10(2.5)^t$ (or equivalent)

8. Approximately how many people will be infected after 4 days have passed?

About 391 people

Unit A1.6 Lesson 1

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.BF.A.1.A
2	1	HSF.BF.A.1.A
3	2	HSF.LE.B.5
4	2	HSF.BF.A.1.A
5	2	HSF.LE.B.5
6	1	HSF.BF.A.1.A
7	1	HSF.BF.A.1.A
8	1	HSF.LE.B.5

Notes:

Additional Practice

A1.6.02

Problems 1–3: These tables show the number of blue and green globs each day.

1. How many of each type of glob will there be on day 4?

Day	0	1	2	3	4
Blue Globs	4	8	16	32	64

Day	0	1	2	3	4
Green Globs	12	24	36	48	60

2. Will there be more blue or green globs on day 10? Show or explain your thinking.

There will be more blue globs on day 10 because they are increasing at a faster rate than the green globs.

3. Which group of globs changes by a constant rate of change? Show or explain how you know.

The green globs are increasing at a constant rate of 12 globs per day. $24 - 12 = 12$ and $48 - 36 = 12$, which shows a constant rate. The blue globs double each day, which represents a constant ratio.

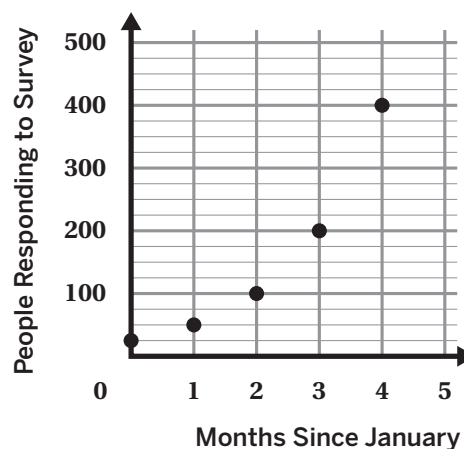
Problems 4–5: This graph shows the number of people who responded to an online survey about a new neighborhood park since January.

4. How many people completed the survey in January?

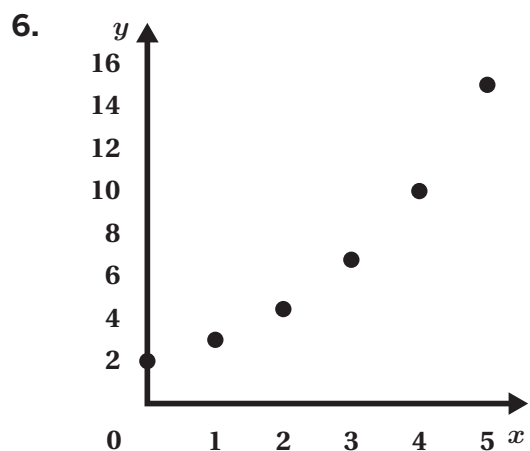
About 25 people

5. Does the number of people responding to the survey grow by a constant difference? Show or explain how you know.

No. Explanations vary. The graph of this relationship is a curve that is growing steeper with each passing month. The graph of a constant difference would look like a straight line.

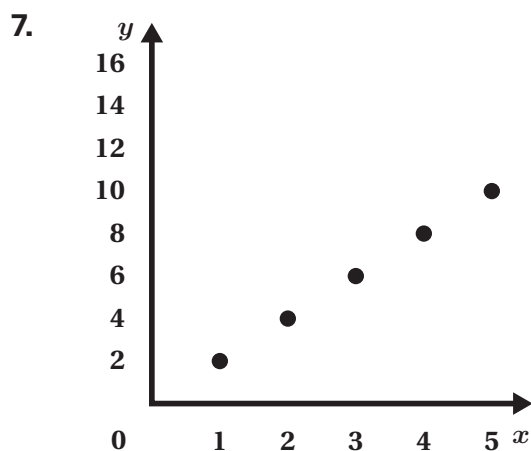


Problems 6–7: Determine whether each graph shows a constant rate of change or a constant growth rate. Circle your choice.



Constant rate
of change

Constant growth
rate



Constant rate
of change

Constant growth
rate

Problems 8–9: Determine whether each table shows a linear or exponential function. Circle your choice.

8.

x	1	2	3	4
y	1	8	15	22

Linear

Exponential

9.

x	1	2	3	4
y	9	18	36	72

Linear

Exponential

Name: _____ Date: _____ Period: _____

Additional Practice

A1.6.02

Problems 1–3: These tables show the number of blue and green globs each day.

1. How many of each type of glob will there be on day 4?

Day	0	1	2	3	4
Blue Globs	4	8	16	32	64

Day	0	1	2	3	4
Green Globs	12	24	36	48	60

2. Will there be more blue or green globs on day 10? Show or explain your thinking.

There will be more blue globs on day 10 because they are increasing at a faster rate than the green globs.

3. Which group of globs changes by a constant rate of change? Show or explain how you know.

The green globs are increasing at a constant rate of 12 globs per day. $24 - 12 = 12$ and $48 - 36 = 12$, which shows a constant rate. The blue globs double each day, which represents a constant ratio.

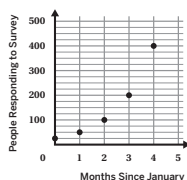
Problems 4–5: This graph shows the number of people who responded to an online survey about a new neighborhood park since January.

4. How many people completed the survey in January?

About 25 people

5. Does the number of people responding to the survey grow by a constant difference? Show or explain how you know.

No. Explanations vary. The graph of this relationship is a curve that is growing steeper with each passing month. The graph of a constant difference would look like a straight line.



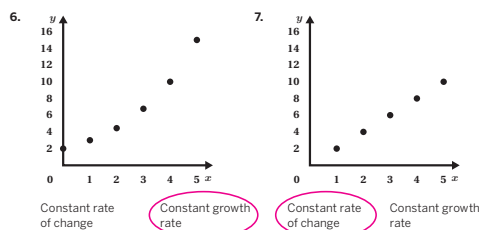
Unit A1.6 Lesson 2

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Name: _____ Date: _____ Period: _____

Problems 6–7: Determine whether each graph shows a constant rate of change or a constant growth rate. Circle your choice.



Constant rate of change

Constant growth rate

Constant rate of change

Constant growth rate

Problems 8–9: Determine whether each table shows a linear or exponential function. Circle your choice.

x	1	2	3	4
y	1	8	15	22

Linear

Exponential

x	1	2	3	4
y	9	18	36	72

Linear

Exponential

Unit A1.6 Lesson 2

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.B.4
2	2	HSF.LE.A.3
3	2	HSF.LE.A.3
4	1	HSF.IF.B.4
5	2	HSF.LE.A.3
6	1	HSF.LE.A.3
7	1	HSF.LE.A.3
8	1	HSF.LE.A.3
9	1	HSF.LE.A.3

Notes:

Additional Practice

A1.6.03

Problems 1–3: Determine the value of each expression when $x = 3$.

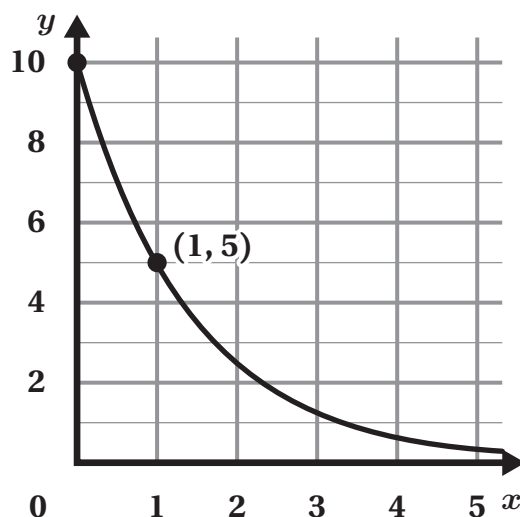
1. $4^x = 64$

2. $\left(\frac{1}{2}\right)^x = \frac{1}{8}$ (or equivalent)

3. $4(2)^x = 32$

4. Here is a graph of $y = 10\left(\frac{1}{2}\right)^x$. Explain where you can see the 10 and the $\frac{1}{2}$ in the graph.

The 10 is the y -intercept of the function. The $\frac{1}{2}$ is the growth rate because at $x = 1$, the y -value is 5, which is $\frac{1}{2}$ of 10.



Problems 5–7: Match each equation to the graph that represents it.

Equation A

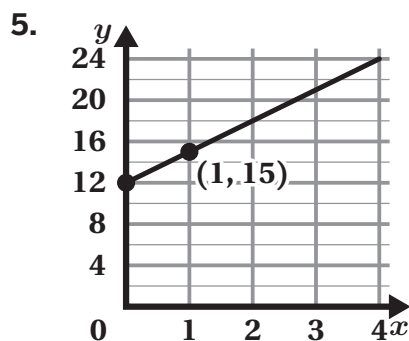
$$y = 12\left(\frac{1}{3}\right)^x$$

Equation B

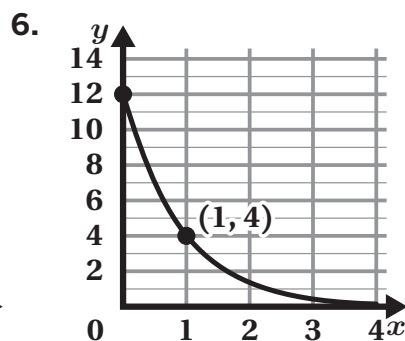
$$y = 12(3)^x$$

Equation C

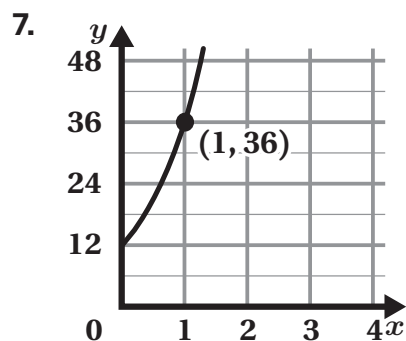
$$y = 12 + 3x$$



Equation C



Equation A



Equation B

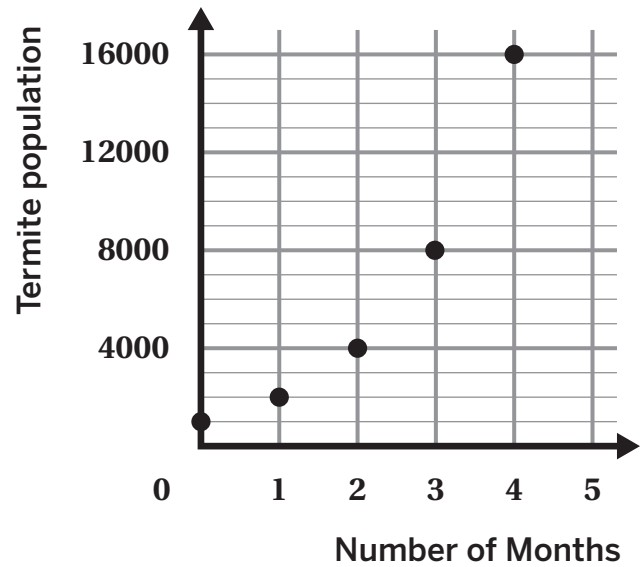
8. Explain how you determined which equation to match with the graph in Problem 7.

Responses vary. I looked for the function that had a growth rate of 3.

Problems 9–10: The number of termites in a colony is measured each month and the results are plotted on the graph.

9. What was the termite population when it was first measured?

Approximately 1,000 termites



10. What equation represents the termite population, t , for m months after it was first measured?

$$t = 1000(2)^t$$

Name: _____ Date: _____ Period: _____

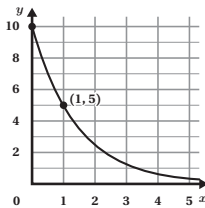
Additional Practice

A1.6.03

Problems 1–3: Determine the value of each expression when $x = 3$.

1. $4^x = 64$ 2. $\left(\frac{1}{2}\right)^x = \frac{1}{8}$ (or equivalent) 3. $4(2)^x = 32$

4. Here is a graph of $y = 10\left(\frac{1}{2}\right)^x$. Explain where you can see the 10 and the $\frac{1}{2}$ in the graph.
The 10 is the y -intercept of the function. The $\frac{1}{2}$ is the growth rate because at $x = 1$, the y -value is 5, which is $\frac{1}{2}$ of 10.

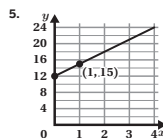


Problems 5–7: Match each equation to the graph that represents it.

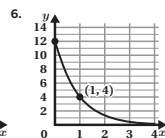
Equation A
 $y = 12\left(\frac{1}{3}\right)^x$

Equation B
 $y = 12(3)^x$

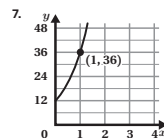
Equation C
 $y = 12 + 3x$



Equation C



Equation A



Equation B

8. Explain how you determined which equation to match with the graph in Problem 7.
Responses vary. I looked for the function that had a growth rate of 3.

Unit A1.6 Lesson 3

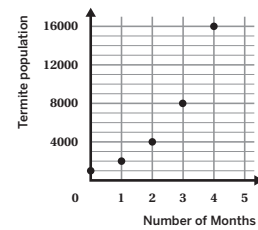
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Name: _____ Date: _____ Period: _____

Problems 9–10: The number of termites in a colony is measured each month and the results are plotted on the graph.

9. What was the termite population when it was first measured?
Approximately 1,000 termites



10. What equation represents the termite population, t , for m months after it was first measured?

$t = 1000(2)^m$

Unit A1.6 Lesson 3

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSN.Q.A.1
2	1	HSN.Q.A.1
3	1	HSN.Q.A.1
4	1	HSF.IF.B.4
5	1	HSF.IF.B.4, HSF.LE.A.2, HSF.LE.A.3
6	1	HSF.IF.B.4, HSF.LE.A.2, HSF.LE.A.3
7	1	HSF.IF.B.4, HSF.LE.A.2, HSF.LE.A.3
8	2	HSF.IF.B.4
9	1	HSF.IF.B.4
10	1	HSA.CED.A.2, HSF.LE.A.2

Notes:

Additional Practice

A1.6.05

1. Which equation best models the data in the table?

A. $f(x) = 48(1.25)^x$

B. $f(x) = 36 + 0.75$

C. $f(x) = 48(0.75)^x$

D. $f(x) = 36(0.75)^x$

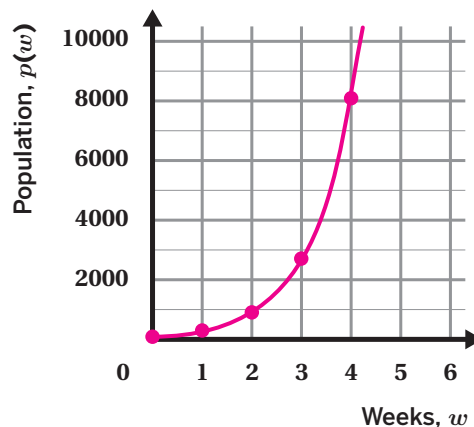
x	$f(x)$
1	36
2	27
3	20.25
4	15.1875

Problems 2–4: The equation $p(w) = 100 \cdot 3^w$ models a butterfly population, $p(w)$, where w is the number of weeks after the population was first measured.

2. Complete the table.

Weeks, w	Population $p(w)$
0	100
1	300
2	900
3	2,700
4	8,100

3. Graph the situation.



4. Where on the graph do you see the 100 from the equation?

The 100 in the equation is represented by the y -intercept.

5. Determine the value of $p(-3)$ and explain what it means in this situation.

$p(-3) \approx 3.7$. This means that 3 weeks before the population was initially measured, there were around 3–4 butterflies.

Problems 6–8: A walrus population, $p(t)$, is modeled by the equation $p(t) = 300 \cdot \left(\frac{4}{3}\right)^t$ where t is the time, in years, since the population was first measured in 2018.

6. How many walrus were in the population in 2018? Show or explain your thinking.

300 walrus. Explanations vary. I substituted 0 for t in the equation.

$$p(0) = 300 \cdot \left(\frac{4}{3}\right)^0 = 300 \text{ walrus}$$

7. How many walrus were in the population in 2022? Show or explain your thinking.

948 walrus. Explanations vary. I substituted 4 for t in the equation.

$$p(4) = 300 \cdot \left(\frac{4}{3}\right)^4 \approx 948 \text{ walrus}$$

8. How many walrus were in the population in 2016? Show or explain your thinking.

About 169 walrus. Explanations vary. I substituted -2 for t in the equation.

$$p(-2) = 300 \cdot \left(\frac{4}{3}\right)^{-2} \approx 169 \text{ walrus}$$

Name: _____ Date: _____ Period: _____

Additional Practice

A1.6.05

1. Which equation best models the data in the table?

- A. $f(x) = 48(1.25)^x$
 B. $f(x) = 36 + 0.75$
C. $f(x) = 48(0.75)^x$
 D. $f(x) = 36(0.75)^x$

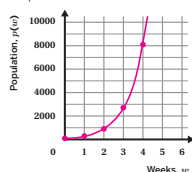
x	$f(x)$
1	36
2	27
3	20.25
4	15.1875

Problems 2–4: The equation $p(w) = 100 \cdot 3^w$ models a butterfly population, $p(w)$, where w is the number of weeks after the population was first measured.

2. Complete the table.

Weeks, w	Population $p(w)$
0	100
1	300
2	900
3	2,700
4	8,100

3. Graph the situation.



4. Where on the graph do you see the 100 from the equation?

The 100 in the equation is represented by the y -intercept.

5. Determine the value of $p(-3)$ and explain what it means in this situation.

$p(-3) \approx 3.7$. This means that 3 weeks before the population was initially measured, there were around 3–4 butterflies.

Unit A1.6 Lesson 5

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Name: _____ Date: _____ Period: _____

Problems 6–8: A walrus population, $p(t)$, is modeled by the equation $p(t) = 300 \cdot \left(\frac{4}{3}\right)^t$, where t is the time, in years, since the population was first measured in 2018.

6. How many walrus were in the population in 2018? Show or explain your thinking.

300 walrus. *Explanations vary. I substituted 0 for t in the equation.*
 $p(0) = 300 \cdot \left(\frac{4}{3}\right)^0 = 300$ walrus

7. How many walrus were in the population in 2022? Show or explain your thinking.

948 walrus. *Explanations vary. I substituted 4 for t in the equation.*
 $p(4) = 300 \cdot \left(\frac{4}{3}\right)^4 \approx 948$ walrus

8. How many walrus were in the population in 2016? Show or explain your thinking.

About 169 walrus. *Explanations vary. I substituted -2 for t in the equation.*
 $p(-2) = 300 \cdot \left(\frac{4}{3}\right)^{-2} \approx 169$ walrus

Unit A1.6 Lesson 5

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA-CED.A.2, HSF.LE.A.2
2	1	HSF.IF.A.2
3	1	HSA-CED.A.2
4	1	HSF.IF.B.4
5	2	HSF.IF.A.2
6	2	HSF.IF.A.2
7	2	HSF.IF.A.2
8	2	HSF.IF.A.2

Notes:

Additional Practice**A1.6.07**

Problems 1–3: A group of ornithologists were researching the hummingbird population in a state park over several years. There were 500 hummingbirds when they first started keeping track. The population has increased by 20% each year.

1. How many hummingbirds are in the state park 1 year after the ornithologists first counted?

600 hummingbirds

2. Write an expression that represents the hummingbird population after 3 years.

$500 \cdot 1.20^3$ (or equivalent)

3. Write an expression that represents the hummingbird population after n years.

$500 \cdot 1.20^n$ (or equivalent)

4. Last year, the attendance at a water park was 825,000 people. This year, attendance increased by 7%. Select *all* the expressions that represent the attendance at the water park this year.

☐ A. $825000 + 0.07$

☒ B. $825000(1.07)$

☒ C. $825000(1 + 0.07)$

☐ D. $825000(1 - 0.07)$

☒ E. $825000 + 0.07(825000)$

Problems 5–6: Kendra deposited \$1,200 in her savings account with an annual interest rate of 3.5%.

5. Write a function, $f(t)$, to represent the amount Kendra will have in her account, in dollars, after t years.

$$f(t) = 1200 \cdot 1.035^t$$

6. Complete the table to determine how much money Kendra will have in her savings account over time, if no additional deposits or withdrawals are made.

Time (yr)	Amount in Account (\$)
0	1200
1	1242
2	1285.47
3	1330.46
4	1377.03

Problems 7–9: Three cities have the same initial population and different percent increases each year. Match each function $p(t)$, representing the population after t years, with its correct description.

$$p(t) = 12000 \cdot (1.45)^t$$

$$p(t) = 12000 \cdot (1.045)^t$$

$$p(t) = 12000 \cdot (1.0045)^t$$

7. City A has a 4.5% annual increase in population.
8. City B has a 0.45% annual increase in population.
9. City C has a 45% annual increase in population.

$$p(t) = 12000 \cdot (1.045)^t$$

$$p(t) = 12000 \cdot (1.0045)^t$$

$$p(t) = 12000 \cdot (1.45)^t$$

Name: _____ Date: _____ Period: _____

Additional Practice

A1.6.07

Problems 1–3: A group of ornithologists were researching the hummingbird population in a state park over several years. There were 500 hummingbirds when they first started keeping track. The population has increased by 20% each year.

- How many hummingbirds are in the state park 1 year after the ornithologists first counted?
600 hummingbirds
- Write an expression that represents the hummingbird population after 3 years.
 $500 \cdot 1.20^3$ (or equivalent)
- Write an expression that represents the hummingbird population after n years.
 $500 \cdot 1.20^n$ (or equivalent)
- Last year, the attendance at a water park was 825,000 people. This year, attendance increased by 7%. Select *all* the expressions that represent the attendance at the water park this year.
 - ☐ A. $825000 + 0.07$
 - ☒ B. $825000(1.07)$
 - ☒ C. $825000(1 + 0.07)$
 - ☐ D. $825000(1 - 0.07)$
 - ☒ E. $825000 + 0.07(825000)$

Unit A1.6 Lesson 7

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Name: _____ Date: _____ Period: _____

Problems 5–6: Kendra deposited \$1,200 in her savings account with an annual interest rate of 3.5%.

- Write a function, $f(t)$, to represent the amount Kendra will have in her account, in dollars, after t years.
 $f(t) = 1200 \cdot 1.035^t$

- Complete the table to determine how much money Kendra will have in her savings account over time, if no additional deposits or withdrawals are made.

Time (yr)	Amount in Account (\$)
0	1200
1	1242
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3	1330.46
4	1377.03

Problems 7–9: Three cities have the same initial population and different percent increases each year. Match each function $p(t)$, representing the population after t years, with its correct description.

$$p(t) = 12000 \cdot (1.45)^t$$

$$p(t) = 12000 \cdot (1.045)^t$$

$$p(t) = 12000 \cdot (1.0045)^t$$

- City A has a 4.5% annual increase in population.
 $p(t) = 12000 \cdot (1.045)^t$
- City B has a 0.45% annual increase in population.
 $p(t) = 12000 \cdot (1.0045)^t$
- City C has a 45% annual increase in population.
 $p(t) = 12000 \cdot (1.45)^t$

Unit A1.6 Lesson 7

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.CED.A.2
2	1	HSA.CED.A.2
3	1	HSA.CED.A.2
4	2	HSA.SSE.A.1.A, HSA.SSE.A.1.B
5	2	HSF.LE.A.2
6	2	HSF.LE.A.2
7	1	HSF.LE.A.2
8	1	HSF.LE.A.2
9	1	HSF.LE.A.2

Notes:

Additional Practice

A1.6.10

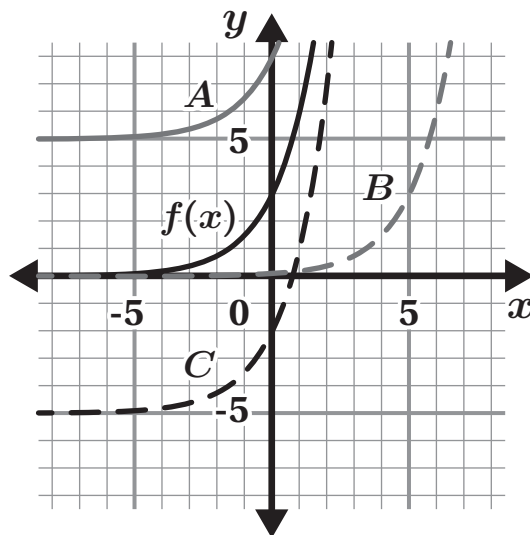
1. Here is a graph of $f(x) = 3 \cdot 2^x$.

Match each function with its graph.

$f(x) = 3 \cdot 2^x - 5$ C

$f(x) = 3 \cdot 2^x + 5$ A

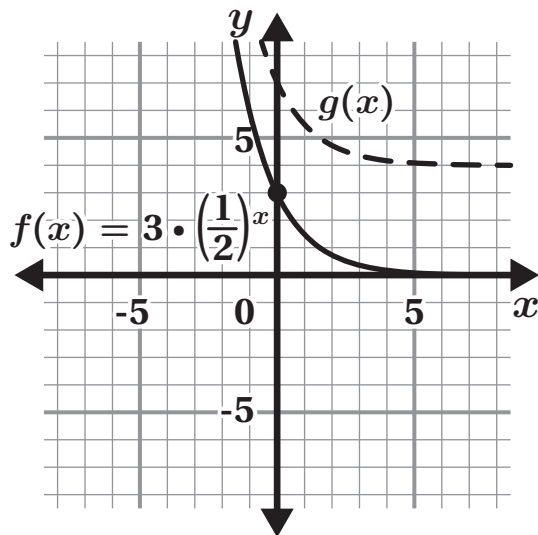
$f(x) = 3 \cdot 2^{x-5}$ B



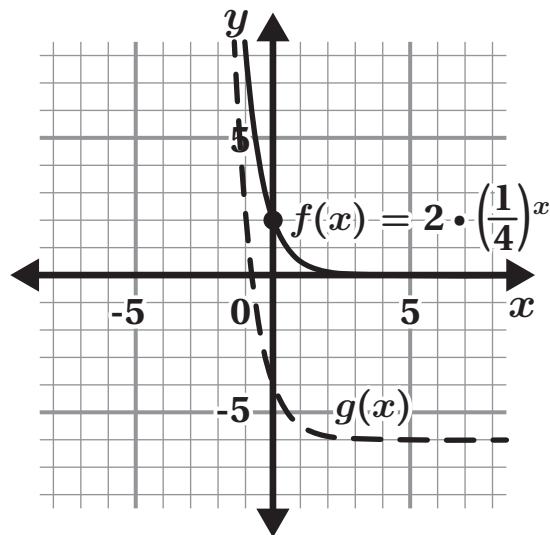
Problems 2–3: For each set of graphs shown, write an equation for the dotted curve, $g(x)$.

2. $f(x) = 3 \cdot \left(\frac{1}{2}\right)^x$

3. $f(x) = 2 \cdot \left(\frac{1}{4}\right)^x$



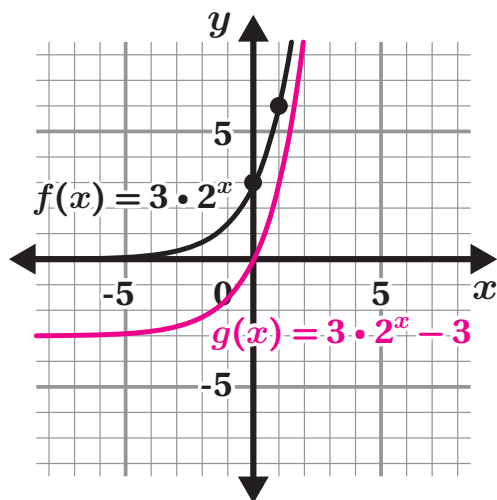
$g(x) = 3 \cdot \left(\frac{1}{2}\right)^x + 4$



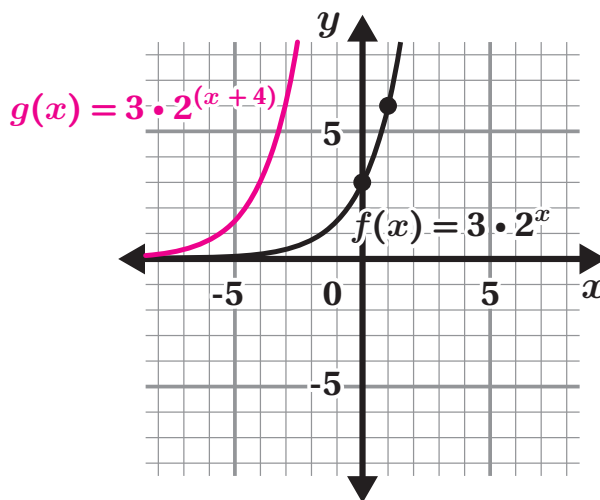
$g(x) = 2 \cdot \left(\frac{1}{4}\right)^x - 6$

Problems 4–5: The function $g(x)$ is a transformation of $f(x) = 3 \cdot 2^x$.

4. Graph $g(x) = 3 \cdot 2^x - 3$



5. Graph $g(x) = 3 \cdot 2^{(x+4)}$



Problems 6–8: The function $g(x)$ is a transformation of $f(x) = 4 \cdot \left(\frac{1}{2}\right)^x$.

6. Melanie says $g(x) = 4 \cdot \left(\frac{1}{2}\right)^{x-4}$ because the point $(0, 4)$ on $f(x)$ shifted to the left 4 units. Explain why Melanie's thinking is incorrect.

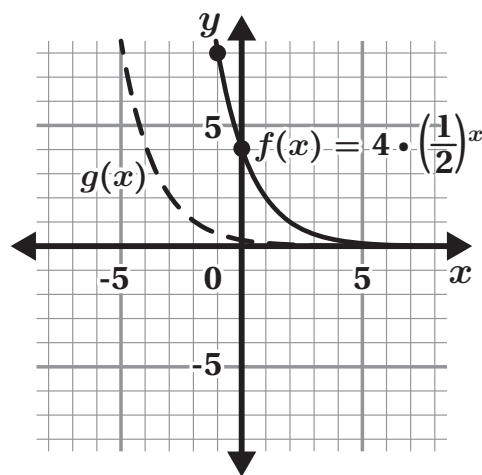
Explanations vary. Melanie is correct that the function $g(x)$ is a horizontal translation 4 units to the left, but this would result in an exponent of $(x - (-4))$ or $(x + 4)$.

7. Yousef says $f(x)$ transformed using a vertical translation because the y -intercept shifted down about 3.5 units. Explain why Yousef's thinking is incorrect.

Explanations vary. If $f(x)$ was shifted down by about 3.5 units, then every point on $f(x)$ would be shifted down by about 3.5 units, and this is not the case. For example, the point $(2, 1)$ on $f(x)$ shifted down only about 1 unit.

8. Write the correct equation for $g(x)$.

$$g(x) = 4 \cdot \left(\frac{1}{2}\right)^{x+4}$$



Name: _____ Date: _____ Period: _____

Additional Practice

A1.6.10

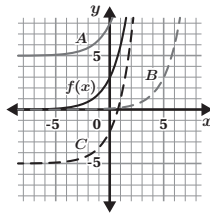
1. Here is a graph of $f(x) = 3 \cdot 2^x$.

Match each function with its graph.

$f(x) = 3 \cdot 2^x - 5$ **C**

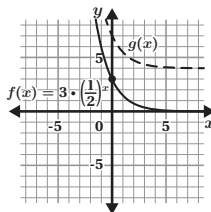
$f(x) = 3 \cdot 2^x + 5$ **A**

$f(x) = 3 \cdot 2^{x-5}$ **B**



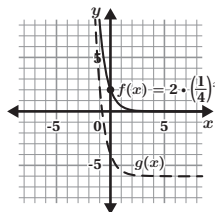
Problems 2–3: For each set of graphs shown, write an equation for the dotted curve, $g(x)$.

2. $f(x) = 3 \cdot \left(\frac{1}{2}\right)^x$



$g(x) = 3 \cdot \left(\frac{1}{2}\right)^x + 4$

3. $f(x) = 2 \cdot \left(\frac{1}{4}\right)^x$



$g(x) = 2 \cdot \left(\frac{1}{4}\right)^x - 6$

Unit A1.6 Lesson 10

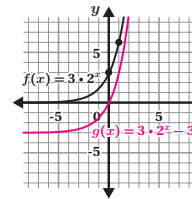
165

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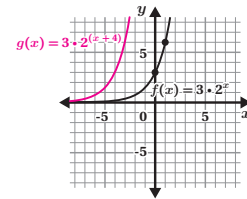
Name: _____ Date: _____ Period: _____

Problems 4–5: The function $g(x)$ is a transformation of $f(x) = 3 \cdot 2^x$.

4. Graph $g(x) = 3 \cdot 2^x - 3$



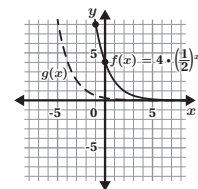
5. Graph $g(x) = 3 \cdot 2^{x+4}$



Problems 6–8: The function $g(x)$ is a transformation of $f(x) = 4 \cdot \left(\frac{1}{2}\right)^x$.

6. Melanie says $g(x) = 4 \cdot \left(\frac{1}{2}\right)^{x-4}$ because the point $(0, 4)$ on $f(x)$ shifted to the left 4 units. Explain why Melanie's thinking is incorrect.

Explanations vary. Melanie is correct that the function $g(x)$ is a horizontal translation 4 units to the left, but this would result in an exponent of $(x - (-4))$ or $(x + 4)$.



7. Yousef says $f(x)$ transformed using a vertical translation because the y -intercept shifted down about 3.5 units. Explain why Yousef's thinking is incorrect.

Explanations vary. If $f(x)$ was shifted down by about 3.5 units, then every point on $f(x)$ would be shifted down by about 3.5 units, and this is not the case. For example, the point $(2, 1)$ on $f(x)$ shifted down only about 1 unit.

8. Write the correct equation for $g(x)$.

$g(x) = 4 \cdot \left(\frac{1}{2}\right)^{x+4}$

Unit A1.6 Lesson 10

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.BF.B.3, HSF.IF.C.7.E*
2	2	HSF.BF.B.3
3	2	HSF.BF.B.3
4	2	HSF.BF.B.3, HSF.IF.C.7.E*
5	2	HSF.BF.B.3, HSF.IF.C.7.E*
6	2	HSF.BF.B.3
7	2	HSF.BF.B.3
8	2	HSF.BF.B.3

Notes:

Additional Practice

A1.6.13

Problems 1–4: Determine if each equation or table represents simple or compound interest. Circle your choice.

1. $b(t) = 500 + 25t$

Simple

Compound

2. $b(t) = 500(1.05)^t$

Simple

Compound

3.

Time (yr)	Account Balance (\$)
0	100
1	125
2	156.25

Simple

Compound

4.

Time (yr)	Account Balance (\$)
0	100
1	125
2	150

Simple

Compound

Problems 5–7: Helen invests \$2,000 in an account that earns 3.5% compound interest per year.

5. Complete the table.

6. Which function represents the amount of money in Helen's account after x years?

A. $f(x) = 2000 + 1.035x$

B. $f(x) = 2000(1.035x)$

C. $f(x) = 2000 + 1.035^x$

D. $f(x) = 2000(1.035)^x$

Time (yr)	Account Balance (\$)
0	2000
1	2070
2	2142.45
3	2217.44
4	2295.05

7. What will the balance of the account be after 8 years?

\$2,633.62

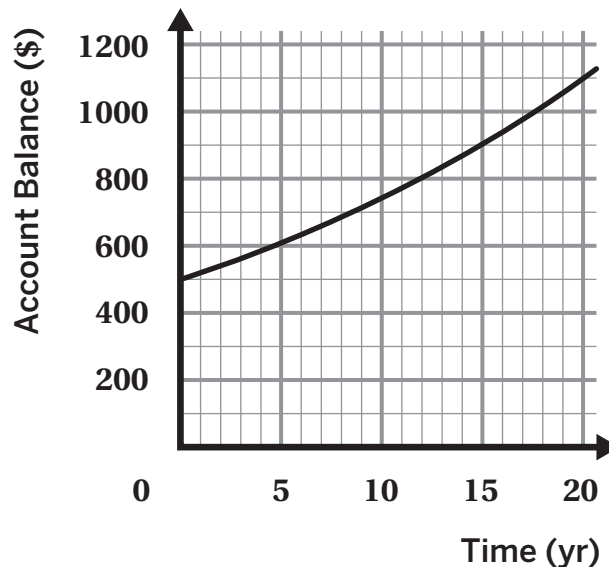
Problems 8–9: Samuel invests \$500 in an account that earns 4% compound interest per year. The graph shows the function $f(t) = 500(1.04)^t$, which gives Samuel's account balance after t years.

8. About how many years will it take for his account balance to reach \$1,000?

About 18 years

9. Use the graph to determine the value of $f(12)$.

$f(12)$ is equal to about 800.



What does that tell you about the situation?

This tells me that after 12 years, Samuel's account balance will be about \$800.

10. You just won a contest and have two prize options:

Option A: Getting a penny that doubles every day for 30 days

Option B: One payment of \$10 million dollars

Which option would you choose? Explain your choice.

Responses and explanations vary.

- I would choose Option A. I calculated the value after 30 days and it was \$10,737,418.24, which is more than Option B.
- I would choose Option B because I would want the money all at once.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.6.13

Problems 1–4: Determine if each equation or table represents simple or compound interest. Circle your choice.

1. $b(t) = 500 + 25t$

Simple

Compound

2. $b(t) = 500(1.05)^t$

Simple

Compound

3.

Time (yr)	Account Balance (\$)
0	100
1	125
2	156.25

Simple

Compound

4.

Time (yr)	Account Balance (\$)
0	100
1	125
2	150

Simple

Compound

Problems 5–7: Helen invests \$2,000 in an account that earns 3.5% compound interest per year.

5. Complete the table.

Time (yr)	Account Balance (\$)
0	2000
1	2070
2	2142.45
3	2217.44
4	2295.05

6. Which function represents the amount of money in Helen's account after x years?

A. $f(x) = 2000 + 1.035x$

B. $f(x) = 2000(1.035x)$

C. $f(x) = 2000 + 1.035^x$

D. $f(x) = 2000(1.035)^x$

7. What will the balance of the account be after 8 years?
\$2,633.62

Unit A1.6 Lesson 13

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Name: _____ Date: _____ Period: _____

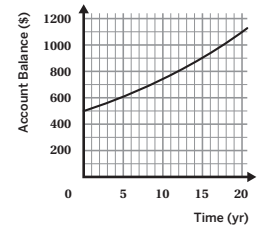
Problems 8–9: Samuel invests \$500 in an account that earns 4% compound interest per year. The graph shows the function $f(t) = 500(1.04)^t$, which gives Samuel's account balance after t years.

8. About how many years will it take for his account balance to reach \$1,000?

About 18 years

9. Use the graph to determine the value of $f(12)$.

$f(12)$ is equal to about 800.



What does that tell you about the situation?

This tells me that after 12 years, Samuel's account balance will be about \$800.

10. You just won a contest and have two prize options:

Option A: Getting a penny that doubles every day for 30 days

Option B: One payment of \$10 million dollars

Which option would you choose? Explain your choice.

Responses and explanations vary.

• I would choose Option A. I calculated the value after 30 days and it was \$10,737,418.24, which is more than Option B.

• I would choose Option B because I would want the money all at once.

Unit A1.6 Lesson 13

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.LE.A.1, HSF.LE.B.5
2	1	HSF.LE.A.1, HSF.LE.B.5
3	1	HSF.LE.A.1, HSF.LE.B.5
4	1	HSF.LE.A.1, HSF.LE.B.5
5	2	HSF.LE.A.1, HSF.LE.B.5
6	2	HSF.LE.A.1, HSF.LE.B.5
7	2	HSF.LE.A.1, HSF.LE.B.5
8	2	HSF.LE.A.1, HSF.LE.B.5
9	2	HSF.LE.A.1, HSF.LE.B.5
10	2	HSF.LE.A.3, HSF.LE.B.5

Notes:

Additional Practice

A1.6.14

Problems 1–2: Sree takes out a \$2,500 loan with a monthly interest rate of 4.5%. She makes no additional payments, deposits, or withdrawals.

1. Select *all* the expressions that can be used to calculate her balance after t years.

- ☐ A. $2500 \cdot 1.045^t$
- ☒ B. $2500 \cdot (1.045^{12})^t$
- ☒ C. $2500 \cdot 1.045^{12t}$
- ☐ D. $2500(1.6959)$
- ☒ E. $2500 \cdot 1.6959^t$

2. What is the interest rate per year for this loan?

69.59%

Problems 3–7: Aki invests money into a savings account to purchase a car in the future. He writes the expression $2000(1.055^{12})^4$ to help him calculate what the account balance will be in 4 years.

3. Explain what 2000 represents in the expression.

2000 represents the initial amount Aki invested.

4. Explain what 1.055 represents in the expression.

1.055 represents the growth factor of the investment, or 5.5% interest earned on the investment.

5. Explain what 12 represents in the expression.

12 represents that this is a monthly interest rate.

6. Explain what 4 represents in the expression.

4 represents the number of years.

7. Write an equivalent expression that could represent Aki's account balance in 4 years.

$2000 \cdot 1.055^{48}$ or $2000(1.9012)^4$ (or equivalent)

Problems 8–9: Jaxson is considering taking out a credit card loan that has a 24% monthly interest rate.

8. Complete the table.

Monthly Interest Rate	24%
Monthly Growth Factor	1.24
Growth Factor Per Year	13.2148
Interest Rate Per Year	1221.48%

9. If Jaxson takes out a \$100 credit card loan, how much will he owe after 3 years, if he made no additional payments?

\$230,770.70

10. Varsha takes out a \$10,000 federal student loan with a monthly interest rate of 0.64%. Write a function, $f(t)$, to calculate the amount Varsha owes after making no payments for t years.

$$f(t) = 10000 \cdot (1.0064)^{12t}$$

Name: _____ Date: _____ Period: _____

Additional Practice

A1.6.14

Problems 1–2: Sree takes out a \$2,500 loan with a monthly interest rate of 4.5%. She makes no additional payments, deposits, or withdrawals.

1. Select *all* the expressions that can be used to calculate her balance after t years.

- ☐ A. $2500 \cdot 1.045^t$
☒ B. $2500 \cdot (1.045^{12})^t$
☒ C. $2500 \cdot 1.045^{12t}$
☐ D. $2500(1.6959)^t$
☒ E. $2500 \cdot 1.6959^t$

2. What is the interest rate per year for this loan?

69.59%

Problems 3–7: Aki invests money into a savings account to purchase a car in the future. He writes the expression $2000(1.055^{12})^t$ to help him calculate what the account balance will be in t years.

3. Explain what 2000 represents in the expression.

2000 represents the initial amount Aki invested.

4. Explain what 1.055 represents in the expression.

1.055 represents the growth factor of the investment, or 5.5% interest earned on the investment.

5. Explain what 12 represents in the expression.

12 represents that this is a monthly interest rate.

6. Explain what 4 represents in the expression.

4 represents the number of years.

7. Write an equivalent expression that could represent Aki's account balance in 4 years.

$2000 \cdot 1.055^{48}$ or $2000(1.9012)^4$ (or equivalent)

Unit A1.6 Lesson 14

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Name: _____ Date: _____ Period: _____

Problems 8–9: Jaxson is considering taking out a credit card loan that has a 24% monthly interest rate.

8. Complete the table.

Monthly Interest Rate	24%
Monthly Growth Factor	1.24
Growth Factor Per Year	13.2148
Interest Rate Per Year	1221.48%

9. If Jaxson takes out a \$100 credit card loan, how much will he owe after 3 years, if he made no additional payments?

\$230,770.70

10. Varsha takes out a \$10,000 federal student loan with a monthly interest rate of 0.64%. Write a function, $f(t)$, to calculate the amount Varsha owes after making no payments for t years.

$f(t) = 10000 \cdot (1.0064)^{12t}$

Unit A1.6 Lesson 14

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSA.SSE.B.3.C, HSF.IF.C.8.B
2	2	HSA.SSE.A.1.A, HSF.IF.C.8.B
3	1	HSA.SSE.A.1.A
4	1	HSA.SSE.A.1.A
5	1	HSA.SSE.A.1.A
6	1	HSA.SSE.A.1.A
7	2	HSA.SSE.B.3.C
8	2	HSA.SSE.A.1.A
9	2	HSA.SSE.B.3.C
10	1	HSF.IF.C.8

Notes:

Additional Practice

A1.6.15

1. Meegan put \$3,000 into a savings account with a 3.25% annual interest rate, compounded quarterly. She made no additional payments, deposits, or withdrawals.

Select *all* the expressions that can be used to calculate her balance after 5 years.

☐ A. $3000(1 + 0.0325)^5$

☐ B. $3000(1 + 0.0325)^{20}$

☒ C. $3000\left(1 + \frac{0.0325}{4}\right)^{20}$

☒ D. $3000(1.008)^{4 \cdot 5}$

☐ E. $3000\left(1 + \frac{0.0325}{4}\right)^4$

Problems 2–4: Amir wants to take out a \$1,200 loan to pay for a new bike. The bank offers him the loan with a 12.5% annual interest rate, compounded daily.

Amir wrote this expression to calculate the balance of the loan in 3 years, but he made an error.

$$1200\left(1 + \frac{0.125}{365}\right)^3$$

2. Find the error and explain why it is incorrect.

Explanations vary. Since the loan is compounded daily, the exponent should be $3 \cdot 365$, or 1095.

3. Write a correct expression to represent Amir's balance after 3 years.

$1200\left(1 + \frac{0.125}{365}\right)^{1095}$ (or equivalent)

4. What will his balance be in 3 years?

\$1,745.88

Problems 5–6: A credit card company offers a \$2,500 loan with a 28.8% annual interest rate.

5. If no other charges or payments are made, what will the balance of the loan be after 1 year at each compounding period?

Compounding Period	Balance (\$)
Annually	3220
Semiannually	3271.84
Quarterly	3301.56

6. Describe how changing the compounding period affects the balance of the loan.

Responses vary. The more often a loan is compounded, the higher the balance will be.

Problems 7–8: Hannah has \$3,000 to invest and is choosing between three investment options:

- **Option A:** 11.5% annual interest rate, compounded quarterly
 - **Option B:** 11.25% annual interest rate, compounded monthly
 - **Option C:** 10.5% annual interest rate, compounded daily
7. Write an expression for each account to represent Hannah's balance after t years.

Option A: $f(t) = 3000\left(1 + \frac{0.115}{4}\right)^{4t}$ (or equivalent)

Option B: $g(t) = 3000\left(1 + \frac{0.1125}{12}\right)^{12t}$ (or equivalent)

Option C: $h(t) = 3000\left(1 + \frac{0.105}{365}\right)^{365t}$ (or equivalent)

8. If she makes no deposits and no withdrawals for 6 years, which option will give her the largest balance after 6 years? Show or explain your thinking.

Option A. Explanations vary.

Option A: $3000\left(1 + \frac{0.115}{4}\right)^{24} = \5923.22

Option B: $3000\left(1 + \frac{0.1125}{12}\right)^{72} = \5873.60

Option C: $3000\left(1 + \frac{0.105}{365}\right)^{2190} = \5632.32

Name: _____ Date: _____ Period: _____

Additional Practice

A1.6.15

1. Meegan put \$3,000 into a savings account with a 3.25% annual interest rate, compounded quarterly. She made no additional payments, deposits, or withdrawals.

Select *all* the expressions that can be used to calculate her balance after 5 years.

- ☐ A. $3000(1 + 0.0325)^5$ ☐ B. $3000(1 + 0.0325)^{20}$
☒ C. $3000\left(1 + \frac{0.0325}{4}\right)^{20}$ ☒ D. $3000(1.008)^{20}$
☐ E. $3000\left(1 + \frac{0.0325}{4}\right)^4$

- Problems 2–4: Amir wants to take out a \$1,200 loan to pay for a new bike. The bank offers him the loan with a 12.5% annual interest rate, compounded daily.

Amir wrote this expression to calculate the balance of the loan in 3 years, but he made an error.

$$1200\left(1 + \frac{0.125}{365}\right)^3$$

2. Find the error and explain why it is incorrect.

Explanations vary. Since the loan is compounded daily, the exponent should be $3 \cdot 365$, or 1095.

3. Write a correct expression to represent Amir's balance after 3 years.

$$1200\left(1 + \frac{0.125}{365}\right)^{1095} \text{ (or equivalent)}$$

4. What will his balance be in 3 years?

\$1,745.88

Unit A1.6 Lesson 15

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Name: _____ Date: _____ Period: _____

- Problems 5–6: A credit card company offers a \$2,500 loan with a 28.8% annual interest rate.

Compounding Period	Balance (\$)
Annually	3220
Semiannually	3271.84
Quarterly	3301.56

5. If no other charges or payments are made, what will the balance of the loan be after 1 year at each compounding period?

6. Describe how changing the compounding period affects the balance of the loan.
Responses vary. The more often a loan is compounded, the higher the balance will be.

- Problems 7–8: Hannah has \$3,000 to invest and is choosing between three investment options:

- **Option A:** 11.5% annual interest rate, compounded quarterly
 - **Option B:** 11.25% annual interest rate, compounded monthly
 - **Option C:** 10.5% annual interest rate, compounded daily
7. Write an expression for each account to represent Hannah's balance after t years.
Option A: $f(t) = 3000\left(1 + \frac{0.115}{4}\right)^{4t}$ (or equivalent)
Option B: $g(t) = 3000\left(1 + \frac{0.1125}{12}\right)^{12t}$ (or equivalent)
Option C: $h(t) = 3000\left(1 + \frac{0.105}{365}\right)^{365t}$ (or equivalent)

8. If she makes no deposits and no withdrawals for 6 years, which option will give her the largest balance after 6 years? Show or explain your thinking.

Option A. Explanations vary.

$$\text{Option A: } 3000\left(1 + \frac{0.115}{4}\right)^{24} = \$5923.22$$

$$\text{Option B: } 3000\left(1 + \frac{0.1125}{12}\right)^{72} = \$5873.60$$

$$\text{Option C: } 3000\left(1 + \frac{0.105}{365}\right)^{2190} = \$5632.32$$

Unit A1.6 Lesson 15

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Additional Practice

Practice Problem Analysis

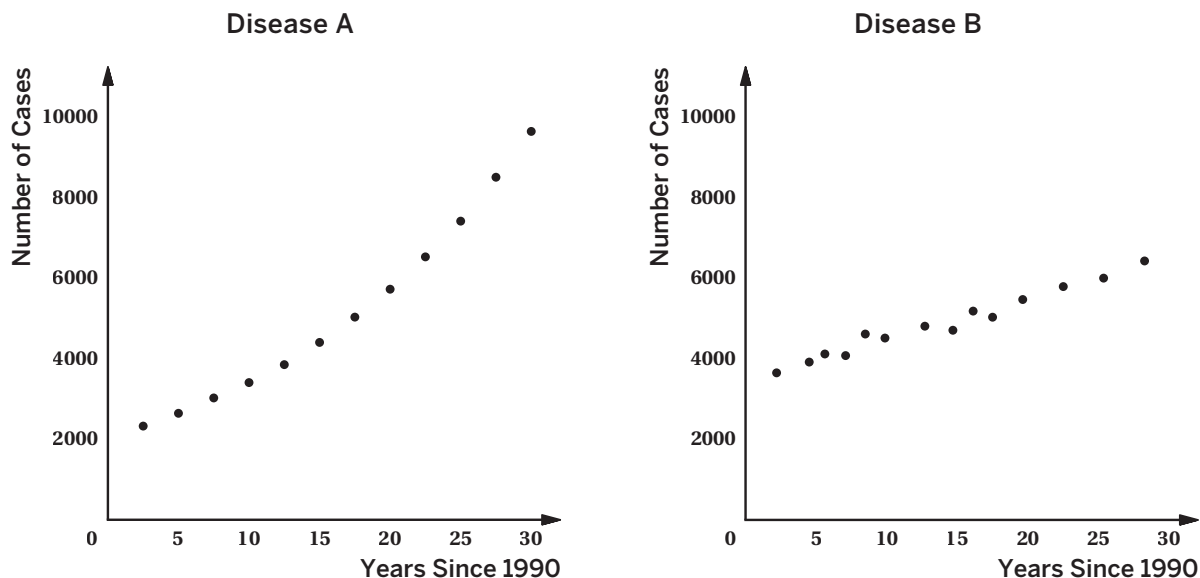
Problem	DOK	Standard(s)
1	2	HSF.BF.A.1, HSF.IF.C.8.B
2	2	HSA.SSE.A.1.A, HSF.BF.A.1
3	2	HSA.SSE.A.1.A, HSF.BF.A.1
4	2	HSA.SSE.A.1.A, HSF.BF.A.1
5	2	HSF.IF.C.8.B
6	2	HSF.IF.C.8.B
7	2	HSF.BF.A.1, HSF.IF.C.8.B
8	2	HSF.BF.A.1, HSF.IF.C.8.B

Notes:

Additional Practice

A1.6.16

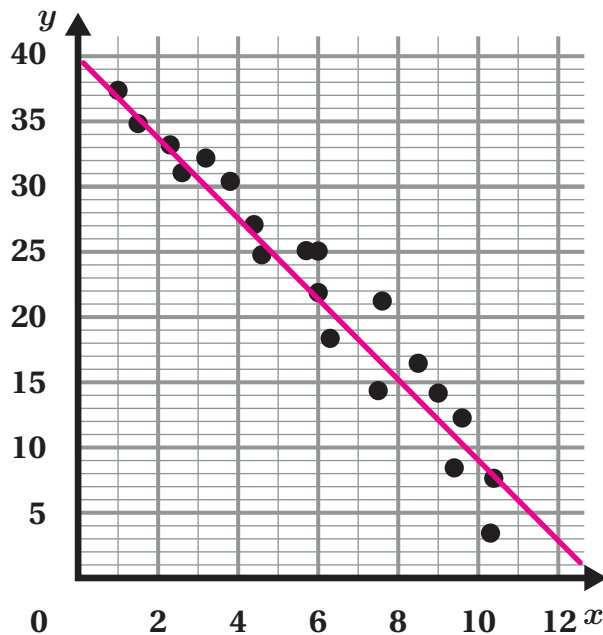
1. Each graph models the spread of a disease.



What kind of function would best model the data for each disease — *linear*, *exponential*, or *neither*? Explain your thinking.

Disease A: Exponential, Disease B: Linear. Explanations vary. For Disease A, the number of cases increases by a greater number as time goes on. For Disease B, the number of cases appears to follow along a straight line.

Problems 2–3: The graph shows data Isabella collected for a research project.



2. Sketch a line or exponential curve of best fit to model the data.

Responses vary. Sample shown on graph.

3. Isabella believes that an exponential function is the best model for the data. Do you agree with Isabella? Circle one.

Yes

No

Explain your thinking.

Explanations vary. An exponential model does not work for the data presented here because the data seems to generally decrease at a more constant rate. In an exponential decay model, the data would decrease faster on the left of the graph than on the right.

Problems 4–7: The graph shows data from a project in Theo's class. Theo modeled the given data with a linear function.

4. Describe an advantage of using Theo's model.

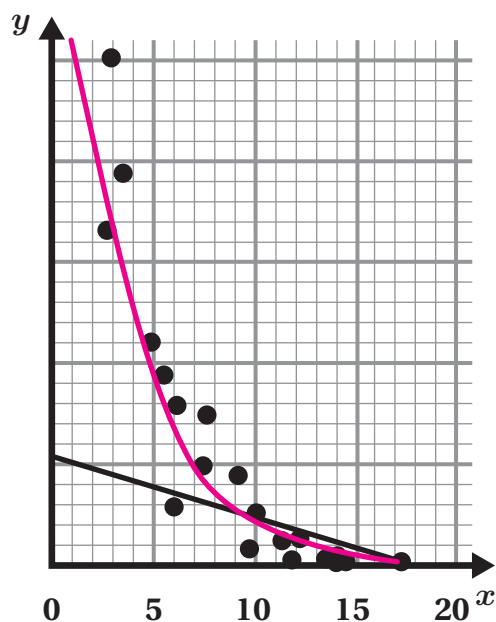
Responses vary. The model does a good job of representing the more constant rate of change between inputs of 10 and 15.

5. Describe a disadvantage of using Theo's model.

Responses vary. The model does not represent the majority of the given data.

6. Provide a suggestion for Theo to improve his model.

Responses vary. I would suggest for Theo to use an exponential model to model the data at the beginning of the graph.



7. Sketch a line or exponential curve of best fit to model the data. Explain your thinking.

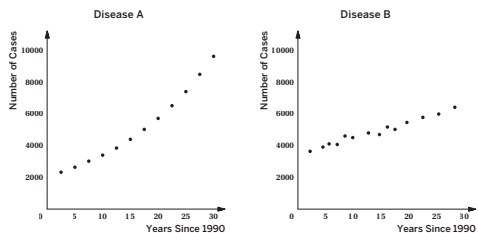
Responses and explanations vary. Sample shown on graph. I chose an exponential curve to model the data because at the beginning of the graph, the points are going down drastically. I did not choose a linear model because the data points are not changing by a more constant rate.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.6.16

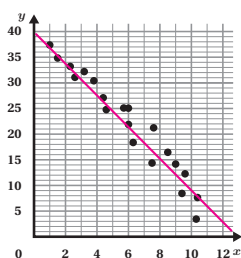
1. Each graph models the spread of a disease.



What kind of function would best model the data for each disease — linear, exponential, or neither? Explain your thinking.

Disease A: Exponential. **Disease B:** Linear. **Explanations vary.** For Disease A, the number of cases increases by a greater number as time goes on. For Disease B, the number of cases appears to follow along a straight line.

Problems 2–3: The graph shows data Isabella collected for a research project.



Unit A1.6 Lesson 16

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Name: _____ Date: _____ Period: _____

2. Sketch a line or exponential curve of best fit to model the data.

Responses vary. Sample shown on graph.

3. Isabella believes that an exponential function is the best model for the data. Do you agree with Isabella? Circle one.

Yes

No

Explain your thinking.

Explanations vary. An exponential model does not work for the data presented here because the data seems to generally decrease at a more constant rate. In an exponential decay model, the data would decrease faster on the left of the graph than on the right.

Problems 4–7: The graph shows data from a project in Theo's class. Theo modeled the given data with a linear function.

4. Describe an advantage of using Theo's model.

Responses vary. The model does a good job of representing the more constant rate of change between inputs of 10 and 15.

5. Describe a disadvantage of using Theo's model.

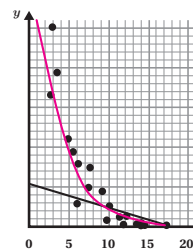
Responses vary. The model does not represent the majority of the given data.

6. Provide a suggestion for Theo to improve his model.

Responses vary. I would suggest for Theo to use an exponential model to model the data at the beginning of the graph.

7. Sketch a line or exponential curve of best fit to model the data. Explain your thinking.

Responses and explanations vary. Sample shown on graph. I chose an exponential curve to model the data because at the beginning of the graph, the points are going down drastically. I did not choose a linear model because the data points are not changing by a more constant rate.



Unit A1.6 Lesson 16

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Additional Practice

Practice Problem Analysis

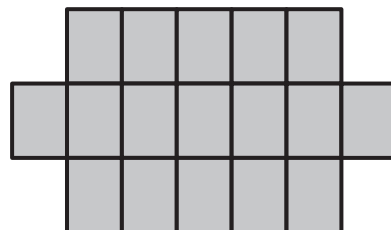
Problem	DOK	Standard(s)
1	1	HSS.ID.B.6.A
2	2	HSS.ID.B.6.A
3	2	HSS.ID.B.6.A
4	2	HSS.ID.B.6, HSS.ID.B.6.A
5	2	HSS.ID.B.6, HSS.ID.B.6.A
6	2	HSS.ID.B.6, HSS.ID.B.6.A
7	2	HSS.ID.B.6.A

Notes:

Additional Practice

A1.7.01

1. Select *all* of the expressions that could represent the number of tiles in this diagram.



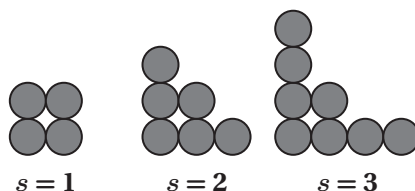
- ☐ A. $7 \cdot 3$
- ☒ B. $7 \cdot 3 - 4$
- ☐ C. $5 \cdot 3 + 4$
- ☒ D. $5 \cdot 3 + 2$
- ☒ E. $2(5) + 1(7)$

2. Here are the first three steps in a pattern. How many dots will there be when $s = 8$?

18.

Explain your thinking.

Explanations vary. I saw that each step is a 2-by-2 square of dots in the bottom left corner and then $s - 1$ dots going up vertically and $s - 1$ dots going to the right horizontally. So $s = 8$ would have $4 + (8 - 1) + (8 - 1)$ dots, or 18 dots.



3. What type of relationship does the pattern in the table represent? Circle one.

Linear Exponential Neither

Explain your thinking.

Explanations vary. As s increases by 1, the number of tiles doesn't change by a common ratio or a common difference.

s	Number of Tiles
1	5
2	20
3	45
4	80

Problems 4–6: A teacher gives the class a table with only the first two rows in a tile pattern.

4. Momo says the pattern is a linear relationship.
 Gerald says there is not enough information to be sure.
 Whose thinking is correct? Explain your thinking.

Gerald's. Explanations vary. If the number of tiles is increasing by 7 with each step, then the pattern is linear. If the number of tiles is being multiplied by 8 with each step, then the pattern is exponential.

s	Number of Tiles
1	1
2	8

5. How many tiles would there be in the next step if the relationship were *linear*?
15 tiles
6. How many tiles would there be in the next step if the relationship were *exponential*?
64 tiles

Problems 7–8: The table shows the relationship between the step number, s , and number of tiles in a pattern.

Heather notices that the number of tiles is increasing in an interesting pattern: The number of tiles increases by 3, then by 5, then by 7, and then by 9.

Aaron notices that the number of tiles in each row is the square of the step number.

7. Based on these observations, is this pattern linear, exponential, or neither? Explain your thinking.

Neither. Explanations vary. As s increases by 1, the number of tiles doesn't change by a common ratio or a common difference.

8. Use both Aaron and Heather's observations to predict the next number in this pattern. Is it the same value? Show or explain your thinking.

Yes. Explanations vary. Using Heather's observation, I added 11 to 25 and got 36 tiles for step 6. Using Aaron's observation, I squared 6 and also got 36 tiles for step 6.

s	Number of Tiles
1	1
2	4
3	9
4	16
5	25

Name: _____ Date: _____ Period: _____

Additional Practice

A1.7.O1

1. Select *all* of the expressions that could represent the number of tiles in this diagram.

- ☐ A. $7 \cdot 3$
☒ B. $7 \cdot 3 - 4$
☐ C. $5 \cdot 3 + 4$
☒ D. $5 \cdot 3 + 2$
☒ E. $2(5) + 1(7)$

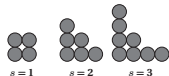


2. Here are the first three steps in a pattern. How many dots will there be when $s = 8$?

18.

Explain your thinking.

Explanations vary. I saw that each step is a 2-by-2 square of dots in the bottom left corner and then $s - 1$ dots going up vertically and $s - 1$ dots going to the right horizontally. So $s = 8$ would have $4 + (8 - 1) + (8 - 1)$ dots, or 18 dots.



3. What type of relationship does the pattern in the table represent? Circle one.

Linear Exponential Neither

Explain your thinking.

Explanations vary. As s increases by 1, the number of tiles doesn't change by a common ratio or a common difference.

s	Number of Tiles
1	5
2	20
3	45
4	80

Unit 7 Lesson 1

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Name: _____ Date: _____ Period: _____

- Problems 4–6: A teacher gives the class a table with only the first two rows in a tile pattern.

4. Momo says the pattern is a linear relationship. Gerald says there is not enough information to be sure. Whose thinking is correct? Explain your thinking.

Gerald's. Explanations vary. If the number of tiles is increasing by 7 with each step, then the pattern is linear. If the number of tiles is being multiplied by 8 with each step, then the pattern is exponential.

s	Number of Tiles
1	1
2	8

5. How many tiles would there be in the next step if the relationship were linear?

15 tiles

6. How many tiles would there be in the next step if the relationship were exponential?

64 tiles

- Problems 7–8: The table shows the relationship between the step number, s , and number of tiles in a pattern.

Heather notices that the number of tiles is increasing in an interesting pattern: The number of tiles increases by 3, then by 5, then by 7, and then by 9.

Aaron notices that the number of tiles in each row is the square of the step number.

s	Number of Tiles
1	1
2	4
3	9
4	16
5	25

7. Based on these observations, is this pattern linear, exponential, or neither? Explain your thinking.

Neither. Explanations vary. As s increases by 1, the number of tiles doesn't change by a common ratio or a common difference.

8. Use both Aaron and Heather's observations to predict the next number in this pattern. Is it the same value? Show or explain your thinking.

Yes. Explanations vary. Using Heather's observation, I added 11 to 25 and got 36 tiles for step 6. Using Aaron's observation, I squared 6 and also got 36 tiles for step 6.

Unit 7 Lesson 1

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSF.BF.A.1.A*
2	2	HSF.BF.A.1.A
3	2	HSF.BF.A.1.A
4	3	HSF.BF.A.1.A
5	2	HSF.BF.A.1.A
6	2	HSF.BF.A.1.A
7	2	HSF.BF.A.1.A
8	2	HSF.BF.A.1.A

*This problem builds toward the standard shown.

Notes:

Additional Practice

A1.7.02

Problems 1–4: Circle whether each expression represents a quadratic relationship, a linear relationship, or neither.

- | | | | |
|------------------|-----------|--------|---------|
| 1. $2 \cdot 3^x$ | Quadratic | Linear | Neither |
| 2. $3n^3$ | Quadratic | Linear | Neither |
| 3. $3n^2 - 2n$ | Quadratic | Linear | Neither |
| 4. $n(5n)$ | Quadratic | Linear | Neither |

5. Does this pattern show a quadratic relationship? Explain your thinking.

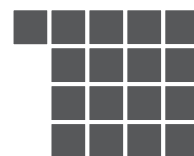
Yes. Explanations vary. This pattern is quadratic because each figure contains a square.



$n = 1$



$n = 2$



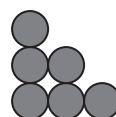
$n = 3$

6. Write an expression to represent the relationship between the figure number, n , and the total number of tiles.

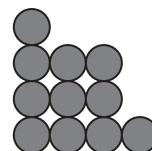
$n^2 + 2$



$n = 1$

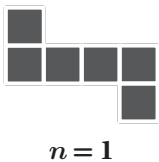


$n = 2$



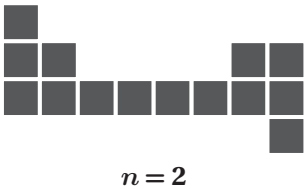
$n = 3$

7. Chidi wrote the expression $2n^2 + 2n + 2$ to represent this pattern.
- Denali wrote the expression $2(n^2 + n + 1)$ to represent this pattern.



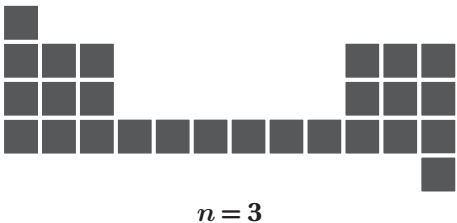
Whose thinking is correct?

Chidi Denali **Both**



Explain your thinking

Explanations vary. Both Chidi and Denali are correct because their expressions are equivalent.



8. Here is a table with the first two rows of a pattern.
- Does this pattern represent a quadratic relationship?

s	Number of Tiles
1	10
2	17

Yes No **Not enough information**

Explain your thinking.

Explanations vary. The table could represent a quadratic pattern because 10 is 1 more than 9, a perfect square and 17 is 1 more than 16, a perfect square. However, without the next step in the pattern, I cannot be sure that the relationship is quadratic. The relationship could also be linear by adding 7 to each step.

Name: _____ Date: _____ Period: _____

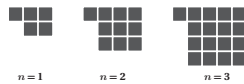
Additional Practice

A1.7.02

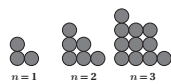
Problems 1–4: Circle whether each expression represents a quadratic relationship, a linear relationship, or neither.

- | | | | |
|------------------|-----------|--------|---------|
| 1. $2 \cdot 3^x$ | Quadratic | Linear | Neither |
| 2. $3n^3$ | Quadratic | Linear | Neither |
| 3. $3n^2 - 2n$ | Quadratic | Linear | Neither |
| 4. $n(5n)$ | Quadratic | Linear | Neither |

5. Does this pattern show a quadratic relationship? Explain your thinking.
Yes. *Explanations vary. This pattern is quadratic because each figure contains a square.*



6. Write an expression to represent the relationship between the figure number, n , and the total number of tiles.
 $n^2 + 2$



Unit 7 Lesson 2

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Name: _____ Date: _____ Period: _____

7. Chidi wrote the expression $2n^2 + 2n + 2$ to represent this pattern.

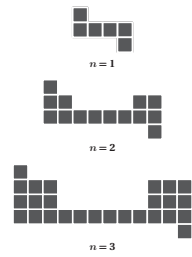
Denali wrote the expression $2(n^2 + n + 1)$ to represent this pattern.

Whose thinking is correct?

Chidi Denali Both

Explain your thinking.

Explanations vary. Both Chidi and Denali are correct because their expressions are equivalent.



8. Here is a table with the first two rows of a pattern.

Does this pattern represent a quadratic relationship?

Yes No Not enough information

Explain your thinking.

Explanations vary. The table could represent a quadratic pattern because 10 is 1 more than 9, a perfect square and 17 is 1 more than 16, a perfect square. However, without the next step in the pattern, I cannot be sure that the relationship is quadratic. The relationship could also be linear by adding 7 to each step.

n	Number of Tiles
1	10
2	17

Unit 7 Lesson 2

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.SSE.A.1.A
2	1	HSA.SSE.A.1.A
3	1	HSA.SSE.A.1.A
4	1	HSA.SSE.A.1.A
5	2	HSA.SSE.A.1.A
6	2	HSA.SSE.A.1.A, HSA.SSE.B.3
7	2	HSA.SSE.A.1.A, HSA.SSE.B.3
8	3	HSF.BF.A.1.A

Notes:

Additional Practice

A1.7.04

Problems 1–3: For each pair of symmetrical points on a parabola, determine the equation for the line of symmetry.

1. $(0, 0)$ and $(-12, 0)$

$x = -6$

2. $(5, 8)$ and $(21, 8)$

$x = 13$

3. $(10, -3)$ and $(-13, -3)$

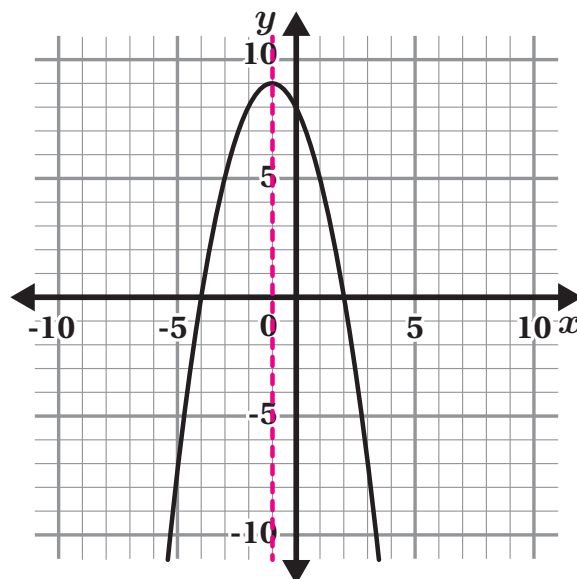
$x = -1.5$

Problems 4–5: Here is a graph of a parabola.

4. Draw the *line of symmetry* where you think it is located on this parabola.

5. Write the equation for the line of symmetry.

$x = -1$



Problems 6–7: Here is an incomplete table that could represent several types of functions.

6. Circle a function type and determine the number of tiles that would be in figure 2. *Responses vary.*

Linear Quadratic Exponential

Figure	Number of Tiles
1	4
2	<i>Responses vary.</i>
3	16

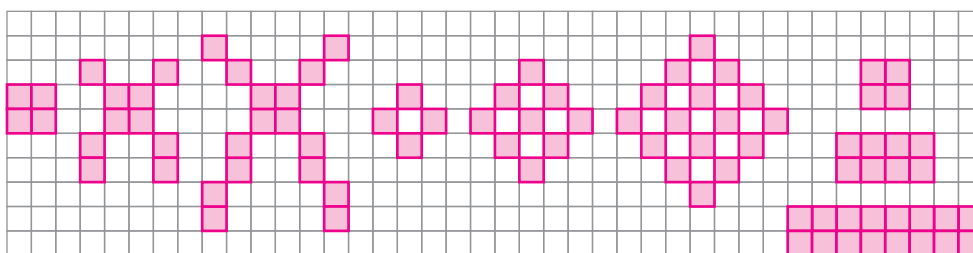
7. Draw three figures to match the pattern in the table.

Responses vary.

Linear

Quadratic

Exponential



Problems 8–10: Here are a few points that belong to a function $f(x)$.

8. Does $f(x)$ represent a quadratic relationship?

Circle one.

Yes

No

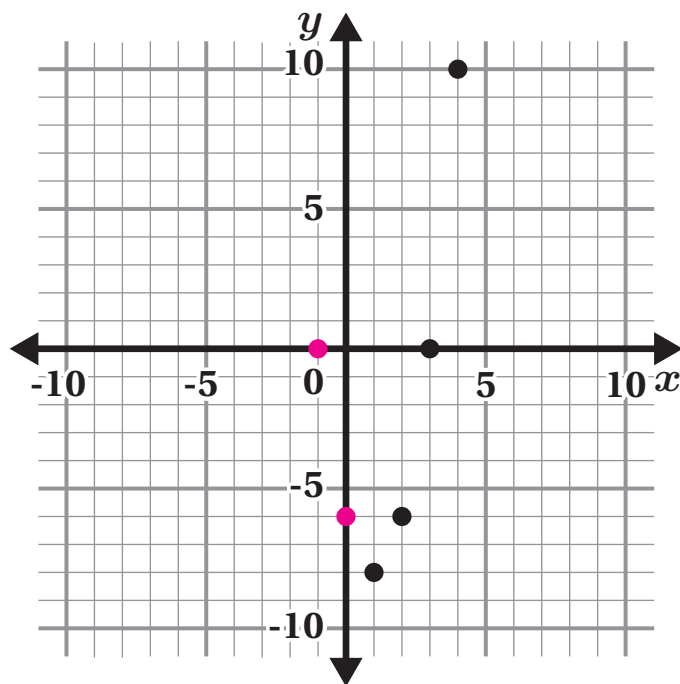
Not enough
information

Explain your thinking.

Explanations vary. I know that $f(x)$ is quadratic because the second differences of the values in the table are all 4. Quadratic relationships have constant second differences.

9. Complete the table for $f(x)$ and plot the new points on the graph, if possible.

x	$f(x)$
-1	0
0	-6
1	-8
2	-6
3	0
4	10
5	24
6	42
7	64



10. Write the equation for the line of symmetry.

$x = \underline{2}$

Name: _____ Date: _____ Period: _____

Additional Practice

A1.7.04

Problems 1–3: For each pair of symmetrical points on a parabola, determine the equation for the line of symmetry.

1. (0, 0) and (–12, 0) 2. (5, 8) and (21, 8) 3. (10, –3) and (–13, –3)

$x = -6$

$x = 13$

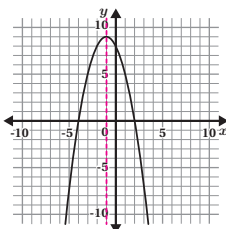
$x = -1.5$

Problems 4–5: Here is a graph of a parabola.

4. Draw the line of symmetry where you think it is located on this parabola.

5. Write the equation for the line of symmetry.

$x = -1$



Problems 6–7: Here is an incomplete table that could represent several types of functions.

6. Circle a function type and determine the number of tiles that would be in figure 2. **Responses vary.**

Linear Quadratic Exponential

Figure	Number of Tiles
1	4
2	Responses vary.
3	16

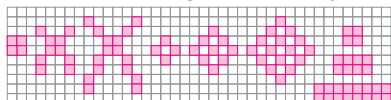
7. Draw three figures to match the pattern in the table.

Responses vary.

Linear

Quadratic

Exponential



Unit 7 Lesson 4

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Name: _____ Date: _____ Period: _____

Problems 8–10: Here are a few points that belong to a function $f(x)$.

8. Does $f(x)$ represent a quadratic relationship? Circle one.

☒ Yes

☐ No

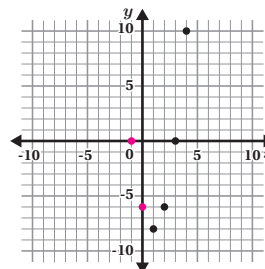
☐ Not enough information

Explain your thinking.

Explanations vary. I know that $f(x)$ is quadratic because the second differences of the values in the table are all 4. Quadratic relationships have constant second differences.

9. Complete the table for $f(x)$ and plot the new points on the graph, if possible.

x	$f(x)$
–1	0
0	–6
1	–8
2	–6
3	0
4	10
5	24
6	42
7	64



10. Write the equation for the line of symmetry.

$x = 1$

Unit 7 Lesson 4

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.B.4
2	1	HSF.IF.B.4
3	1	HSF.IF.B.4
4	1	HSF.IF.B.4
5	1	HSF.IF.B.4
6	3	HSF.IF.B.4
7	3	HSF.IF.B.4
8	2	HSF.IF.B.4
9	2	HSF.IF.B.4
10	2	HSF.IF.B.4

Notes:

Additional Practice

A1.7.05

Problems 1–3: This table shows the height of a stomp rocket at various times.

Time (sec)	0	1	2	3	4	5	6
Height (ft)	8	28	40	44	40	28	8

1. What is the maximum height the stomp rocket reaches?

44 feet

2. What is the height of the rocket at launch?

8 feet

3. Issa says the stomp rocket reaches the ground at 6 seconds.

Samnang says the stomp rocket reaches the ground at 7 seconds.

Whose thinking is correct?

Issa

Samnang

Neither

Explain your thinking.

Explanations vary. The stomp rocket reaches the ground between 6 and 7 seconds. Issa is incorrect because at 6 seconds, the stomp rocket would still be 8 feet in the air. If I continue the table, at 7 seconds, the rocket would be at –20 feet. This means that the stomp rocket reaches the ground sometime between 6 and 7 seconds.

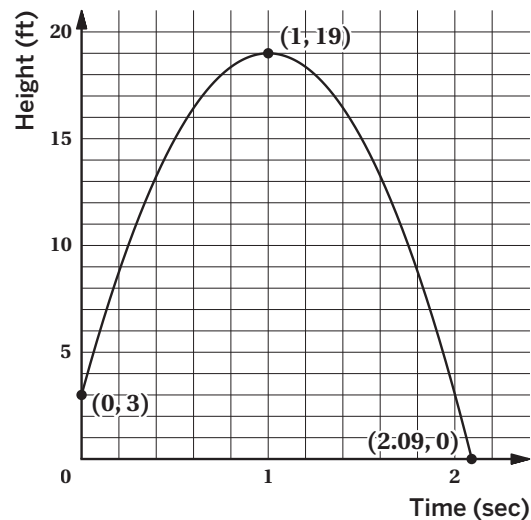
4. $f(x)$ is a quadratic function. Complete the table.

x	0	1	2	3	4	5	6	7
$f(x)$	108	90	76	66	60	58	60	66

Describe the strategy you used to complete the table.

Responses vary. I looked for the second difference in the table which is 4. Then I applied the second difference to find the first difference to complete the table.

Problems 5–7: Shawn throws a ball up in the air. The graph shows the height of the ball, in feet, above the ground as a function of time, in seconds.



5. How high above the ground was the ball when Shawn threw it?

3 feet

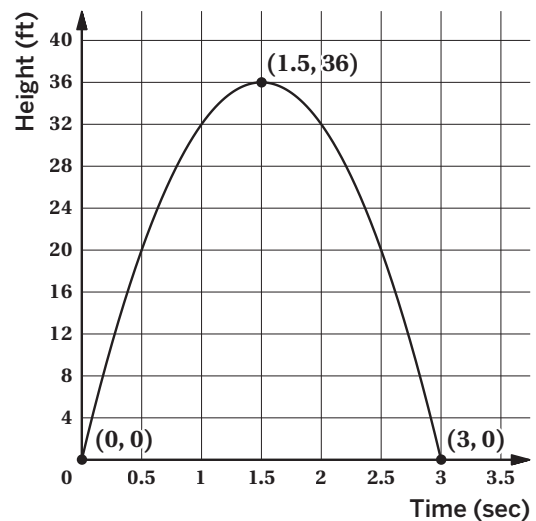
6. When did the ball reach its maximum height? How high was it?

1 second after it was thrown; 19 feet

7. When did the ball hit the ground?

2.09 seconds

8. A disc is launched into the air. The graph shows the height of the disc, in feet, above the ground as a function of time, in seconds. Select *all* the true statements about the situation.



- ☒ A. The disc was launched from the ground.
- ☐ B. The disc was launched 4 feet from the ground.
- ☒ C. The maximum height of the disc was 36 feet.
- ☒ D. The disc was in the air for 3 seconds.
- ☐ E. The disc was in the air for 36 seconds.
- ☒ F. The disc reached its maximum height 1.5 seconds after it was launched.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.7.05

Problems 1–3: This table shows the height of a stomp rocket at various times.

Time (sec)	0	1	2	3	4	5	6
Height (ft)	8	28	40	44	40	28	8

- What is the maximum height the stomp rocket reaches?
44 feet
- What is the height of the rocket at launch?
8 feet
- Issa says the stomp rocket reaches the ground at 6 seconds.
Samnang says the stomp rocket reaches the ground at 7 seconds.
Whose thinking is correct?
Issa Samnang **Neither**

Explain your thinking.

Explanations vary. The stomp rocket reaches the ground between 6 and 7 seconds. Issa is incorrect because at 6 seconds, the stomp rocket would still be 8 feet in the air. If I continue the table, at 7 seconds, the rocket would be at -20 feet. This means that the stomp rocket reaches the ground sometime between 6 and 7 seconds.

- $f(x)$ is a quadratic function. Complete the table.

x	0	1	2	3	4	5	6	7
$f(x)$	108	90	76	66	60	58	60	66

Describe the strategy you used to complete the table.

Responses vary. I looked for the second difference in the table which is 4. Then I applied the second difference to find the first difference to complete the table.

Unit 7 Lesson 5

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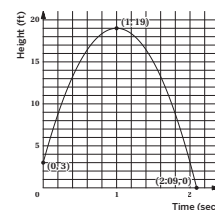
Name: _____ Date: _____ Period: _____

Problems 5–7: Shawn throws a ball up in the air. The graph shows the height of the ball, in feet, above the ground as a function of time, in seconds.

- How high above the ground was the ball when Shawn threw it?
3 feet

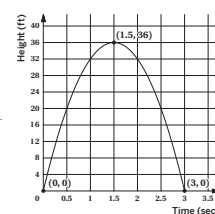
- When did the ball reach its maximum height? How high was it?
1 second after it was thrown; 19 feet

- When did the ball hit the ground?
2.09 seconds



- A disc is launched into the air. The graph shows the height of the disc, in feet, above the ground as a function of time, in seconds. Select *all* the true statements about the situation.

- ☒ A. The disc was launched from the ground.
- ☐ B. The disc was launched 4 feet from the ground.
- ☒ C. The maximum height of the disc was 36 feet.
- ☒ D. The disc was in the air for 3 seconds.
- ☐ E. The disc was in the air for 36 seconds.
- ☒ F. The disc reached its maximum height 1.5 seconds after it was launched.



Unit 7 Lesson 5

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.B.4
2	1	HSF.IF.B.4
3	2	HSF.IF.B.4
4	2	HSF.IF.B.4
5	1	HSF.IF.B.4
6	1	HSF.IF.B.4
7	1	HSF.IF.B.4
8	2	HSF.IF.B.4

Notes:

Additional Practice

A1.7.06

- The key features of this parabola are labeled A , B , C , and D .

Match each key feature with a term from the word bank.

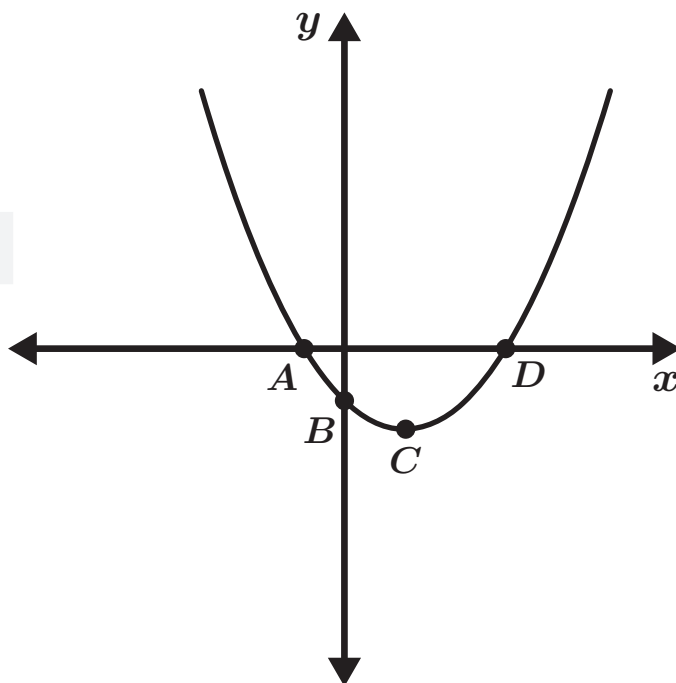
vertex x -intercept y -intercept

A: x -intercept

B: y -intercept

C: vertex

D: x -intercept

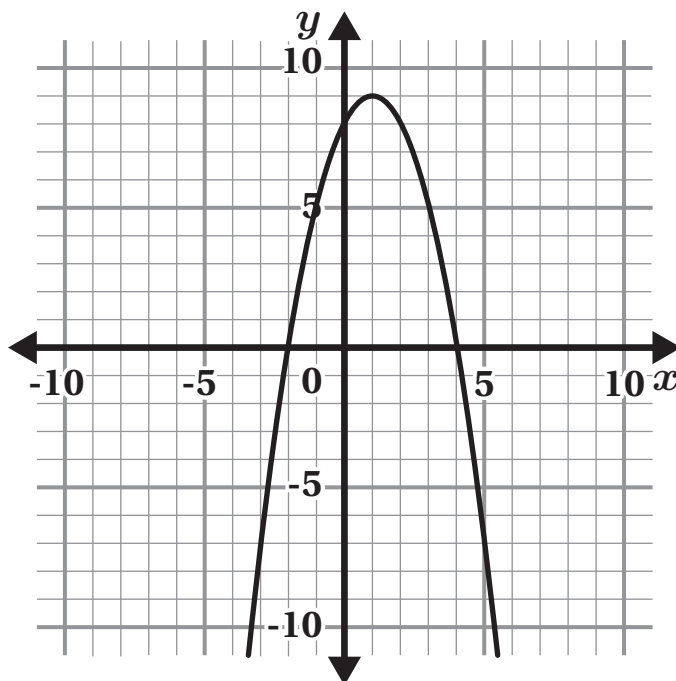


Problems 2–4: Use the graph to determine the coordinates of each key feature.

- Vertex: $(1, 9)$

- x -intercept(s): $(-2, 0)$ and $(4, 0)$

- y -intercept: $(0, 8)$

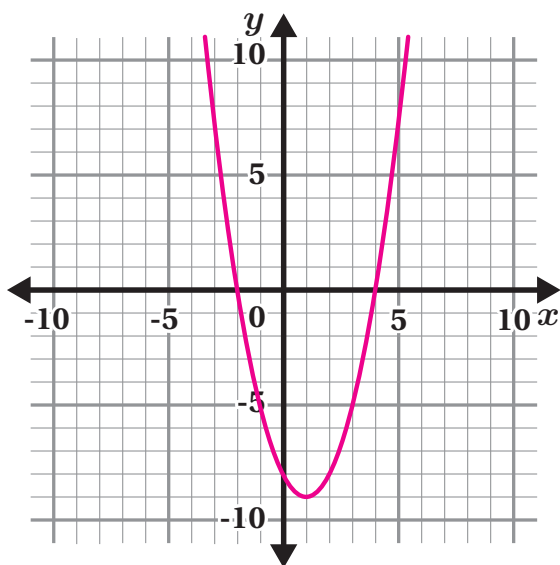


- A parabola has a vertex at $(-6, -2)$. Give two possible coordinates for its x -intercepts.

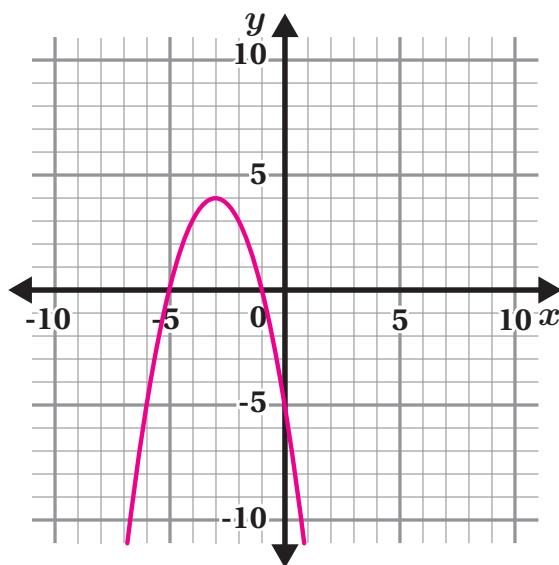
Responses vary. $(-5, 0)$ and $(-7, 0)$. Correct responses will be an equal distance from $x = -6$.

Problems 6–9: Graph a parabola that fits each description. *Responses vary.*

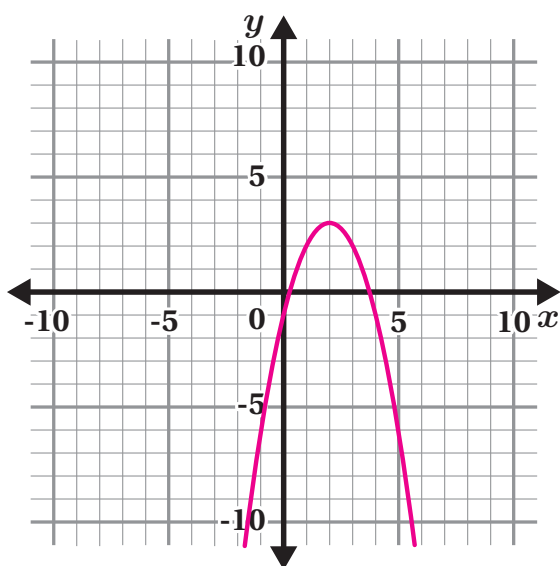
- 6.** Concave up with a negative y -intercept



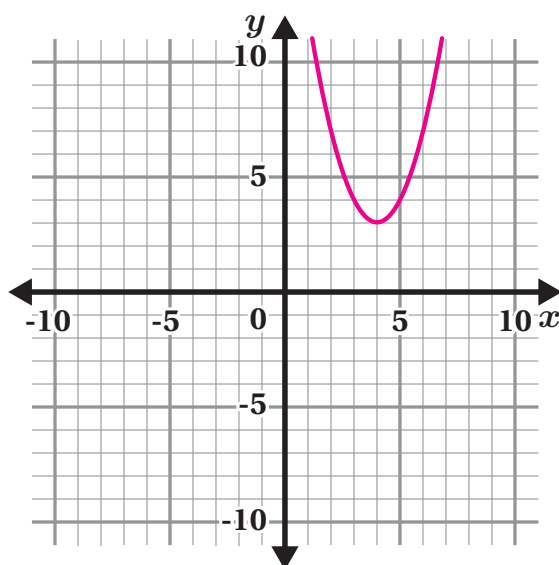
- 7.** Concave down and vertex at $(-3, 4)$



- 8.** Concave down and line of symmetry of $x = 2$



- 9.** Concave up with no x -intercepts



Name: _____ Date: _____ Period: _____

Additional Practice

A1.7.06

1. The key features of this parabola are labeled A, B, C, and D.

Match each key feature with a term from the word bank.

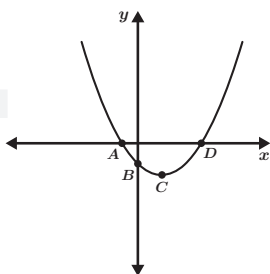
vertex x -intercept y -intercept

A: x -intercept

B: y -intercept

C: vertex

D: x -intercept

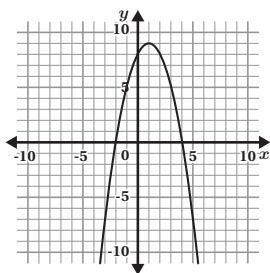


Problems 2–4: Use the graph to determine the coordinates of each key feature.

2. Vertex: (1, 9)

3. x -intercept(s): (-2, 0) and (4, 0)

4. y -intercept: (0, 8)



5. A parabola has a vertex at $(-6, -2)$. Give two possible coordinates for its x -intercepts.

Responses vary: $(-5, 0)$ and $(-7, 0)$. Correct responses will be an equal distance from $x = -6$.

Unit 7 Lesson 6

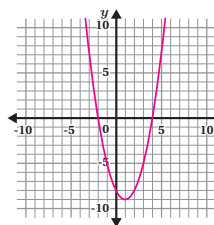
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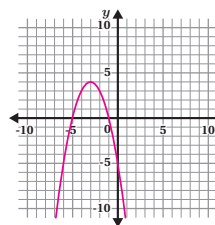
Name: _____ Date: _____ Period: _____

Problems 6–9: Graph a parabola that fits each description. Responses vary.

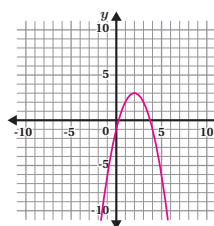
6. Concave up with a negative y -intercept



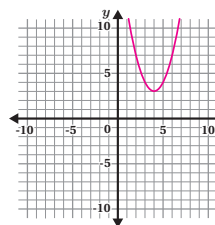
7. Concave down and vertex at $(-3, 4)$



8. Concave down and line of symmetry of $x = 2$



9. Concave up with no x -intercepts



Unit 7 Lesson 6

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSF.IF.B.4
2	1	HSF.IF.B.4
3	1	HSF.IF.B.4
4	1	HSF.IF.B.4
5	2	HSF.IF.B.4
6	2	HSF.IF.B.4
7	2	HSF.IF.B.4
8	2	HSF.IF.B.4
9	2	HSF.IF.B.4

Notes:

Additional Practice

A1.7.10

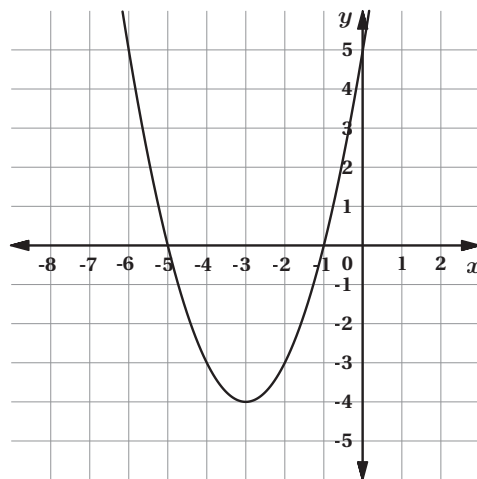
1. Here is the graph of $k(x) = (x + 1)(x + 5)$ is shown.

a What are the x -intercepts?

$(-5, 0)$ and $(-1, 0)$

b What is the y -intercept?

$(0, 5)$



2. What is the y -intercept of the function $j(x) = x^2 + 4$?

A. $(0, -4)$

B. $(0, 0)$

C. $(0, 1)$

$(0, 4)$

3. What are the x -intercepts of the function $f(x) = (x + 3)(x - 1)$?

A. $(3, 0)$ and $(1, 0)$

$(-3, 0)$ and $(1, 0)$

C. $(3, 0)$ and $(-1, 0)$

D. $(-3, 0)$ and $(-1, 0)$

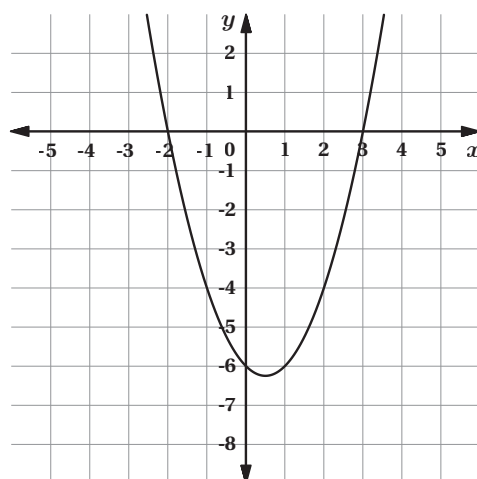
4. Here is a graph of a quadratic function. Which function could this be?

A. $g(x) = (x - 2)(x - 3)$

B. $g(x) = (x - 2)(x + 3)$

$g(x) = (x + 2)(x - 3)$

D. $g(x) = (x + 2)(x + 3)$



5. Select *all* the quadratic functions that have a y -intercept of $(0, -9)$.

☐ A. $f(x) = x(x - 9)$

☐ B. $f(x) = x^2 + 6x + 9$

☒ C. $f(x) = x^2 - 5x - 9$

☒ D. $f(x) = (x - 3)(x + 3)$

☒ E. $f(x) = (x - 1)(x + 9)$

6. Here is a function: $g(x) = x^2 - 2x - 24$.

a What is the y -intercept of $g(x)$?

$(0, -24)$

b Here is the function written a different way: $g(x) = (x - 6)(x + 4)$.
What are the x -intercepts of $g(x)$?

$(6, 0)$ and $(-4, 0)$

7. Determine the x - and y -intercepts of the function $p(x) = (x - 5)(x + 5)$.

x -intercepts: **$(5, 0)$** and **$(-5, 0)$**

y -intercept: **$(0, -25)$**

8. Here is a quadratic function: $r(x) = x(6 - x)$. The table shows where Jada and Elena think the x - and y -intercepts are located. Is either person's work completely correct? Explain your thinking.

	x -intercepts	y -intercept
Jada	$(-6, 0)$ and $(6, 0)$	$(0, 0)$
Elena	$(0, 0)$ and $(6, 0)$	$(0, -6)$

Neither person's work is completely correct. Explanations vary. Jada correctly determined the y -intercept and Elena correctly determined the x -intercepts.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.7.10

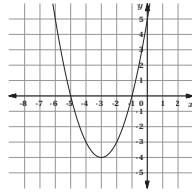
1. Here is the graph of $k(x) = (x + 1)(x + 5)$ is shown.

a. What are the x -intercepts?

(-5, 0) and (-1, 0)

b. What is the y -intercept?

(0, 5)



2. What is the y -intercept of the function $j(x) = x^2 + 4$?

A. (0, -4)

B. (0, 0)

C. (0, 1)

D. (0, 4)

3. What are the x -intercepts of the function $f(x) = (x + 3)(x - 1)$?

A. (3, 0) and (1, 0)

B. (-3, 0) and (1, 0)

C. (3, 0) and (-1, 0)

D. (-3, 0) and (-1, 0)

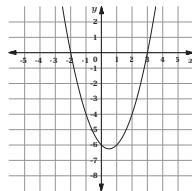
4. Here is a graph of a quadratic function. Which function could this be?

A. $g(x) = (x - 2)(x - 3)$

B. $g(x) = (x - 2)(x + 3)$

C. $g(x) = (x + 2)(x - 3)$

D. $g(x) = (x + 2)(x + 3)$



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Name: _____ Date: _____ Period: _____

5. Select *all* the quadratic functions that have a y -intercept of (0, -9).

☐ A. $f(x) = x(x - 9)$

☐ B. $f(x) = x^2 + 6x + 9$

☒ C. $f(x) = x^2 - 5x - 9$

☒ D. $f(x) = (x - 3)(x + 3)$

☒ E. $f(x) = (x - 1)(x + 9)$

6. Here is a function: $g(x) = x^2 - 2x - 24$.

a. What is the y -intercept of $g(x)$?

(0, -24)

b. Here is the function written a different way: $g(x) = (x - 6)(x + 4)$. What are the x -intercepts of $g(x)$?

(6, 0) and (-4, 0)

7. Determine the x - and y -intercepts of the function $p(x) = (x - 5)(x + 5)$.

x -intercepts: **(5, 0)** and **(-5, 0)**

y -intercept: **(0, -25)**

8. Here is a quadratic function: $r(x) = x(6 - x)$. The table shows where Jada and Elena think the x - and y -intercepts are located. Is either person's work completely correct? Explain your thinking.

	x -intercepts	y -intercept
Jada	(-6, 0) and (6, 0)	(0, 0)
Elena	(0, 0) and (6, 0)	(0, -6)

Neither person's work is completely correct. Explanations vary. Jada correctly determined the y -intercept and Elena correctly determined the x -intercepts.

Unit 7 Lesson 10

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.B.4
2	1	HSF.IF.B.4
3	2	HSF.IF.B.4
4	2	HSF.IF.B.4
5	2	HSF.IF.B.4
6	2	HSF.IF.B.4
7	2	HSF.IF.B.4
8	2	HSF.IF.B.4

Notes:

Additional Practice

A1.7.11

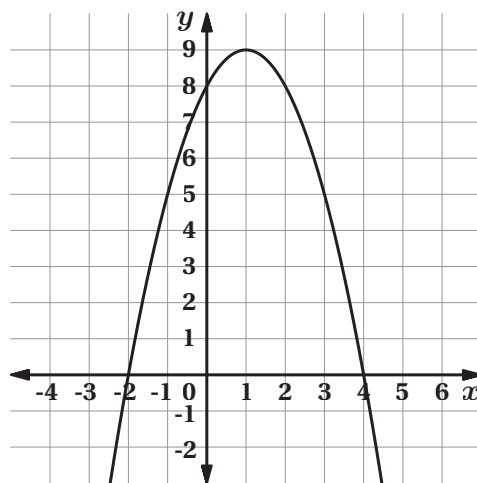
1. Here is the graph of a quadratic function.

a What is the vertex?

(1, 9)

b What is the equation of the line of symmetry?

$x = 1$



2. A quadratic function has a vertex of $(-2, -8)$.

What is the equation of the line of symmetry?

☒ A. $x = -2$

☐ B. $x = -8$

☐ C. $x = 2$

☐ D. $x = 8$

3. Select *all* the true statements about the function $f(x) = x(x + 10)$.

☒ A. The x -coordinate of its vertex is -5 .

☐ B. The x -coordinate of its vertex is 5 .

☐ C. The x -coordinate of its vertex is 10 .

☒ D. It has exactly two x -intercepts.

☐ E. The x -intercepts are $(-1, 0)$ and $(-10, 0)$.

☒ F. The x -intercepts are $(0, 0)$ and $(-10, 0)$.

☐ G. The only x -intercept is $(10, 0)$.

4. Select *all* the functions that have a vertex with an x -coordinate of -3 .

☐ A. $g(x) = (x - 3)(x + 3)$

☒ B. $g(x) = (x + 2)(x + 4)$

☐ C. $g(x) = (x - 1)(x - 5)$

☒ D. $g(x) = x(x + 6)$

☐ E. $g(x) = x(x - 6)$

5. Determine the x -intercepts and the line of symmetry of the function

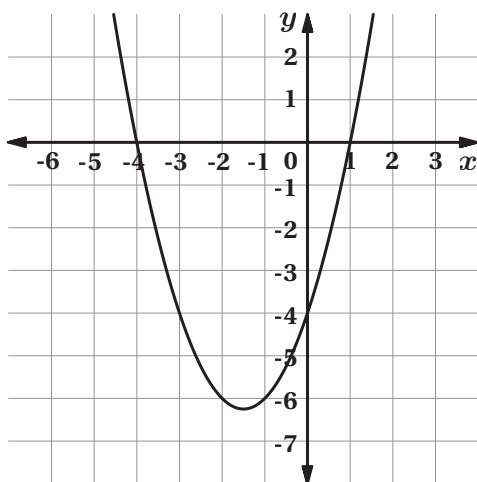
$$j(x) = (x + 6)(x + 2).$$

x -intercepts: $(-6, 0)$ and $(-2, 0)$

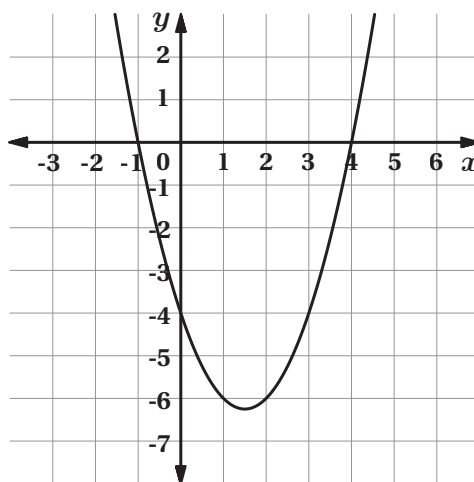
Line of symmetry: $x = -4$

6. Here are the graphs of two quadratic functions. Which is the graph of $w(x) = (x + 4)(x - 1)$? Explain your thinking.

Graph A



Graph B



Graph A. *Explanations vary.* $w(x)$ has x -intercepts at $(-4, 0)$ and $(1, 0)$, and so does Graph A.

7. The quadratic function $r(x)$ has x -intercepts at $(-6, 0)$ and $(1, 0)$. What is the equation for the axis of symmetry?

$$x = -2.5$$

8. Here are the two quadratic functions:

$$f(x) = (x + 3)(x - 11) \qquad g(x) = 2x(x - 8)$$

Kiran claims that the axis of symmetry for both functions is $x = 4$. Do you agree with Kiran's claim? Show or explain your thinking.

Yes. *Explanations vary.* The x -intercepts of $f(x)$ are $(-3, 0)$ and $(11, 0)$, and right between -3 and 11 is 4 . The x -intercepts of $g(x)$ are $(0, 0)$ and $(8, 0)$, and right between 0 and 8 is 4 .

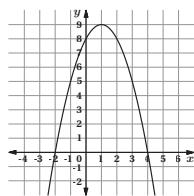
Name: _____ Date: _____ Period: _____

Additional Practice

A1.7.11

1. Here is the graph of a quadratic function.

- a. What is the vertex?
(1, 9)
- b. What is the equation of the line of symmetry?
 $x = 1$



2. A quadratic function has a vertex of $(-2, -8)$.

What is the equation of the line of symmetry?

- A.** $x = -2$
B. $x = -8$
C. $x = 2$
D. $x = 8$

3. Select *all* the true statements about the function $f(x) = x(x + 10)$.

- ☒ A. The x -coordinate of its vertex is -5 .
☐ B. The x -coordinate of its vertex is 5 .
☐ C. The x -coordinate of its vertex is 10 .
☒ D. It has exactly two x -intercepts.
☐ E. The x -intercepts are $(-1, 0)$ and $(-10, 0)$.
☒ F. The x -intercepts are $(0, 0)$ and $(-10, 0)$.
☐ G. The only x -intercept is $(10, 0)$.

4. Select *all* the functions that have a vertex with an x -coordinate of -3 .

- ☐ A. $g(x) = (x - 3)(x + 3)$ ☒ B. $g(x) = (x + 2)(x + 4)$
☐ C. $g(x) = (x - 1)(x - 5)$ ☒ D. $g(x) = x(x + 6)$
☐ E. $g(x) = x(x - 6)$

Unit 7 Lesson 11

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Name: _____ Date: _____ Period: _____

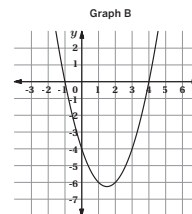
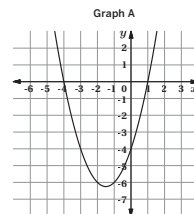
5. Determine the x -intercepts and the line of symmetry of the function

$$j(x) = (x + 6)(x + 2).$$

x -intercepts: **$(-6, 0)$ and $(-2, 0)$**

Line of symmetry: $x =$ **-4**

6. Here are the graphs of two quadratic functions. Which is the graph of $w(x) = (x + 4)(x - 1)$? Explain your thinking.



Graph A. Explanations vary. $w(x)$ has x -intercepts at $(-4, 0)$ and $(1, 0)$, and so does Graph A.

7. The quadratic function $r(x)$ has x -intercepts at $(-6, 0)$ and $(1, 0)$. What is the equation for the axis of symmetry?

$x = -2.5$

8. Here are the two quadratic functions:

$$f(x) = (x + 3)(x - 11) \quad g(x) = 2x(x - 8)$$

Kiran claims that the axis of symmetry for both functions is $x = 4$. Do you agree with Kiran's claim? Show or explain your thinking.

Yes. Explanations vary. The x -intercepts of $f(x)$ are $(-3, 0)$ and $(11, 0)$, and right between -3 and 11 is 4 . The x -intercepts of $g(x)$ are $(0, 0)$ and $(8, 0)$, and right between 0 and 8 is 4 .

Unit 7 Lesson 11

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.C.7.A
2	2	HSF.IF.C.7.A
3	2	HSF.IF.C.7.A
4	2	HSF.IF.C.7.A
5	2	HSF.IF.C.7.A
6	2	HSF.IF.C.7.A
7	2	HSF.IF.C.7.A
8	2	HSF.IF.C.7.A

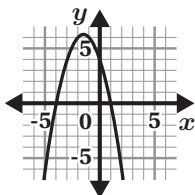
Notes:

Additional Practice

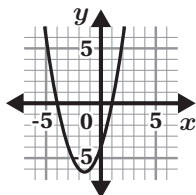
A1.7.12

1. Which graph shows the function $y = -(x - 4)(x + 1)$?

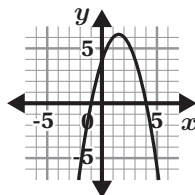
A.



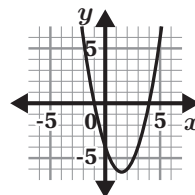
B.



C.



D.



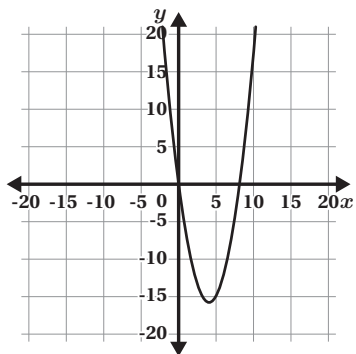
2. Match each equation to the graph it represents. One equation will have no match.

$$y = -x(x - 8)$$

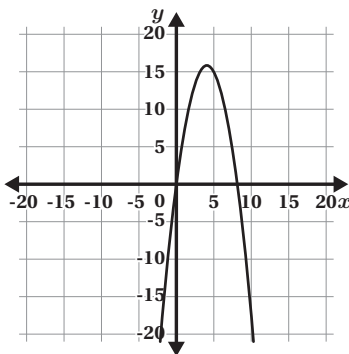
$$y = x(x - 8)$$

$$y = x(x + 8)$$

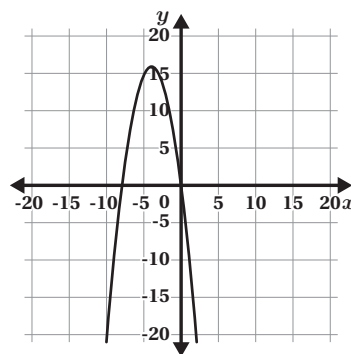
$$y = -x(x + 8)$$



$$y = x(x - 8)$$



$$y = -x(x - 8)$$



$$y = -x(x + 8)$$

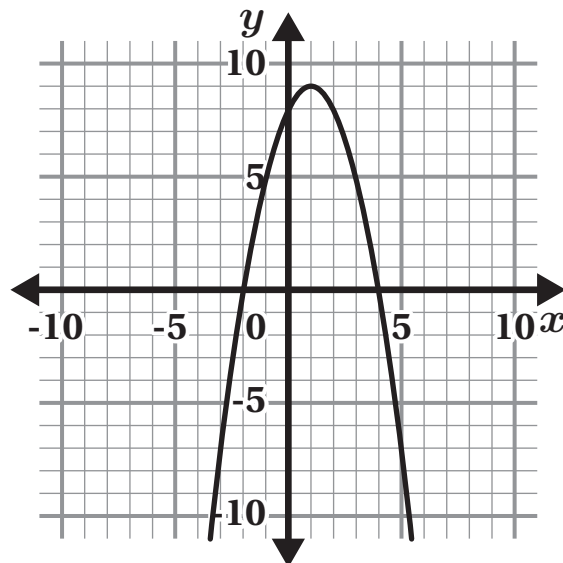
Problems 3–4: Here is a graph of $y = -1(x + 2)(x - 4)$. Change the equation so the vertex goes through:

3. $(1, -9)$

$$y = (x + 2)(x - 4)$$

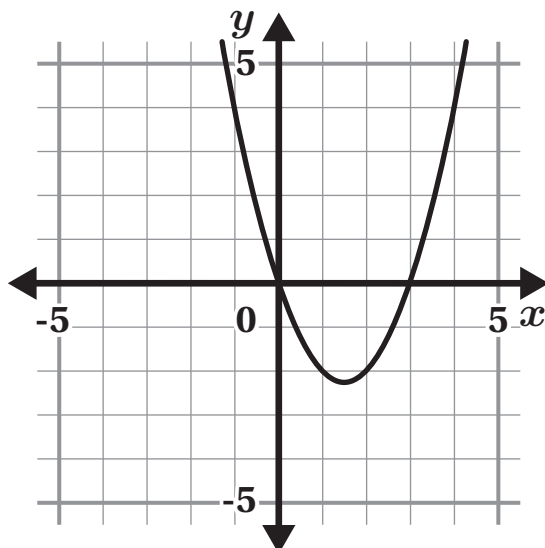
4. $(1, 4.5)$

$$y = -\frac{1}{2}(x + 2)(x - 4)$$

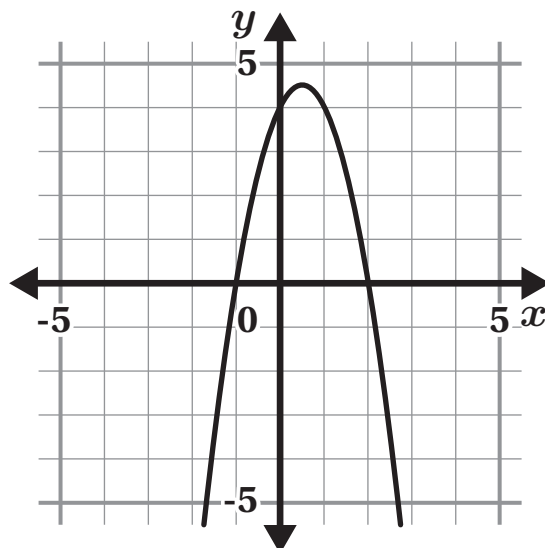


Problems 5–6: Write an equation to match each graph.

5. Equation: $y = x(x - 3)$



6. Equation: $y = -2(x + 1)(x - 2)$



7. Write the equations of three different quadratic functions that have the same x -intercepts but different y -intercepts.

Responses vary.

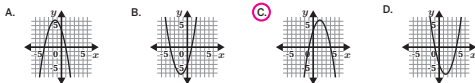
Equation 1	Equation 2	Equation 3
$y = (x - 3)(x + 5)$	$y = -(x - 3)(x + 5)$	$y = 2(x - 3)(x + 5)$

Name: _____ Date: _____ Period: _____

Additional Practice

A1.7.12

1. Which graph shows the function $y = -(x - 4)(x + 1)$?



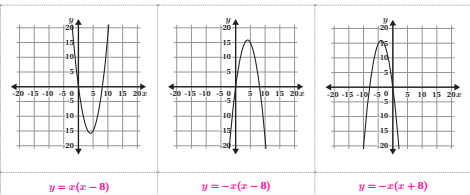
2. Match each equation to the graph it represents. One equation will have no match.

$$y = -x(x - 8)$$

$$y = x(x - 8)$$

$$y = x(x + 8)$$

$$y = -x(x + 8)$$



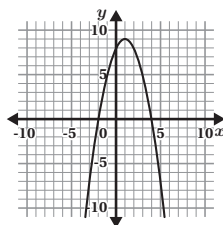
- Problems 3–4: Here is a graph of $y = -1(x + 2)(x - 4)$. Change the equation so the vertex goes through:

3. $(1, -9)$

$$y = (x + 2)(x - 4)$$

4. $(1, 4.5)$

$$y = -\frac{1}{2}(x + 2)(x - 4)$$



Unit 7 Lesson 12

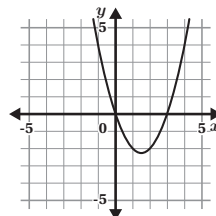
203

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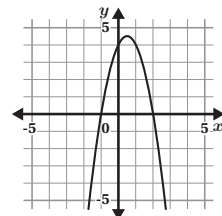
Name: _____ Date: _____ Period: _____

- Problems 5–6: Write an equation to match each graph.

5. Equation: $y = x(x - 3)$



6. Equation: $y = -2(x + 1)(x - 2)$



7. Write the equations of three different quadratic functions that have the same x-intercepts but different y-intercepts.

Responses vary.

Equation 1	Equation 2	Equation 3
$y = (x - 3)(x + 5)$	$y = -(x - 3)(x + 5)$	$y = 2(x - 3)(x + 5)$

Unit 7 Lesson 12

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.C.7.A
2	2	HSF.IF.C.7.A
3	2	HSF.IF.C.7.A, HSA.SSE.A.1.B
4	2	HSF.IF.C.7.A, HSA.SSE.A.1.B
5	2	HSF.BF.A.1, HSF.IF.C.7.A
6	2	HSF.BF.A.1, HSF.IF.C.7.A
7	3	HSF.BF.A.1, HSF.IF.C.7.A

Notes:

Additional Practice

A1.7.14

Problems 1–2: Circle if the transformation of $y = x^2$ is a *horizontal translation*, *vertical translation*, or *neither*.

1. $y = x^2 - 7$

Horizontal
translation

Vertical
translation

Neither

2. $y = 6x^2$

Horizontal
translation

Vertical
translation

Neither

3. These parabolas are translations of $y = x^2$.

Select *all* of the equations shown on the graph.

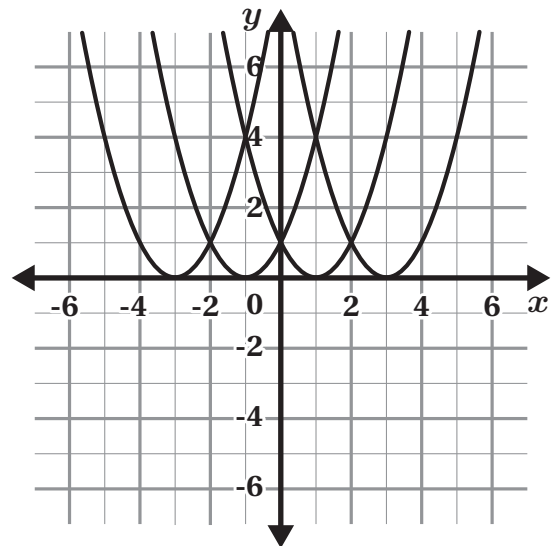
☐ A. $y = x^2 + 3$

☒ B. $y = (x - 3)^2$

☒ C. $y = (x + 3)^2$

☐ D. $y = x^2 - 1$

☒ E. $y = (x + 1)^2$



4. Match each vertex of a parabola to its equation.

a $y = x^2 - 3$ **b** (3, -4)

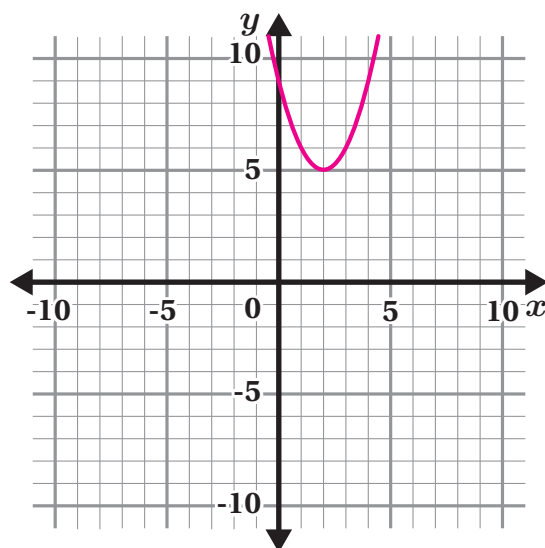
b $y = (x - 3)^2 - 4$ **c** (3, 0)

c $y = (x - 3)^2$ **d** (-3, -4)

d $y = (x + 3)^2 - 4$ **a** (0, -3)

e $y = (x - 4)^2 - 3$ **e** (4, -3)

5. Draw the graph of $y = (x - 2)^2 + 5$.



6. Here are four equations in vertex form. Which function has a graph with a vertex at $(-4, 5)$?

A. $y = (x - 4)^2 + 5$

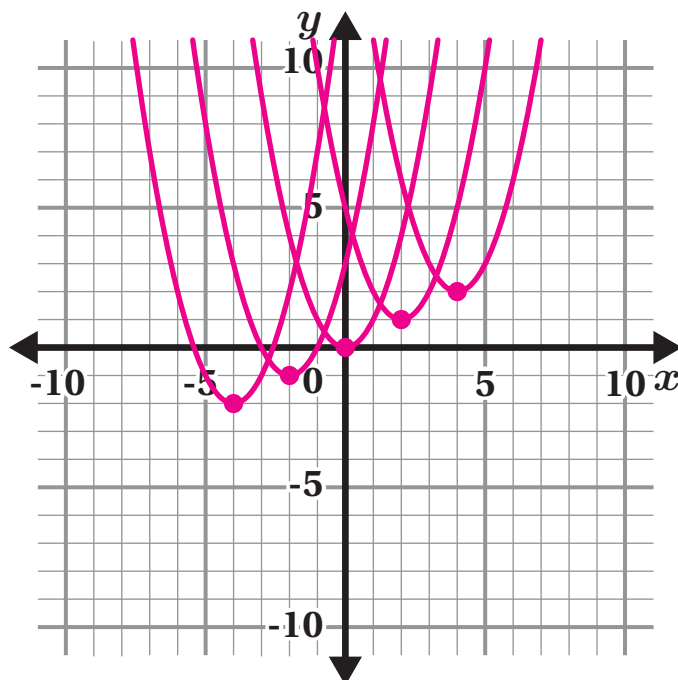
B. $y = (x + 4)^2 - 5$

C. $y = (x - 5)^2 + 4$

D. $y = (x + 4)^2 + 5$

7. Write and graph the equations of five parabolas so that when you connect their vertices, they form the line $y = \frac{1}{2}x$. **Responses vary.**

Equations
$y = x^2$
$y = (x - 2)^2 + 1$
$y = (x - 4)^2 + 2$
$y = (x + 2)^2 - 1$
$y = (x + 4)^2 - 2$



Name: _____ Date: _____ Period: _____

Additional Practice

A1.7.14

Problems 1-2: Circle if the transformation of $y = x^2$ is a horizontal translation, vertical translation, or neither.

1. $y = x^2 - 7$

Horizontal translation ☐ Vertical translation ☒ Neither ☐

2. $y = 6x^2$

Horizontal translation ☐ Vertical translation ☐ Neither ☒

3. These parabolas are translations of $y = x^2$.

Select all of the equations shown on the graph.

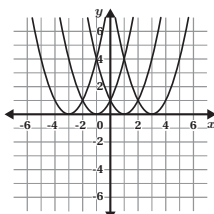
☐ A. $y = x^2 + 3$

☒ B. $y = (x - 3)^2$

☒ C. $y = (x + 3)^2$

☐ D. $y = x^2 - 1$

☒ E. $y = (x + 1)^2$



4. Match each vertex of a parabola to its equation.

a. $y = x^2 - 3$ _____ b. $(3, -4)$

b. $y = (x - 3)^2 - 4$ _____ c. $(3, 0)$

c. $y = (x - 3)^2$ _____ d. $(-3, -4)$

d. $y = (x + 3)^2 - 4$ _____ a. $(0, -3)$

e. $y = (x - 4)^2 - 3$ _____ e. $(4, -3)$

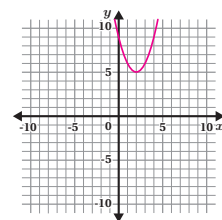
Unit 7 Lesson 14

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Name: _____ Date: _____ Period: _____

5. Draw the graph of $y = (x - 2)^2 + 5$.



6. Here are four equations in vertex form. Which function has a graph with a vertex at $(-4, 5)$?

A. $y = (x - 4)^2 + 5$

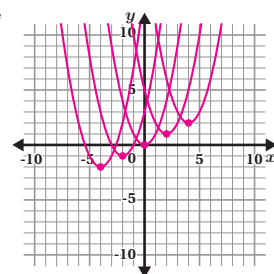
B. $y = (x + 4)^2 - 5$

C. $y = (x - 5)^2 + 4$

☒ D. $y = (x + 4)^2 + 5$

7. Write and graph the equations of five parabolas so that when you connect their vertices, they form the line $y = \frac{1}{2}x$. Responses vary.

Equations
$y = x^2$
$y = (x - 2)^2 + 1$
$y = (x - 4)^2 + 2$
$y = (x + 2)^2 - 1$
$y = (x + 4)^2 - 2$



Unit 7 Lesson 14

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.BF.B.3
2	2	HSF.BF.B.3
3	2	HSF.BF.B.3
4	2	HSF.BF.B.3, HSF.IF.C.7
5	2	HSF.BF.B.3, HSF.IF.C.7
6	2	HSF.BF.B.3
7	3	HSF.BF.B.3

Notes:

Additional Practice

A1.7.15

Problems 1–3: Here's the graph of $y = -2x^2$. Change one number to make the graph:

Responses vary.

1. Wider:

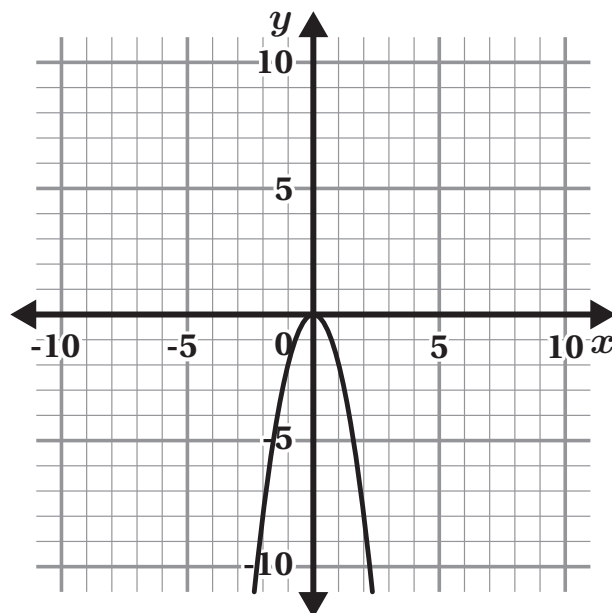
$y = -x^2$ (or equations with $-2 < a < 0$)

2. Narrower:

$y = -4x^2$ (or equations with $a < -2$)

3. Concave up:

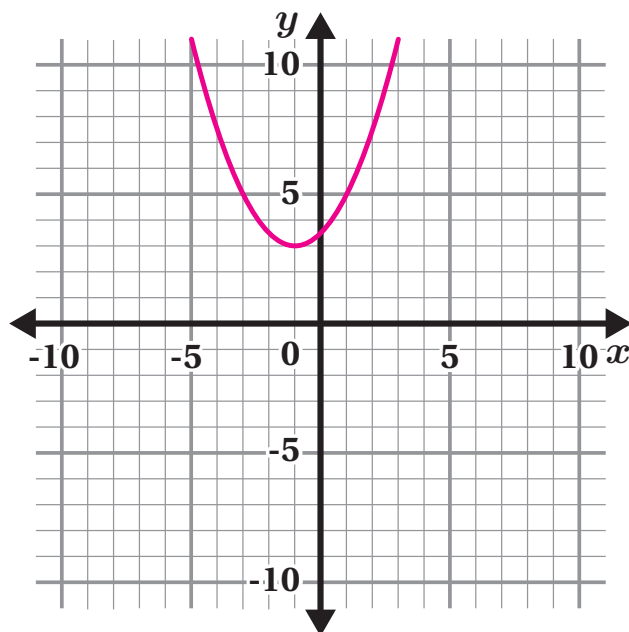
$y = 2x^2$ (or equations with $a > 0$)



4. Describe how the graph of $f(x) = x^2$ compares to $g(x) = 3(x + 1)^2 - 4$.

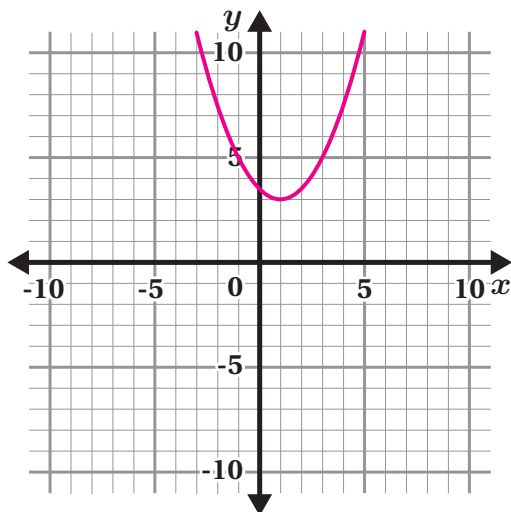
Responses vary. The vertex would be translated to $(-1, -4)$ and the parabola would open up. Since $a = 3$, the vertical distance between the vertex and the next point would be 3 units.

5. Draw a graph of a parabola that has a vertex at $(-1, 3)$ and is scaled vertically by a factor of $\frac{1}{2}$.

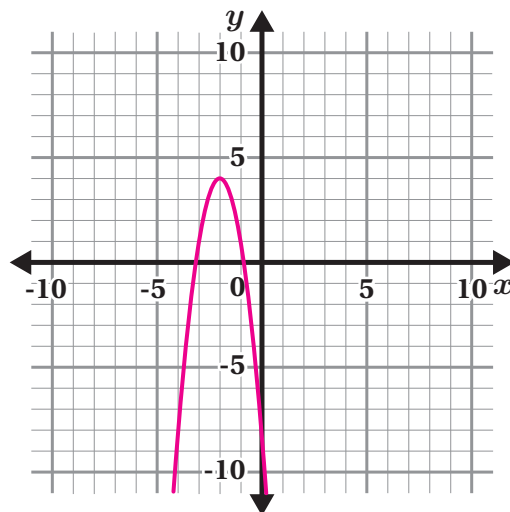


Problems 6–7: Draw a graph for each equation.

6. $c(x) = 0.5(x - 1)^2 + 3$



7. $d(x) = -3(x + 2)^2 + 4$



8. The graph of $h(x) = x^2$ is transformed so that its new equation is $j(x) = -2(x - 9)^2 + 6$. Select *all* of the true statements.

- ☐ A. The graph of $j(x)$ is concave up.
- ☒ B. The graph of $j(x)$ is concave down.
- ☒ C. The graph of $j(x)$ is narrower than $h(x)$.
- ☐ D. The graph of $j(x)$ is wider than $h(x)$.
- ☒ E. The vertex of $h(x)$ was translated 9 units to the right and 6 units up to get to the vertex of $j(x)$.
- ☐ F. The vertex of $h(x)$ was translated 9 units to the left and 6 units down to get to the vertex of $j(x)$.

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Additional Practice

A1.7.15

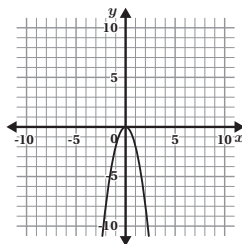
Problems 1–3: Here's the graph of $y = -2x^2$. Change one number to make the graph:

Responses vary.

1. Wider:
 $y = -ax^2$ (or equations with $-2 < a < 0$)

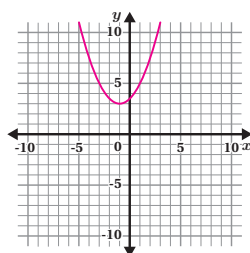
2. Narrower:
 $y = -4x^2$ (or equations with $a < -2$)

3. Concave up:
 $y = 2x^2$ (or equations with $a > 0$)



4. Describe how the graph of $f(x) = x^2$ compares to $g(x) = 3(x + 1)^2 - 4$.
Responses vary. The vertex would be translated to $(-1, -4)$ and the parabola would open up. Since $a = 3$, the vertical distance between the vertex and the next point would be 3 units.

5. Draw a graph of a parabola that has a vertex at $(-1, 3)$ and is scaled vertically by a factor of $\frac{1}{2}$.



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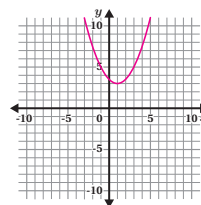
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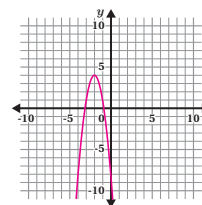
Name: _____ Date: _____ Period: _____

Problems 6–7: Draw a graph for each equation.

6. $c(x) = 0.5(x - 1)^2 + 3$



7. $d(x) = -3(x + 2)^2 + 4$



8. The graph of $h(x) = x^2$ is transformed so that its new equation is $j(x) = -2(x - 9)^2 + 6$. Select *all* of the true statements.

- ☐ A. The graph of $j(x)$ is concave up.
- ☒ B. The graph of $j(x)$ is concave down.
- ☒ C. The graph of $j(x)$ is narrower than $h(x)$.
- ☐ D. The graph of $j(x)$ is wider than $h(x)$.
- ☒ E. The vertex of $h(x)$ was translated 9 units to the right and 6 units up to get to the vertex of $j(x)$.
- ☐ F. The vertex of $h(x)$ was translated 9 units to the left and 6 units down to get to the vertex of $j(x)$.

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.BF.B.3
2	1	HSF.BF.B.3
3	1	HSF.BF.B.3
4	2	HSF.IF.C.7.A, HSF.BF.B.3
5	2	HSF.IF.C.7.A, HSF.BF.B.3
6	2	HSF.IF.C.7.A, HSF.BF.B.3
7	2	HSF.IF.C.7.A, HSF.BF.B.3
8	2	HSF.BF.B.3

Notes:

Additional Practice

A1.7.16

1. Which equation has a graph with a vertex at (5, 8)?

A. $y = (x + 5)^2 + 8$

B. $y = (x - 5)^2 + 8$

C. $y = (x + 8)^2 + 5$

D. $y = (x - 8)^2 + 5$

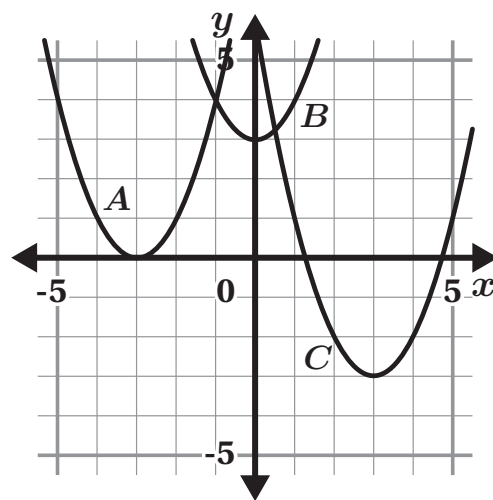
2. Parabolas A, B, and C are translations of $y = x^2$.

Determine the equation of each parabola.

A: $y = (x + 3)^2$

B: $y = x^2 + 3$

C: $y = (x - 3)^2 - 3$



3. Select *all* the equations that match this graph.

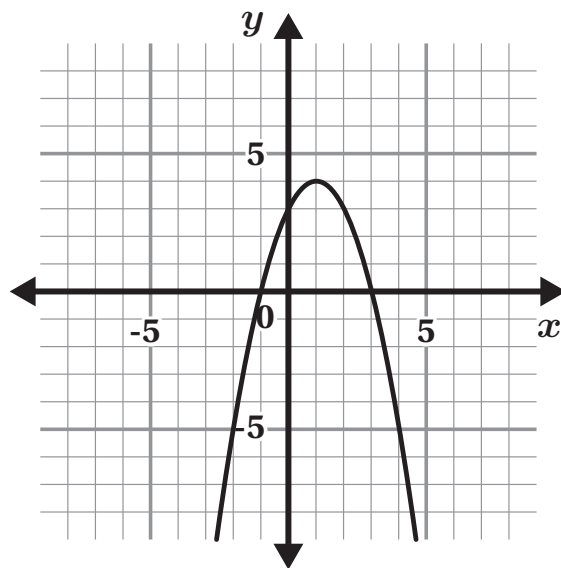
☐ A. $y = -(x + 1)^2 - 4$

☒ B. $y = -(x + 1)(x - 3)$

☒ C. $y = -(x - 1)^2 + 4$

☐ D. $y = (x - 1)(x + 3)$

☐ E. $y = (x - 1)^2 + 4$



4. Write an equation for a quadratic function with x -intercepts at (4, 0) and (-7, 0).

Responses vary. $y = (x - 4)(x + 7)$

Problems 5–6: Write an equation for a quadratic function that matches each description. Use graphing technology to check your work.

5. Concave up with a vertex at $(0, 5)$

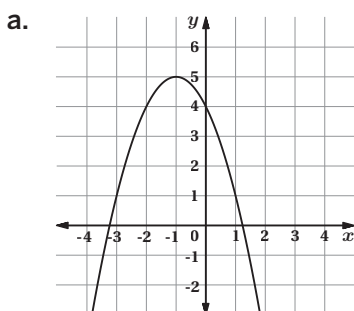
Responses vary. $y = x^2 + 5$

6. Concave down with a vertex at $(-2, 0)$

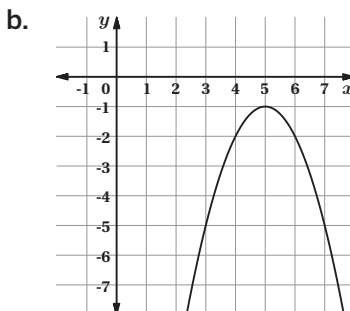
Responses vary. $y = -(x + 2)^2$

7. Match each graph with the equation that it represents.

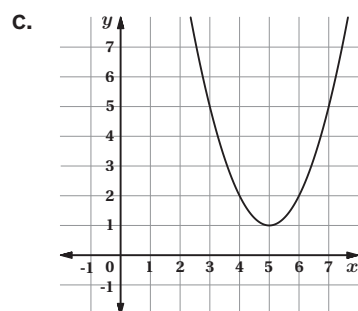
..... **a** $y = -(x + 1)^2 + 5$



..... **c** $y = (x - 5)^2 + 1$



..... **b** $y = -(x - 5)^2 - 1$



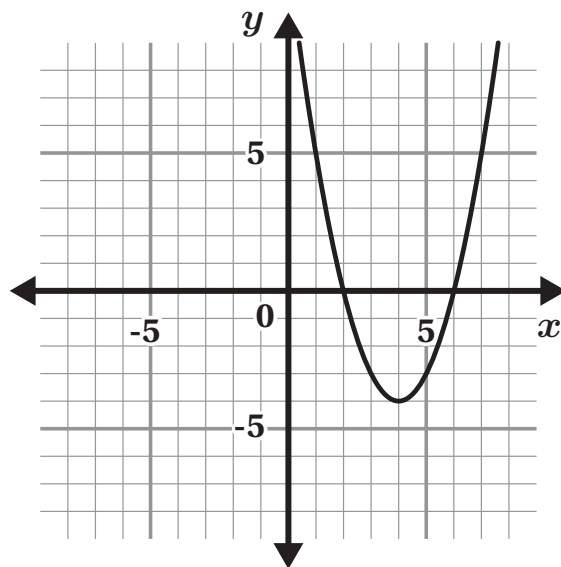
8. Here are two equations:

$$f(x) = (x - 2)(x - 6)$$

$$g(x) = (x - 4)^2 - 4$$

Show or explain how you know that both equations describe this graph.

Responses vary. $f(x)$ is written in factored form, so I know the x -intercepts are at 2 and 6. The vertex is exactly between the x -intercepts, so I plugged in $x = 4$. I found the vertex was at $(4, -4)$. $g(x)$ is written in vertex form and shows the vertex is at $(4, -4)$.



Name: _____ Date: _____ Period: _____

Additional Practice

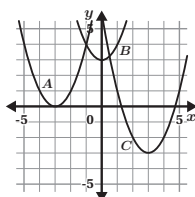
A1.7.16

1. Which equation has a graph with a vertex at (5, 8)?

- A. $y = (x + 5)^2 + 8$
B. $y = (x - 5)^2 + 8$
 C. $y = (x + 8)^2 + 5$
 D. $y = (x - 8)^2 + 5$

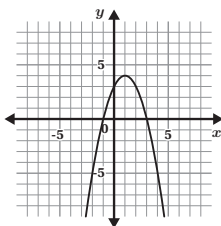
2. Parabolas A, B, and C are translations of $y = x^2$. Determine the equation of each parabola.

- A: $y = (x + 3)^2$
 B: $y = x^2 + 3$
 C: $y = (x - 3)^2 - 3$



3. Select all the equations that match this graph.

- ☐ A. $y = -(x + 1)^2 - 4$
☒ B. $y = -(x + 1)(x - 3)$
☒ C. $y = -(x - 1)^2 + 4$
☐ D. $y = (x - 1)(x + 3)$
☐ E. $y = (x - 1)^2 + 4$



4. Write an equation for a quadratic function with x -intercepts at (4, 0) and (-7, 0).
 Responses vary. $y = (x - 4)(x + 7)$

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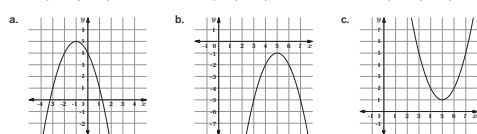
Name: _____ Date: _____ Period: _____

Problems 5–6: Write an equation for a quadratic function that matches each description. Use graphing technology to check your work.

5. Concave up with a vertex at (0, 5)
 Responses vary. $y = x^2 + 5$
6. Concave down with a vertex at (-2, 0)
 Responses vary. $y = -(x + 2)^2$

7. Match each graph with the equation that it represents.

- a. $y = -(x + 1)^2 + 5$ b. $y = (x - 5)^2 + 1$ c. $y = -(x - 5)^2 - 1$



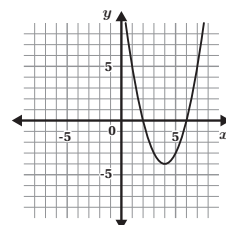
8. Here are two equations:

$$f(x) = (x - 2)(x - 6)$$

$$g(x) = (x - 4)^2 - 4$$

Show or explain how you know that both equations describe this graph.

Responses vary. $f(x)$ is written in factored form, so I know the x -intercepts are at 2 and 6. The vertex is exactly between the x -intercepts, so I plugged in $x = 4$. I found the vertex was at (4, -4). $g(x)$ is written in vertex form and shows the vertex is at (4, -4).



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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSF.BF.A.1, HSF.IF.C.8
2	2	HSF.IF.C.8
3	2	HSF.BF.A.1, HSF.IF.C.8
4	2	HSF.IF.C.8, HSF.IF.C.7.A*
5	2	HSF.IF.C.8, HSF.IF.C.7.A*
6	2	HSF.IF.C.8, HSF.IF.C.7.A*
7	2	HSF.IF.C.8
8	2	HSF.IF.C.7.A*

*This problem builds toward the standard shown.

Notes:

Additional Practice

A1.7.17

1. Which function has a graph with a vertex at $(-3, 7)$?

A. $y = (x - 3)^2 + 7$

B. $y = (x - 3)^2 - 7$

☒ C. $y = (x + 3)^2 + 7$

D. $y = (x + 3)^2 - 7$

2. Select *all* the functions whose graphs have an x -intercept at $(-2, 0)$.

☐ A. $a(x) = (x - 2)(x - 4)$

☒ B. $b(x) = (x + 2)(x + 4)$

☐ C. $c(x) = 2x(x + 6)$

☒ D. $d(x) = (2x + 4)(x + 3)$

☐ E. $e(x) = (3x - 6)(x + 5)$

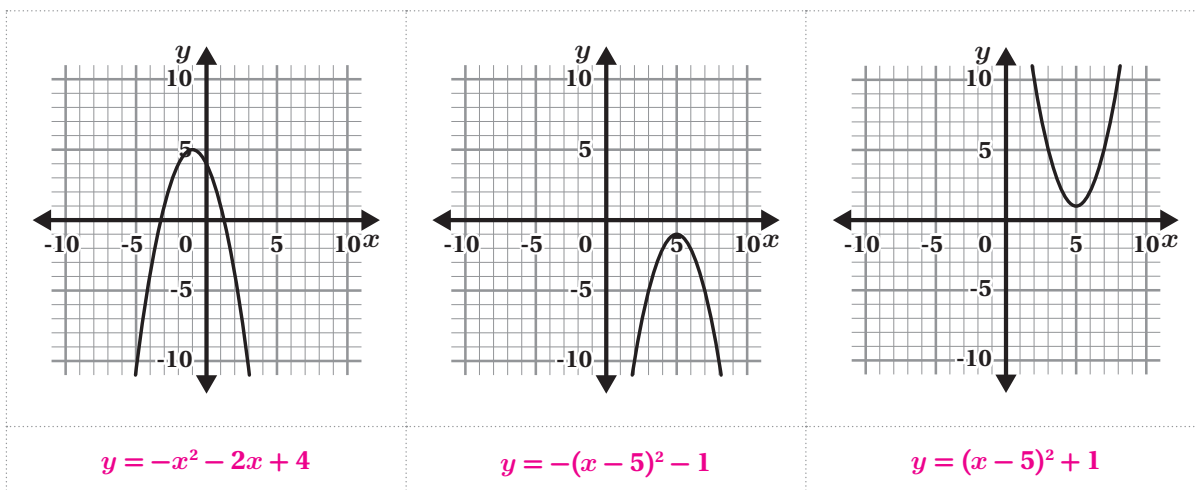
3. Match each equation to the graph it represents. One equation will have no match.

$$y = -(x - 1)^2 + 5$$

$$y = -x^2 - 2x + 4$$

$$y = -(x - 5)^2 - 1$$

$$y = (x - 5)^2 + 1$$



4. Write an equation of a parabola that opens downward and has x -intercepts at $(-4, 0)$ and $(6, 0)$.

Responses vary. $y = -(x + 4)(x - 6)$

Problems 5–6: Terrence and Raj try to write an equation to describe this graph. At least one equation is incorrect.

Terrence

$$y = (x + 2)^2 - 25$$

Raj

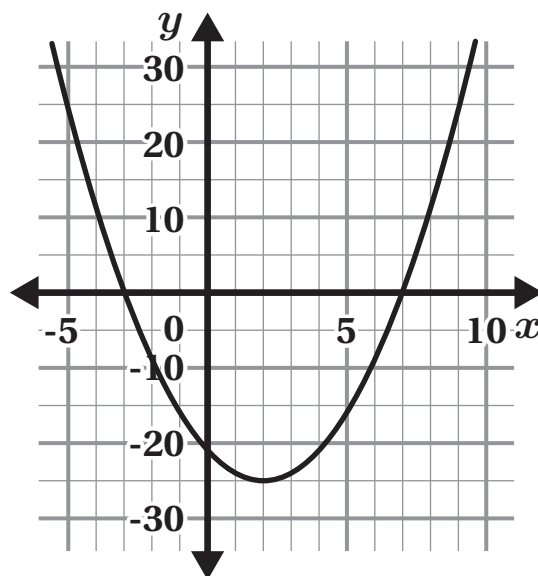
$$y = -(x + 3)(x - 7)$$

5. Whose equation is correct? Circle one.

Terrence

Raj

Neither



6. Explain how you would change the incorrect equation(s) so that it creates this graph.

Responses vary. The graph of Terrence's equation would have a vertex of $(-2, -25)$, so he should change his equation to $y = (x - 2)^2 - 25$. The graph of Raj's equation would be concave down, so he should remove the negative sign in front.

Problems 7–8: Here is a function: $r(x) = -x(x + 4)$.

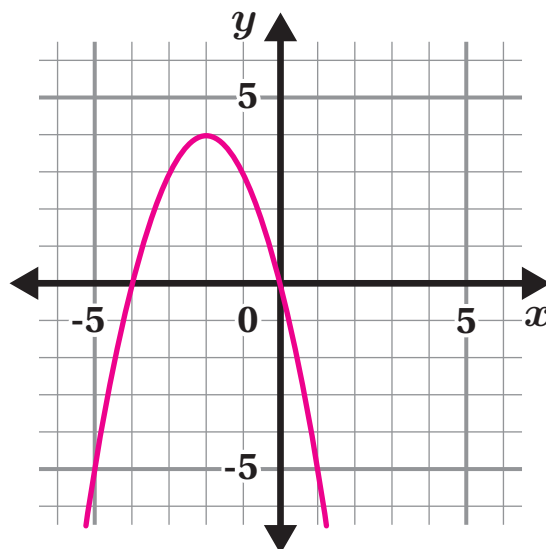
7. Determine the x -intercepts and vertex of $r(x)$.

x -intercept: **$(0, 0)$**

x -intercept: **$(-4, 0)$**

Vertex: **$(-2, 4)$**

8. Draw the graph of the function $r(x)$.



Name: _____ Date: _____ Period: _____

Additional Practice

A1.7.17

1. Which function has a graph with a vertex at $(-3, 7)$?

- A. $y = (x - 3)^2 + 7$
 B. $y = (x - 3)^2 - 7$
☒ C. $y = (x + 3)^2 + 7$
 D. $y = (x + 3)^2 - 7$

2. Select all the functions whose graphs have an x -intercept at $(-2, 0)$.

- ☐ A. $a(x) = (x - 2)(x - 4)$
☒ B. $b(x) = (x + 2)(x + 4)$
☐ C. $c(x) = 2x(x + 6)$
☒ D. $d(x) = (2x + 4)(x + 3)$
☐ E. $e(x) = (3x - 6)(x + 5)$

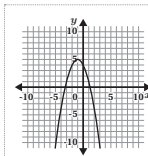
3. Match each equation to the graph it represents. One equation will have no match.

$$y = -(x - 1)^2 + 5$$

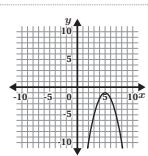
$$y = -x^2 - 2x + 4$$

$$y = -(x - 5)^2 - 1$$

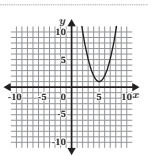
$$y = (x - 5)^2 + 1$$



$$y = -x^2 - 2x + 4$$



$$y = -(x - 5)^2 - 1$$



$$y = (x - 5)^2 + 1$$

Unit 7 Lesson 17

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4. Write an equation of a parabola that opens downward and has x -intercepts at $(-4, 0)$ and $(6, 0)$.

Responses vary. $y = -(x + 4)(x - 6)$

Problems 5–6: Terrence and Raj try to write an equation to describe this graph. At least one equation is incorrect.

Terrence

Raj

$$y = (x + 2)^2 - 25$$

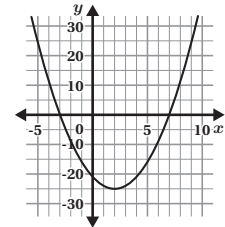
$$y = -(x + 3)(x - 7)$$

5. Whose equation is correct? Circle one.

Terrence

Raj

☒ Neither



6. Explain how you would change the incorrect equation(s) so that it creates this graph.

Responses vary. The graph of Terrence's equation would have a vertex of $(-2, -25)$, so he should change his equation to $y = (x - 2)^2 - 25$. The graph of Raj's equation would be concave down, so he should remove the negative sign in front.

Problems 7–8: Here is a function: $r(x) = -x(x + 4)$.

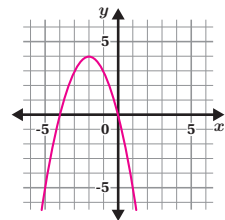
7. Determine the x -intercepts and vertex of $r(x)$.

x -intercept: $(0, 0)$

x -intercept: $(-4, 0)$

Vertex: $(-2, 4)$

8. Draw the graph of the function $r(x)$.



Unit 7 Lesson 17

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.BF.A.1*
2	2	HSF.BF.A.1
3	2	HSF.BF.A.1
4	2	HSF.BF.A.1
5	2	HSF.BF.A.1
6	2	HSF.BF.A.1
7	2	HSF.BF.A.1
8	2	HSF.BF.A.1

*This problem builds toward the standard shown.

Notes:

Additional Practice

A1.8.02

1. Match each expression to its equivalent expression in standard form.

- | | |
|----------------------|--|
| a. $(2x + 3)(x + 4)$ | c $x^2 + 10x + 16$ |
| b. $(x + 3)(x + 2)$ | a $2x^2 + 11x + 12$ |
| c. $(x + 8)(x + 2)$ | b $x^2 + 5x + 6$ |
| d. $(x + 3)(2x + 1)$ | d $2x^2 + 7x + 3$ |

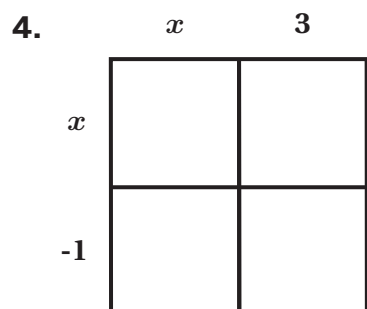
2. Complete the table by writing each expression in standard form.

Factored Form	Standard Form
$(x - 2)(x - 4)$	$x^2 - 6x + 8$
$(x + 5)(x + 2)$	$x^2 + 7x + 10$

3. Which expression is the standard form of $(x + 1)(2x - 3)$?

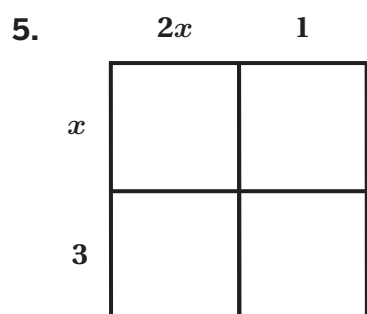
- A. $2x^2 - 2x - 6$
- B. $2x^2 - x - 3$**
- C. $3x^2 - 2x - 6$
- D. $3x^2 - x - 3$

Problems 4–6: For each area model, write two expressions, one in factored form and one in standard form.



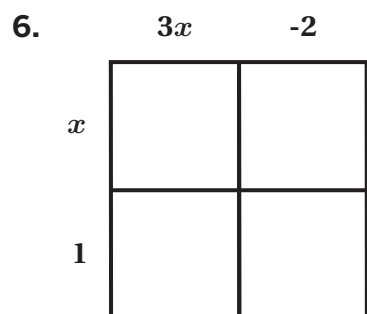
Factored form: $(x + 3)(x - 1)$

Standard form: $x^2 + 2x - 3$



Factored form: $(2x + 1)(x + 3)$

Standard form: $2x^2 + 7x + 3$



Factored form: $(3x - 2)(x + 1)$

Standard form: $3x^2 + x - 2$

Name: _____ Date: _____ Period: _____

Additional Practice

A1.8.02

1. Match each expression to its equivalent expression in standard form.

- a. $(2x + 3)(x + 4)$ **c.** $x^2 + 10x + 16$
 b. $(x + 3)(x + 2)$ **a.** $2x^2 + 11x + 12$
 c. $(x + 8)(x + 2)$ **b.** $x^2 + 5x + 6$
 d. $(x + 3)(2x + 1)$ **d.** $2x^2 + 7x + 3$

2. Complete the table by writing each expression in standard form.

Factored Form	Standard Form
$(x - 2)(x - 4)$	$x^2 - 6x + 8$
$(x + 5)(x + 2)$	$x^2 + 7x + 10$

3. Which expression is the standard form of $(x + 1)(2x - 3)$?

- A. $2x^2 - 2x - 6$
B. $2x^2 - x - 3$
 C. $3x^2 - 2x - 6$
 D. $3x^2 - x - 3$

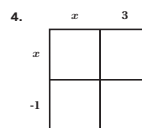
Unit 8 Lesson 2

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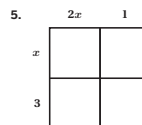
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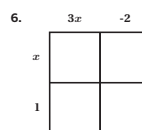
Problems 4–6: For each area model, write two expressions, one in factored form and one in standard form.



Factored form: $(x + 3)(x - 1)$
 Standard form: $x^2 + 2x - 3$



Factored form: $(2x + 1)(x + 3)$
 Standard form: $2x^2 + 7x + 3$



Factored form: $(3x - 2)(x + 1)$
 Standard form: $3x^2 + x - 2$

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSA.SSE.A.2
2	2	HSA.SSE.A.2
3	1	HSA.SSE.A.2
4	2	HSA.SSE.A.2
5	2	HSA.SSE.A.2
6	2	HSA.SSE.A.2

Notes:

Additional Practice

A1.8.03

1. Is the quadratic expression $2x(x - 1)$ in standard form or in factored form?

Factored form

2. Which quadratic expression is written in standard form?

A. $(x + 3)^2 - 2$

B. $(x - 8)3x$

C. $4x^2 + 7(x - 1)$

D. $-2x^2 + 7x + 4$

3. Consider the quadratic expression $3x^2 - x$.

- a Is the expression in standard form? Explain your thinking.

Yes. Explanations vary. It is the sum of a squared variable term and a linear term.

- b Is the expression equivalent to $x(3x - 1)$? Explain your thinking.

Yes. Explanations vary. Applying the distributive property gives $3x^2 - x$.

4. Which expression is equivalent to $(4x - 2)(2x + 3)$?

A. $8x^2 - 4x + 6$

B. $8x^2 + 12x + 6$

C. $8x^2 + 8x - 6$

D. $8x^2 + 10x - 6$

5. Match each quadratic expression in factored form with an equivalent expanded expression in standard form.

Factored Form

Standard Form

a. $(x + 3)(x + 8)$

d $x^2 - 8x + 12$

b. $(2x + 4)(x - 3)$

a $x^2 + 11x + 24$

c. $(2x + 6)(x + 4)$

b $2x^2 - 2x - 12$

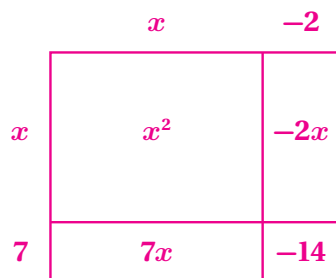
d. $(x - 2)(x - 6)$

c $2x^2 + 14x + 24$

6. For each expression in factored form, write an equivalent expression in standard form. Draw a diagram to show your thinking. *Diagrams vary.*

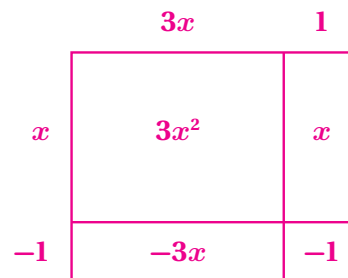
a $(x - 2)(x + 7)$

$$x^2 + 5x - 14$$



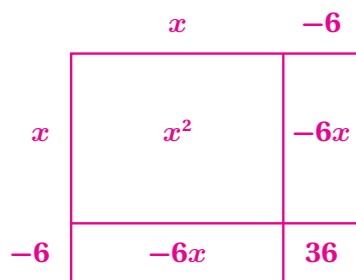
b $(3x + 1)(x - 1)$

$$3x^2 - 2x - 1$$



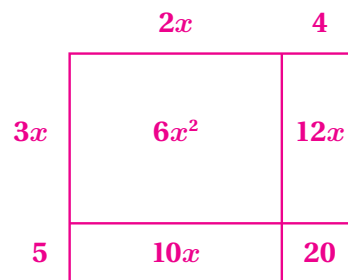
c $(x - 6)^2$

$$x^2 - 12x + 36$$



d $(2x + 4)(3x + 5)$

$$6x^2 + 22x + 20$$



7. Here are two quadratic expressions:

$$(x + 3)^2 - 1$$

$$(x + 4)(x + 2)$$

Are the expressions equivalent? Explain your thinking.

Yes. Explanations vary. When in standard form, both expressions can be written as $x^2 + 6x + 8$.

8. Here is the quadratic expression $(2x - 6)(-x + 5)$. Mai claims that when the expression is written in standard form, it has two negative terms. Han claims that when the expression is written in standard form, it has three negative terms. Who is correct? Show or explain your thinking.

Mai is correct. Explanations vary. When I use the distributive property, the result is $-2x^2 + 16x - 30$, which I can rewrite as $-2x^2 + 16x + (-30)$. Only the squared variable term and the constant term are negative.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.8.03

1. Is the quadratic expression $2x(x - 1)$ in standard form or in factored form?

Factored form

2. Which quadratic expression is written in standard form?

A. $(x + 3)^2 - 2$

B. $(x - 8)3x$

C. $4x^2 + 7(x - 1)$

D. $-2x^2 + 7x + 4$

3. Consider the quadratic expression $3x^2 - x$.

- a. Is the expression in standard form? Explain your thinking.

Yes. Explanations vary. It is the sum of a squared variable term and a linear term.

- b. Is the expression equivalent to $x(3x - 1)$? Explain your thinking.

Yes. Explanations vary. Applying the distributive property gives $3x^2 - x$.

4. Which expression is equivalent to $(4x - 2)(2x + 3)$?

A. $8x^2 - 4x + 6$

B. $8x^2 + 12x + 6$

C. $8x^2 + 8x - 6$

D. $8x^2 + 10x - 6$

5. Match each quadratic expression in factored form with an equivalent expanded expression in standard form.

Factored Form

Standard Form

a. $(x + 3)(x + 8)$

d. $x^2 - 8x + 12$

b. $(2x + 4)(x - 3)$

a. $x^2 + 11x + 24$

c. $(2x + 6)(x + 4)$

b. $2x^2 - 2x - 12$

d. $(x - 2)(x - 6)$

c. $2x^2 + 14x + 24$

Unit 8 Lesson 3

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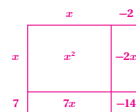
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6. For each expression in factored form, write an equivalent expression in standard form.

Draw a diagram to show your thinking. **Diagrams vary.**

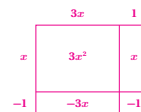
a. $(x - 2)(x + 7)$

$x^2 + 5x - 14$



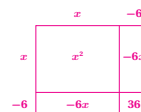
b. $(3x + 1)(x - 1)$

$3x^2 - 2x - 1$



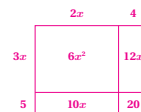
c. $(x - 6)^2$

$x^2 - 12x + 36$



d. $(2x + 4)(3x + 5)$

$6x^2 + 22x + 20$



7. Here are two quadratic expressions:

$(x + 3)^2 - 1$

$(x + 4)(x + 2)$

Are the expressions equivalent? Explain your thinking.

Yes. Explanations vary. When in standard form, both expressions can be written as $x^2 + 6x + 8$.

8. Here is the quadratic expression $(2x - 6)(-x + 5)$. Mai claims that when the expression is written in standard form, it has two negative terms. Han claims that when the expression is written in standard form, it has three negative terms. Who is correct? Show or explain your thinking.

Mai is correct. Explanations vary. When I use the distributive property, the result is $-2x^2 + 16x - 30$, which I can rewrite as $-2x^2 + 16x + (-30)$. Only the squared variable term and the constant term are negative.

Unit 8 Lesson 3

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.SSE.A.2
2	2	HSA.SSE.A.2
3	2	HSA.SSE.A.2
4	2	HSA.SSE.A.2
5	2	HSA.SSE.A.2
6	2	HSA.SSE.A.2
7	2	HSA.SSE.A.2
8	2	HSA.SSE.A.2

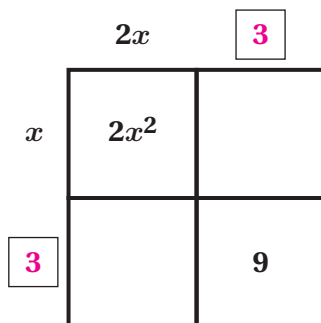
Notes:

Additional Practice

A1.8.04

1. Write two possible constants that could complete the outside of the diagram.

Responses vary. Sample shown on diagram.



2. Rewrite each expression in factored or standard form.

Factored Form	Standard Form
$(3x + 1)(3x - 1)$	$9x^2 - 1$
$(4x + 1)(x - 2)$	$4x^2 - 7x - 2$
$(2x + 3)(x - 6)$	$2x^2 - 9x - 18$
$(x + 3)(x + 3)$	$x^2 + 6x + 9$

3. Determine values that make the equation true.

$$2x^2 + x - \underline{3} = (\underline{1}x - 1)(2x + \underline{3})$$

Problems 4–5: Complete the diagram puzzles and expressions.

4.

	$3x$	2
x	$3x^2$	$2x$
-4	$-12x$	-8

Factored form: $(3x + 2)(x - 4)$

Standard form: $3x^2 - 10x - 8$

5.

	$4x$	-3
x	$4x^2$	$-3x$
2	$8x$	-6

Factored form: $(4x - 3)(x + 2)$

Standard form: $4x^2 + 5x - 6$

- 6.** This quadratic expression in standard form has an unknown c -value. Given that we know the expression can be factored, fill in the c -value.

$$3x^2 - 7x - \underline{6}$$

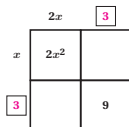
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Additional Practice

A1.8.04

1. Write two possible constants that could complete the outside of the diagram.

Responses vary. Sample shown on diagram.



2. Rewrite each expression in factored or standard form.

Factored Form	Standard Form
$(3x + 1)(3x - 1)$	$9x^2 - 1$
$(4x + 1)(x - 2)$	$4x^2 - 7x - 2$
$(2x + 3)(x - 6)$	$2x^2 - 9x - 18$
$(x + 3)(x + 3)$	$x^2 + 6x + 9$

3. Determine values that make the equation true.

$$2x^2 + x - 3 = (1x - 1)(2x + 3)$$

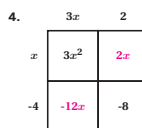
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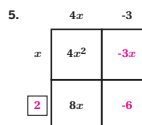
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Problems 4–5: Complete the diagram puzzles and expressions.



Factored form: $(3x + 2)(x - 4)$

Standard form: $3x^2 - 10x - 8$



Factored form: $(4x - 3)(x + 2)$

Standard form: $4x^2 + 5x - 6$

6. This quadratic expression in standard form has an unknown c -value. Given that we know the expression can be factored, fill in the c -value.

$$3x^2 - 7x - 6$$

Unit 8 Lesson 4

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.SSE.A.2
2	1	HSA.SSE.A.2
3	2	HSA.SSE.A.2
4	2	HSA.SSE.A.2
5	2	HSA.SSE.A.2
6	3	HSA.SSE.A.2

Notes:

Additional Practice

A1.8.05

1. Select *all* expressions that are equivalent to the expression $x + (-7)$.

☐ A. $7 + x$

☒ B. $x - 7$

☐ C. $7 - x$

☒ D. $-7 + x$

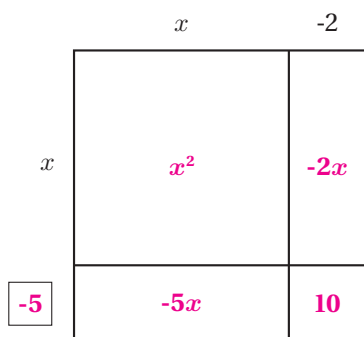
☐ E. $7 + (-x)$

☒ F. $-7 - (-x)$

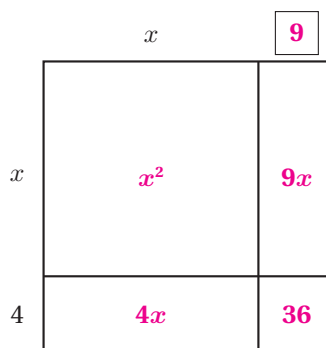
☐ G. $-x - (-7)$

2. Complete the diagrams to show that each pair of expressions is equivalent.

a $(x - 2)(x - 5)$ and $x^2 - 7x + 10$



b $(x + 9)(x + 4)$ and $x^2 + 13x + 36$



3. Fill in the blanks to make each equation true. Use the diagrams if they help your thinking.

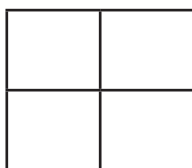
a $x^2 + 9x + 8 = (x + 8)(x + 1)$



b $x^2 + 5x + 6 = (x + 2)(x + 3)$



c $x^2 - 16x + 63 = (x - 9)(x - 7)$



d $x^2 - 9x + 20 = (x - 5)(x - 4)$



4. Rewrite each quadratic expression in standard form.

a $(x + 6)(x + 8)$

b $(x - 1)(x - 3)$

5. Factor each expression.

a $x^2 - 12x + 20$
 $(x - 2)(x - 10)$

b $x^2 + 14x + 49$
 $(x + 7)(x + 7)$

c $x^2 - 17x + 72$
 $(x - 8)(x - 9)$

d $x^2 + 9x + 20$
 $(x + 4)(x + 5)$

6. Rewrite each expression in factored or standard form.

Factored Form	Standard Form
$x(x + 4)$	$x^2 + 4x$
$x(x - 7)$	$x^2 - 7x$
$(x + 11)(x + 2)$	$x^2 + 13x + 22$
$(x - 5)(x - 5)$	$x^2 - 10x + 25$
$(x - 9)(x - 3)$	$x^2 - 12x + 27$

7. Here are two expressions: $(x + n)(x + n)$ and $x^2 + nx + n^2$. Tyler claims the expressions are equivalent. Shawn claims the expressions are not equivalent. Who is correct? Explain your thinking.

Shawn. Explanations vary. When I use the distributive property, the expression $(x + n)(x + n)$ is equivalent to $x^2 + nx + nx + n^2$ or $x^2 + 2nx + n^2$.

Name: _____ Date: _____ Period: _____

Additional Practice

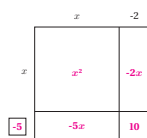
A1.8.05

1. Select *all* expressions that are equivalent to the expression $x + (-7)$.

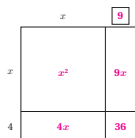
- ☐ A. $7 + x$ ☒ B. $x - 7$
☐ C. $7 - x$ ☒ D. $-7 + x$
☐ E. $7 + (-x)$ ☒ F. $-7 - (-x)$
☐ G. $-x - (-7)$

2. Complete the diagrams to show that each pair of expressions is equivalent.

a. $(x - 2)(x - 5)$ and $x^2 - 7x + 10$



b. $(x + 9)(x + 4)$ and $x^2 + 13x + 36$



3. Fill in the blanks to make each equation true. Use the diagrams if they help your thinking.

a. $x^2 + 9x + 8 = (x + 8)(x + 1)$



b. $x^2 + 5x + 6 = (x + 2)(x + 3)$



c. $x^2 - 16x + 63 = (x - 9)(x - 7)$



d. $x^2 - 9x + 20 = (x - 5)(x - 4)$



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4. Rewrite each quadratic expression in standard form.

a. $(x + 6)(x + 8)$ b. $(x - 1)(x - 3)$

5. Factor each expression.

a. $x^2 - 12x + 20$
 $(x - 2)(x - 10)$

b. $x^2 + 14x + 49$
 $(x + 7)(x + 7)$

c. $x^2 - 17x + 72$
 $(x - 8)(x - 9)$

d. $x^2 + 9x + 20$
 $(x + 4)(x + 5)$

6. Rewrite each expression in factored or standard form.

Factored Form	Standard Form
$x(x + 4)$	$x^2 + 4x$
$x(x - 7)$	$x^2 - 7x$
$(x + 11)(x + 2)$	$x^2 + 13x + 22$
$(x - 5)(x - 5)$	$x^2 - 10x + 25$
$(x - 9)(x - 3)$	$x^2 - 12x + 27$

7. Here are two expressions: $(x + n)(x + n)$ and $x^2 + nx + n^2$. Tyler claims the expressions are equivalent. Shawn claims the expressions are not equivalent. Who is correct? Explain your thinking.

Shawn. *Explanations vary. When I use the distributive property, the expression $(x + n)(x + n)$ is equivalent to $x^2 + nx + nx + n^2$ or $x^2 + 2nx + n^2$.*

Unit 8 Lesson 5

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.SSE.A.2
2	2	HSA.SSE.A.2
3	2	HSA.SSE.A.2
4	2	HSA.SSE.A.2
5	2	HSA.SSE.A.2
6	2	HSA.SSE.A.2
7	3	HSA.SSE.A.2

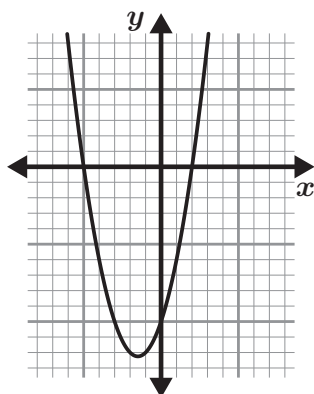
Notes:

Additional Practice

A1.8.06

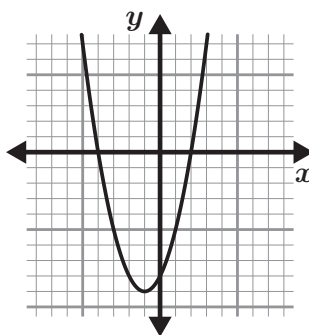
1. What are the x -intercepts of the function $f(x) = (x + 5)(x - 2)$?

- ☒ A. $(-5, 0)$ and $(2, 0)$
- ☐ B. $(5, 0)$ and $(-2, 0)$
- ☐ C. $(0, 5)$ and $(0, 2)$
- ☐ D. $(2, 5)$ and $(0, 0)$



Problems 2–3: What are the x -intercepts of the function?

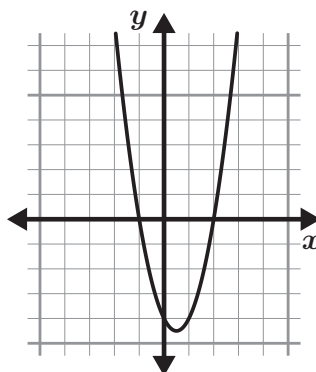
2. $f(x) = x^2 + 2x - 8$



x -intercept 1: $(-4, 0)$

x -intercept 2: $(2, 0)$

3. $g(x) = 2x^2 - 2x - 4$



x -intercept 1: $(-1, 0)$

x -intercept 2: $(2, 0)$

4. Select *all* the functions that have $(3, 0)$ and $(-2, 0)$ as their x -intercepts.

☐ A. $f(x) = (x + 3)(x - 2)$

☒ B. $g(x) = (x - 3)(x + 2)$

☒ C. $h(x) = x^2 - x - 6$

☐ D. $j(x) = 2x^2 - x - 6$

☐ E. $k(x) = (2x + 3)(2x - 2)$

5. How many x -intercepts does the function $g(x) = x^2 + 6x + 9$ have?

A. 0

☒ B. 1

C. 2

D. There is not enough information.

Explain your thinking.

When $x = -3$, the factor $x + 3 = 0$, so $f(-3) = 0$. There is no other x -value that makes $f(x) = 0$ true, so there is only one zero at $x = -3$.

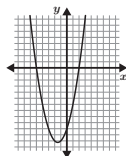
Name: _____ Date: _____ Period: _____

Additional Practice

A1.8.06

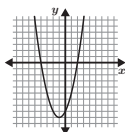
1. What are the x -intercepts of the function $f(x) = (x + 5)(x - 2)$?

- ☒ A. $(-5, 0)$ and $(2, 0)$
- ☐ B. $(5, 0)$ and $(-2, 0)$
- ☐ C. $(0, 5)$ and $(0, 2)$
- ☐ D. $(2, 5)$ and $(0, 0)$



Problems 2–3: What are the x -intercepts of the function?

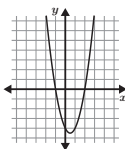
2. $f(x) = x^2 + 2x - 8$



x -intercept 1: $(-4, 0)$

x -intercept 2: $(2, 0)$

3. $g(x) = 2x^2 - 2x - 4$



x -intercept 1: $(-1, 0)$

x -intercept 2: $(2, 0)$

Unit 8 Lesson 6

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Name: _____ Date: _____ Period: _____

4. Select *all* the functions that have $(3, 0)$ and $(-2, 0)$ as their x -intercepts.

- ☐ A. $f(x) = (x + 3)(x - 2)$
- ☒ B. $g(x) = (x - 3)(x + 2)$
- ☒ C. $h(x) = x^2 - x - 6$
- ☐ D. $j(x) = 2x^2 - x - 6$
- ☐ E. $k(x) = (2x + 3)(2x - 2)$

5. How many x -intercepts does the function $g(x) = x^2 + 6x + 9$ have?

- ☐ A. 0
- ☒ B. 1
- ☐ C. 2
- ☐ D. There is not enough information.

Explain your thinking.

When $x = -3$, the factor $x + 3 = 0$, so $f(-3) = 0$. There is no other x -value that makes $f(x) = 0$ true, so there is only one zero at $x = -3$.

Unit 8 Lesson 6

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSA.SSE.B.3.A
2	2	HSA.SSE.B.3.A
3	2	HSA.SSE.B.3.A
4	2	HSA.SSE.B.3.A
5	2	HSA.SSE.B.3.A

Notes:

Additional Practice

A1.8.07

1. Bard is solving the equation $x^2 - 8x = 20$. Bard wants to rewrite the equation so that one side is equal to 0. What should be Bard's first step?

- A. Factor $x^2 - 8x$.
- B. Divide each side by x .
- C. Add 20 to both sides of the equation.
- D. Subtract 20 from both sides of the equation.**

2. Rewrite each standard form quadratic equation in factored form.

a $x^2 + 2x - 8$
 $(x - 2)(x + 4)$

b $x^2 - 2x - 15$
 $(x + 3)(x - 5)$

3. Solve each equation.

a $n(n + 9) = 0$
 $n = 0$ and $n = -9$

b $(6 + y)(6 - y) = 0$
 $y = -6$ and $y = 6$

c $(3x - 1)(x + 11) = 0$
 $x = \frac{1}{3}$ and $x = -11$

d $(4p + 8)(4p + 8) = 0$
 $p = -2$

e $(b + 7)(10 - b) = 0$
 $b = -7$ and $b = 10$

4. Solve each equation.

a $x^2 - 11x + 10 = 0$
 $x = 1$ and $x = 10$

b $t^2 + 24t + 144 = 0$
 $t = -12$

c $c^2 - c - 72 = 0$
 $c = -8$ and $c = 9$

5. $x = 3$ is one of the solutions to the equation $x^2 - x - 6 = 0$. What is the other solution?
 $x = -2$

6. Write two equations with $x = -7$ and $x = \frac{1}{2}$ as solutions. *Responses vary.*

Equation 1: $(x + 7)\left(x - \frac{1}{2}\right) = 0$

Equation 2: $(x + 7)(2x - 1) = 0$

7. Lin is solving the quadratic equation $x^2 + 8x - 48 = 0$. Her work is shown. Is Lin's work correct or incorrect? If it is incorrect, explain the error and correct it. If it is correct, check Lin's solutions by substituting them into the original equation and showing that the equation is true.

Lin
$x^2 + 8x - 48 = 0$
$(x - 12)(x + 4) = 0$
$x - 12 = 0$ or $x + 4 = 0$
$x = 12$ or $x = -4$

Lin's work is incorrect. *Explanations vary.* Lin incorrectly rewrote the expression in factored form. It should be $(x + 12)(x - 4) = 0$ so that the linear terms have a sum of $8x$. This means $x = -12$ and $x = 4$ are the correct solutions.

8. A square pool has a walkway surrounding it. The total area of the pool and walkway is given by the equation $y = (x + 4)(x + 6)$, where y represents the area in square feet and x represents the side length of the pool in feet. The total area is 360 square feet.

- a Write an equation to represent the total area of the pool and walkway.

$(x + 4)(x + 6) = 360$ (or equivalent)

- b What is the side length of the pool? Show or explain your thinking.

14 feet. Explanations vary.

$(x + 4)(x + 6) = 360$

$x^2 + 10x + 24 - 360 = 0$

$x^2 + 10x - 336 = 0$

$(x - 14)(x + 24) = 0$

$x = 14$ or $x = -24$

Because a side length cannot be negative, the length of the pool must be the positive solution, 14 feet.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.8.07

- Bard is solving the equation $x^2 - 8x = 20$. Bard wants to rewrite the equation so that one side is equal to 0. What should be Bard's first step?
 - Factor $x^2 - 8x$.
 - Divide each side by x .
 - Add 20 to both sides of the equation.
 - Subtract 20 from both sides of the equation.
- Rewrite each standard form quadratic equation in factored form.
 - $x^2 + 2x - 8$
 $(x - 2)(x + 4)$
 - $x^2 - 2x - 15$
 $(x + 3)(x - 5)$
- Solve each equation.
 - $n(n + 9) = 0$
 $n = 0$ and $n = -9$
 - $(6 + y)(6 - y) = 0$
 $y = -6$ and $y = 6$
 - $(3x - 1)(x + 11) = 0$
 $x = \frac{1}{3}$ and $x = -11$
 - $(4p + 8)(4p + 8) = 0$
 $p = -2$
 - $(b + 7)(10 - b) = 0$
 $b = -7$ and $b = 10$
- Solve each equation.
 - $x^2 - 11x + 10 = 0$
 $x = 1$ and $x = 10$
 - $t^2 + 24t + 144 = 0$
 $t = -12$
 - $c^2 - c - 72 = 0$
 $c = -8$ and $c = 9$
- $x = 3$ is one of the solutions to the equation $x^2 - x - 6 = 0$. What is the other solution?
 $x = -2$

Unit 8 Lesson 7

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- Write two equations with $x = -7$ and $x = \frac{1}{2}$ as solutions. *Responses vary.*

Equation 1: $(x + 7)(x - \frac{1}{2}) = 0$

Equation 2: $(x + 7)(2x - 1) = 0$

- Lin is solving the quadratic equation $x^2 + 8x - 48 = 0$. Her work is shown. Is Lin's work correct or incorrect? If it is incorrect, explain the error and correct it. If it is correct, check Lin's solutions by substituting them into the original equation and showing that the equation is true.

Lin

$$x^2 + 8x - 48 = 0$$

$$(x - 12)(x + 4) = 0$$

$$x - 12 = 0 \text{ or } x + 4 = 0$$

$$x = 12 \text{ or } x = -4$$

Lin's work is incorrect. *Explanations vary.* Lin incorrectly rewrote the expression in factored form. It should be $(x + 12)(x - 4) = 0$ so that the linear terms have a sum of $8x$. This means $x = -12$ and $x = 4$ are the correct solutions.

- A square pool has a walkway surrounding it. The total area of the pool and walkway is given by the equation $y = (x + 4)(x + 6)$, where y represents the area in square feet and x represents the side length of the pool in feet. The total area is 360 square feet.

- Write an equation to represent the total area of the pool and walkway.

$$(x + 4)(x + 6) = 360 \text{ (or equivalent)}$$

- What is the side length of the pool? Show or explain your thinking.

14 feet. *Explanations vary.*

$$(x + 4)(x + 6) = 360$$

$$x^2 + 10x + 24 - 360 = 0$$

$$x^2 + 10x - 336 = 0$$

$$(x - 14)(x + 24) = 0$$

$$x = 14 \text{ or } x = -24$$

Because a side length cannot be negative, the length of the pool must be the positive solution, 14 feet.

Unit 8 Lesson 7

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.B.4.B
2	1	HSA.SSE.A.2
3	2	HSA.REI.B.4.B
4	2	HSA.REI.B.4.B
5	2	HSA.REI.B.4.B
6	2	HSA.REI.B.4.B
7	2	HSA.REI.B.4.B
8	2	HSA.REI.B.4.B

Notes:

Additional Practice

A1.8.08

1. How many solutions does the equation have? $(x + 10)^2 = -25$

- ☒ A. No Solutions
- ☐ B. One Solution
- ☐ C. Two Solutions
- ☐ D. There is not enough information.

2. For each equation, determine the number of solutions.

Equation	Number of Solutions
$x^2 + 2 = 2$	One solution
$(x - 3)^2 = 1$	Two solutions
$(x + 8)^2 = -16$	No solutions
$(x + 3)^2 = 0$	One solution
$x^2 - 8 = -4$	Two solutions

3. Which value of x is a solution to the equation $(x - 2)^2 + 27 = 27$?

- ☐ A. $x = -2$
- ☐ B. $x = 0$
- ☒ C. $x = 2$
- ☐ D. There is no solution to the equation.

Problems 4–5: Determine the solution for each equation.

4. $(x + 2)^2 = 0$

$x = -2$

5. $(x + 3)(x + 3) = 0$

$x = -3$

Problems 6–7: Determine the two solutions for each equation. Show your thinking.

6. $x^2 - 20 = -4$

$x = -4$ and $x = 4$. *Work varies.* I combined like terms and got $x^2 = 16$. Then, I took the square root of both sides to get $x = -4$ and $x = 4$.

7. $100 + (x - 3)^2 = 136$

$x = -3$ and $x = 9$. *Work varies.* I combined like terms and got $(x - 3)^2 = 36$. Then, I took the square root of both sides and got $x - 3 = \pm 6$. I solved $x - 3 = -6$ and $x - 3 = 6$ to get $x = -3$ and $x = 9$.

8. Which value for x is a solution to the equation $x^2 + 7x = x - 9$?

A. $x = -9$

B. $x = -3$

C. $x = 0$

D. $x = 3$

Explain your thinking.

Explanations vary. First, I brought all the terms to the left side of the equal sign and got $x^2 + 6x + 9 = 0$. Then, I factored the left side of the equation: $(x + 3)(x + 3) = 0$. This means $x = -3$.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.8.08

1. How many solutions does the equation have? $(x + 10)^2 = -25$

- ☒ A. No Solutions
☐ B. One Solution
☐ C. Two Solutions
☐ D. There is not enough information.

2. For each equation, determine the number of solutions.

Equation	Number of Solutions
$x^2 + 2 = 2$	One solution
$(x - 3)^2 = 1$	Two solutions
$(x + 8)^2 = -16$	No solutions
$(x + 3)^2 = 0$	One solution
$x^2 - 8 = -4$	Two solutions

3. Which value of x is a solution to the equation $(x - 2)^2 + 27 = 27$?

- ☐ A. $x = -2$
☐ B. $x = 0$
☒ C. $x = 2$
☐ D. There is no solution to the equation.

Problems 4–5: Determine the solution for each equation.

4. $(x + 2)^2 = 0$

$x = -2$

5. $(x + 3)(x + 3) = 0$

$x = -3$

Unit 8 Lesson 8

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Name: _____ Date: _____ Period: _____

Problems 6–7: Determine the two solutions for each equation. Show your thinking.

6. $x^2 - 20 = -4$

$x = -4$ and $x = 4$. Work varies. I combined like terms and got $x^2 = 16$. Then, I took the square root of both sides to get $x = -4$ and $x = 4$.

7. $100 + (x - 3)^2 = 136$

$x = -3$ and $x = 9$. Work varies. I combined like terms and got $(x - 3)^2 = 36$. Then, I took the square root of both sides and got $x - 3 = \pm 6$. I solved $x - 3 = -6$ and $x - 3 = 6$ to get $x = -3$ and $x = 9$.

8. Which value for x is a solution to the equation $x^2 + 7x = x - 9$?

- ☐ A. $x = -9$
☒ B. $x = -3$
☐ C. $x = 0$
☐ D. $x = 3$

Explain your thinking.

Explanations vary. First, I brought all the terms to the left side of the equal sign and got $x^2 + 6x + 9 = 0$. Then, I factored the left side of the equation: $(x + 3)(x + 3) = 0$. This means $x = -3$.

Unit 8 Lesson 8

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.B.4, HSA.REI.B.4.B
2	1	HSA.REI.B.4, HSA.REI.B.4.B
3	1	HSA.REI.B.4, HSA.REI.B.4.B
4	1	HSA.REI.B.4, HSA.REI.B.4.B
5	2	HSA.REI.B.4, HSA.REI.B.4.B
6	2	HSA.REI.B.4, HSA.REI.B.4.B
7	2	HSA.REI.B.4, HSA.REI.B.4.B
8	2	HSA.REI.B.4, HSA.REI.B.4.B

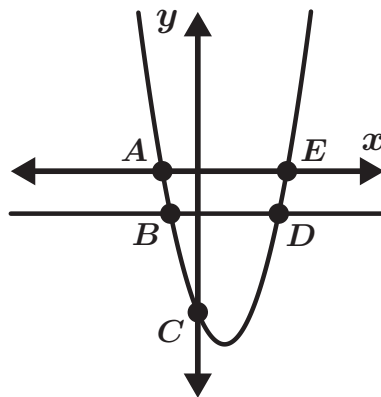
Notes:

Additional Practice

A1.8.09

1. Here is a graph of $y = x^2 - 3x - 10$ and $y = -4$.
Select *all* the points that are solutions to $x^2 - 3x - 10 = -4$.

- ☐ A. Point A
☒ B. Point B
☐ C. Point C
☒ D. Point D
☐ E. Point E



Problems 2–5: Circle how many solutions each equation has. Record any solutions.

2. $3x^2 + 6 = 18$ No solutions One solution Two solutions $x = \underline{\textcolor{red}{-2}}$
 $x = \underline{\textcolor{red}{2}}$

3. $x(x - 5) = -6$ No solutions One solution Two solutions $x = \underline{\textcolor{red}{2}}$
 $x = \underline{\textcolor{red}{3}}$

4. $0 = -(x - 3)(x + 2) - 10$ No solutions One solution Two solutions $x = \underline{\hspace{1cm}}$
 $x = \underline{\hspace{1cm}}$

5. $x(x + 4) = -16$ No solutions One solution Two solutions $x = \underline{\hspace{1cm}}$
 $x = \underline{\hspace{1cm}}$

Problems 6–7: Fill in the blank so the equation has:

6. One solution

$x(x - 2) = \underline{\textcolor{red}{-1}}$

7. No solutions *Responses vary.*

$(x + 3)(x - \underline{\textcolor{red}{1}}) = -7$

Problems 8–9: The graphs of $y = x^2 - 16$, $y = -15$, $y = -12$, and $y = -7$ all intersect at integer values.

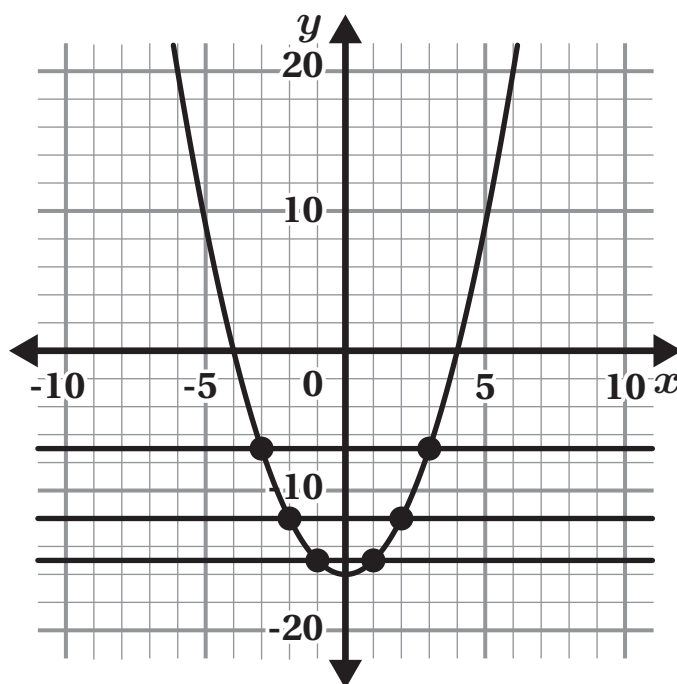
8. Write the equations of three more lines that follow this property.

Responses vary.

$$y = \underline{\quad 0 \quad}$$

$$y = \underline{\quad 9 \quad}$$

$$y = \underline{\quad 20 \quad}$$

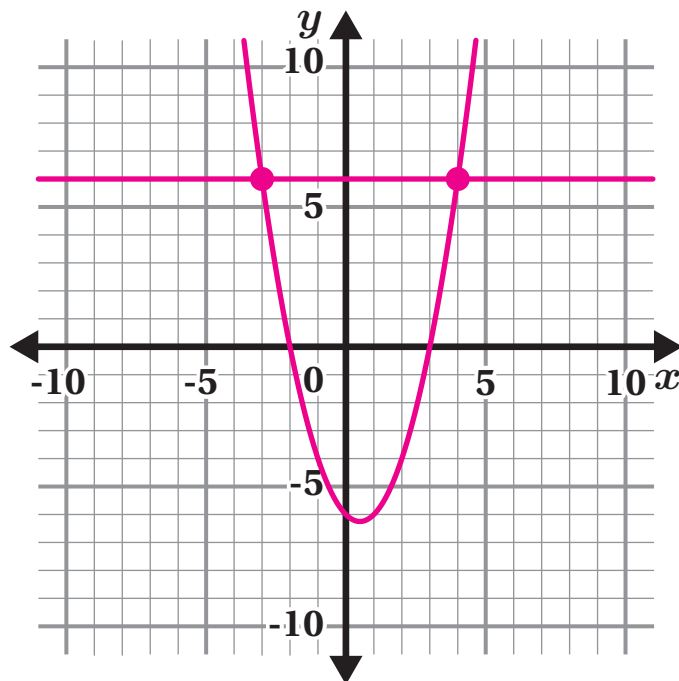


9. What pattern do you notice?

Responses vary. If you add each y -value to 16, it equals a perfect square.

10. Solve $(x - 3)(x + 2) = 6$. Show or explain your thinking. Use the graph if it helps with your thinking.

$x = -3$ and $x = 4$. *Explanations vary.* I wrote the equation as two different functions, $y = (x - 3)(x + 2)$ and $y = 6$, and graphed them. The x -values of the intersection points are -3 and 4 .



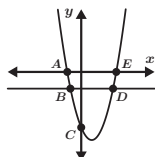
Name: _____ Date: _____ Period: _____

Additional Practice

A1.8.09

1. Here is a graph of $y = x^2 - 3x - 10$ and $y = -4$.
Select all the points that are solutions to $x^2 - 3x - 10 = -4$.

- ☐ A. Point A
☒ B. Point B
☐ C. Point C
☒ D. Point D
☐ E. Point E



Problems 2–5: Circle how many solutions each equation has. Record any solutions.

2. $3x^2 + 6 = 18$ No solutions One solution **Two solutions** $x = -2$
 $x = 2$
3. $x(x - 5) = -6$ No solutions One solution **Two solutions** $x = 2$
 $x = 3$
4. $0 = -(x - 3)(x + 2) - 10$ **No solutions** One solution Two solutions $x =$ _____
 $x =$ _____
5. $x(x + 4) = -16$ **No solutions** One solution Two solutions $x =$ _____
 $x =$ _____

Problems 6–7: Fill in the blank so the equation has:

6. One solution $x(x - 2) = \underline{-1}$
7. No solutions **Responses vary.** $(x + 3)(x - \underline{1}) = -7$

Unit 8 Lesson 9

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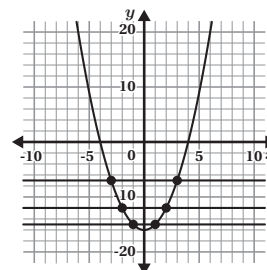
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Name: _____ Date: _____ Period: _____

Problems 8–9: The graphs of $y = x^2 - 16$, $y = -15$, $y = -12$, and $y = -7$ all intersect at integer values.

8. Write the equations of three more lines that follow this property.

Responses vary.
 $y = 0$
 $y = 9$
 $y = 20$

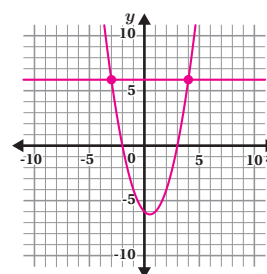


9. What pattern do you notice?

Responses vary. If you add each y-value to 16, it equals a perfect square.

10. Solve $(x - 3)(x + 2) = 6$. Show or explain your thinking. Use the graph if it helps with your thinking.

$x = -3$ and $x = 4$. **Explanations vary.** I wrote the equation as two different functions, $y = (x - 3)(x + 2)$ and $y = 6$, and graphed them. The x-values of the intersection points are -3 and 4 .



Unit 8 Lesson 9

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Additional Practice

Practice Problem Analysis

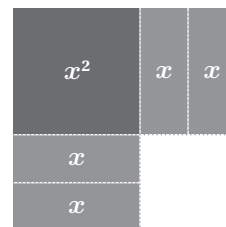
Problem	DOK	Standard(s)
1	1	HSA.REI.D.10
2	2	HSA.REI.B.4
3	2	HSA.REI.B.4
4	2	HSA.REI.B.4
5	2	HSA.REI.B.4
6	2	HSA.REI.B.4
7	2	HSA.REI.D.10
8	3	HSA.REI.D.10
9	3	HSA.REI.D.10
10	2	HSA.REI.D.10

Notes:

Additional Practice

A1.8.11

1. The algebra tiles in the figure represent the expression $x^2 + 4x$. How many 1-tiles are needed to make the figure a square?



1-tile = 

- A. 1 tile
B. 2 tiles
C. 4 tiles
D. 8 tiles

2. Which expression is a perfect square expression?

- A. $x(x - 4)$
B. $(x + 6)^2$
C. $x - 5^2$
D. $x^2 + 1$

3. Here is an incomplete quadratic expression: $x^2 - 14x + \dots$.

- a What value can be added to make the expression a perfect square expression?

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- b Write the perfect square expression in factored form.

$(x - 7)^2$ or $(x - 7)(x - 7)$

4. For each expression, determine the value that, when added to the expression, makes it a perfect square expression. Then write it in standard form and factored form.

Incomplete Expression	Perfect Square Expression in Standard Form	Perfect Square Expression in Factored Form
$x^2 + 12x + \dots$	$x^2 + 12x + 36$	$(x + 6)^2$
$x^2 - 18x + \dots$	$x^2 - 18x + 81$	$(x - 9)^2$
$x^2 + \dots + 121$	$x^2 + 22x + 121$	$(x + 11)^2$
$x^2 - 5x + \dots$	$x^2 - 5x + \frac{25}{4}$ (or equivalent)	$\left(x - \frac{5}{2}\right)^2$ (or equivalent)

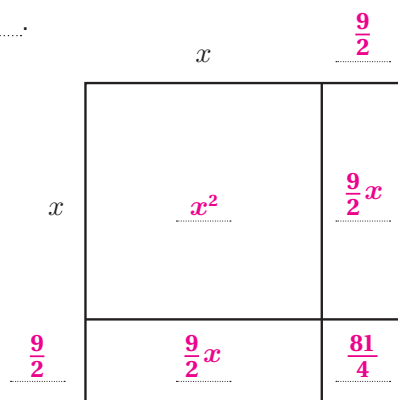
5. Here is an incomplete quadratic expression: $x^2 + 9x + \dots$.

- a Complete the area diagram to turn the expression into a perfect square expression. What is the value of the missing term?

$$\left(\frac{9}{2}\right)^2 \text{ or } \frac{81}{4} \text{ (or equivalent)}$$

- b Write the perfect square expression in factored form.

$$\left(x + \frac{9}{2}\right)^2 \text{ or } \left(x + \frac{9}{2}\right)\left(x + \frac{9}{2}\right) \text{ (or equivalent)}$$



6. Each of the following expressions written in standard form is an incomplete perfect square expression. Determine the missing value. Then match each expression in standard form with an equivalent expression in factored form.

Standard Form

Factored Form

a $x^2 + 14x + \dots$

c $(x - 3)^2$

b $x^2 + \dots x + 4$

a $(x + 7)^2$

c $x^2 - 6x + \dots$

b $(x + 2)^2$

d $x^2 - \dots x + 16$

d $(x - 4)^2$

7. Lin is changing the expression $x^2 + 16x$ so that it will be a perfect square expression. Her work is shown. Complete Lin's missing step.

Lin

$$x^2 + 16x = \dots$$

$$= (x + 8)^2$$

8. Here is a quadratic expression: $x^2 - 30x$. Elena claims that 900 can be added to make it a perfect square expression. Shawn argues that the number 225 should be added instead. Who is correct? Explain your thinking.

Shawn. Explanations vary. To make the expression a perfect square expression, the number $\left(-\frac{30}{2}\right)^2 = (-15)^2$ or 225 must be added. The perfect square expression can be written as $x^2 - 30x + 225$ or $(x - 15)^2$.

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Additional Practice

A1.8.11

1. The algebra tiles in the figure represent the expression $x^2 + 4x$. How many 1-tiles are needed to make the figure a square?

- A. 1 tile
B. 2 tiles
C. 4 tiles
D. 8 tiles



1-tile = 1

2. Which expression is a perfect square expression?

- A. $x(x-4)$
B. $(x+6)^2$
C. $x-5^2$
D. x^2+1

3. Here is an incomplete quadratic expression: $x^2 - 14x + \underline{\hspace{1cm}}$.

- a. What value can be added to make the expression a perfect square expression?
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- b. Write the perfect square expression in factored form.
 $(x-7)^2$ or $(x-7)(x-7)$

4. For each expression, determine the value that, when added to the expression, makes it a perfect square expression. Then write it in standard form and factored form.

Incomplete Expression	Perfect Square Expression in Standard Form	Perfect Square Expression in Factored Form
$x^2 + 12x + \underline{\hspace{1cm}}$	$x^2 + 12x + 36$	$(x+6)^2$
$x^2 - 18x + \underline{\hspace{1cm}}$	$x^2 - 18x + 81$	$(x-9)^2$
$x^2 + \underline{\hspace{1cm}} + 121$	$x^2 + 22x + 121$	$(x+11)^2$
$x^2 - 5x + \underline{\hspace{1cm}}$	$x^2 - 5x + \frac{25}{4}$ (or equivalent)	$\left(x - \frac{5}{2}\right)^2$ (or equivalent)

Unit 8 Lesson 11

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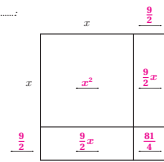
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Name: _____ Date: _____ Period: _____

5. Here is an incomplete quadratic expression: $x^2 + 9x + \underline{\hspace{1cm}}$.

- a. Complete the area diagram to turn the expression into a perfect square expression. What is the value of the missing term?

$\left(\frac{9}{2}\right)^2$ or $\frac{81}{4}$ (or equivalent)



- b. Write the perfect square expression in factored form.

$\left(x + \frac{9}{2}\right)^2$ or $\left(x + \frac{9}{2}\right)\left(x + \frac{9}{2}\right)$ (or equivalent)

6. Each of the following expressions written in standard form is an incomplete perfect square expression. Determine the missing value. Then match each expression in standard form with an equivalent expression in factored form.

Standard Form

Factored Form

- a. $x^2 + 14x + \underline{49}$ c. $(x-3)^2$
b. $x^2 + \underline{4}x + 4$ a. $(x+7)^2$
c. $x^2 - 6x + \underline{9}$ b. $(x+2)^2$
d. $x^2 - \underline{8}x + 16$ d. $(x-4)^2$

7. Lin is changing the expression $x^2 + 16x$ so that it will be a perfect square expression. Her work is shown. Complete Lin's missing step.

Lin
 $x^2 + 16x = \underline{\hspace{1cm}}$
 $= (x+8)^2$

8. Here is a quadratic expression: $x^2 - 30x$. Elena claims that 900 can be added to make it a perfect square expression. Shawn argues that the number 225 should be added instead. Who is correct? Explain your thinking.

Shawn. *Explanations vary. To make the expression a perfect square expression, the number $\left(-\frac{30}{2}\right)^2 = (-15)^2$ or 225 must be added. The perfect square expression can be written as $x^2 - 30x + 225$ or $(x-15)^2$.*

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSA.SSE.A.2
2	2	HSA.SSE.A.2
3	2	HSA.SSE.A.2
4	2	HSA.SSE.A.2
5	2	HSA.SSE.A.2
6	2	HSA.SSE.A.2
7	2	HSA.SSE.A.2
8	2	HSA.SSE.A.2

Notes:

Additional Practice

A1.8.12

1. What number can you add to make each expression a perfect square?

a $x^2 + 10x + 14$

11

b $x^2 - 8x - 3$

19

2. Add the number that would make the expression a perfect square. Then write an equivalent expression in factored form.

a $x^2 - 4x + \boxed{4}$

$(x - 2)^2$

b $x^2 + 10x + \boxed{25}$

$(x + 5)^2$

c $x^2 + 16x + \boxed{64}$

$(x + 8)^2$

3. Jada wants to solve the quadratic equation $x^2 - 10x + 21 = 0$ by completing the square. Her work is shown. She says that there is one solution, $x = 5$. Is Jada correct? Explain your thinking.

No. Explanations vary. Jada made a mistake in solving the equation. She did not add 4 to both sides of the equation. The equation she wrote, $x^2 - 10x + 25 = 0$, is not equivalent to the original equation. The correct solutions are $x = 3$ and $x = 7$.

Jada

$$x^2 - 10x + 21 = 0$$

$$x^2 - 10x + 21 + 4 = 0$$

$$x^2 - 10x + 25 = 0$$

$$(x - 5)^2 = 0$$

4. Solve each equation by completing the square.

a $x^2 - 4x - 3 = 2$

$x = -1$ and $x = 5$

b $x^2 + 8x + 3 = -9$

$x = -6$ and $x = -2$

c $x^2 - 14x + 54 = 5$

$x = 7$

d $7 = x^2 + 6x - 20$

$x = -9$ and $x = 3$

5. Which of the following could be the first step in solving the quadratic equation $x^2 + 8x - 20 = 0$ by completing the square?

A. $x^2 + 8x - 16 = 20$

☒ B. $x^2 + 8x - 20 + 36 = 0 + 36$

C. $x^2 + 8x - 20 = -20$

D. $x^2 + 8x - 20 + 16 = 0$

6. Priya wants to solve the quadratic equation $x^2 + 12x + 11 = 0$ by completing the square. Her work is shown. She says that the solutions are $x = 19$ and $x = -31$. Which of the following statements is true?

A. Priya's work and solutions are correct.

B. Priya made an error in the second line by adding 25 to each side of the equation.

C. Priya made an error in the third line by incorrectly writing the expression in factored form.

☒ D. Priya made an error in the fourth line by not taking the square root of 25.**Priya**

$$x^2 + 12x + 11 = 0$$

$$x^2 + 12x + 36 = 25$$

$$(x + 6)^2 = 25$$

$$x + 6 = 25 \text{ and } x + 6 = -25$$

$$x = 19 \text{ and } x = -31$$

7. Here is the quadratic equation $x^2 + 16x + 38 = 23$. Bard solves the equation by completing the square and says there is only one solution, $x = -1$. Is Bard correct? Explain your thinking.

No. Explanations vary. Bard is missing the second solution. When solving the equation $x^2 + 16x + 38 = 23$ by completing the square, the equation $(x + 8)^2 = 49$ results. Because $x + 8 = 7$ and $x + 8 = -7$, the equation has two solutions, $x = -1$ and $x = -15$.

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Additional Practice

A1.8.12

1. What number can you add to make each expression a perfect square?

a. $x^2 + 10x + 14$ b. $x^2 - 8x - 3$
11 **19**

2. Add the number that would make the expression a perfect square. Then write an equivalent expression in factored form.

a. $x^2 - 4x + \boxed{4}$ b. $x^2 + 10x + \boxed{25}$ c. $x^2 + 16x + \boxed{64}$
 $(x - 2)^2$ $(x + 5)^2$ $(x + 8)^2$

3. Jada wants to solve the quadratic equation $x^2 - 10x + 21 = 0$ by completing the square. Her work is shown. She says that there is one solution, $x = 5$. Is Jada correct? Explain your thinking.

No. Explanations vary. Jada made a mistake in solving the equation. She did not add 4 to both sides of the equation. The equation she wrote, $x^2 - 10x + 25 = 0$, is not equivalent to the original equation. The correct solutions are $x = 3$ and $x = 7$.

Jada

$$\begin{aligned} x^2 - 10x + 21 &= 0 \\ x^2 - 10x + 21 + 4 &= 0 \\ x^2 - 10x + 25 &= 0 \\ (x - 5)^2 &= 0 \end{aligned}$$

4. Solve each equation by completing the square.

a. $x^2 - 4x - 3 = 2$ b. $x^2 + 8x + 3 = -9$
 $x = -1$ and $x = 5$ $x = -6$ and $x = -2$
c. $x^2 - 14x + 54 = 5$ d. $7 = x^2 + 6x - 20$
 $x = 7$ $x = -9$ and $x = 3$

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5. Which of the following could be the first step in solving the quadratic equation $x^2 + 8x - 20 = 0$ by completing the square?

A. $x^2 + 8x - 16 = 20$ **B. $x^2 + 8x - 20 + 36 = 0 + 36$**
C. $x^2 + 8x - 20 = -20$ D. $x^2 + 8x - 20 + 16 = 0$

6. Priya wants to solve the quadratic equation $x^2 + 12x + 11 = 0$ by completing the square. Her work is shown. She says that the solutions are $x = 19$ and $x = -31$. Which of the following statements is true?

Priya

$$\begin{aligned} x^2 + 12x + 11 &= 0 \\ x^2 + 12x + 36 &= 25 \\ (x + 6)^2 &= 25 \\ x + 6 &= 25 \text{ and } x + 6 = -25 \\ x &= 19 \text{ and } x = -31 \end{aligned}$$

- A. Priya's work and solutions are correct.
B. Priya made an error in the second line by adding 25 to each side of the equation.
C. Priya made an error in the third line by incorrectly writing the expression in factored form.
D. Priya made an error in the fourth line by not taking the square root of 25.

7. Here is the quadratic equation $x^2 + 16x + 38 = 23$. Bard solves the equation by completing the square and says there is only one solution, $x = -1$. Is Bard correct? Explain your thinking.

No. Explanations vary. Bard is missing the second solution. When solving the equation $x^2 + 16x + 38 = 23$ by completing the square, the equation $(x + 8)^2 = 49$ results. Because $x + 8 = 7$ and $x + 8 = -7$, the equation has two solutions, $x = -1$ and $x = -15$.

Unit 8 Lesson 12

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.SSE.A.2
2	1	HSA.SSE.A.2
3	2	HSA.REI.B.4, HSA.REI.B.4.A
4	2	HSA.REI.B.4, HSA.REI.B.4.A
5	2	HSA.REI.B.4, HSA.REI.B.4.A
6	2	HSA.REI.B.4, HSA.REI.B.4.A
7	2	HSA.REI.B.4.A

Notes:

Additional Practice

A1.8.14

1. Which of the following is the solution to any quadratic equation $ax^2 + bx + c = 0$?

- A. $x = \frac{b \pm \sqrt{b - 4ac}}{2a}$ B. $x = \frac{-b \pm \sqrt{b^2 - ac}}{2a}$ C. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{a}$
 D. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ E. $y = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$

Problems 2–5: Identify the values for a , b , and c in each of the quadratic equations below.

	Equation	a -term	b -term	c -term
2.	$-2x^2 - 7x + 5 = 0$	-2	-7	5
3.	$8x^2 = 2x + 4$	8	-2	-4
4.	$4x^2 = 12$	4	0	-12
5.	$2x(x - 5) = 0$	2	-10	0

6. Use the quadratic formula to show the equation $x^2 + 6x + 10 = 0$ has no solutions. Show or explain how you know.

$$\begin{aligned}
 x^2 + 6x + 10 &= 0 \\
 a = 1 \quad b = 6 \quad c &= 10 \\
 x &= \frac{-6 \pm \sqrt{6^2 - 4(1)(10)}}{2(1)} \\
 x &= \frac{-6 \pm \sqrt{36 - 40}}{2} \\
 x &= \frac{-6 \pm \sqrt{-4}}{2}
 \end{aligned}$$

There are no solutions for x because the $\sqrt{-4}$ is not a real number.

Problems 7–10: Here is the equation $x^2 - 8x = -7$.

7. Write the equation in standard form.

$$x^2 - 8x + 7 = 0$$

8. Identify the values for a , b , and c .

$$a = \underline{1} \quad b = \underline{-8} \quad c = \underline{7}$$

9. Substitute the values for a , b , and c into the quadratic formula. (You do not need to perform any operations.)

$$x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(7)}}{2(1)}$$

10. Show or explain how the expression you wrote is related to solving $x^2 - 8x = -7$ by completing the square.

Explanations vary. The expression takes all the steps for solving by completing the square and bundles them up into one expression. Instead of having to evaluate every step, you can do the calculation all at once.

$$x^2 - 8x + 7 = 0$$

$$x^2 - 8x + 16 - 16 + 7 = 0$$

$$(x - 4)^2 - 9 = 0$$

$$(x - 4)^2 = 9$$

$$\sqrt{(x - 4)^2} = \sqrt{9}$$

$$x - 4 = \pm \sqrt{9}$$

$$x = 4 \pm \sqrt{9} \quad \leftarrow \text{This is a simpler form of the values in the quadratic formula.}$$

$$x = 4 \pm 3$$

$$x = 7 \text{ and } x = 1$$

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Additional Practice

A1.8.14

1. Which of the following is the solution to any quadratic equation $ax^2 + bx + c = 0$?

A. $x = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$ B. $x = \frac{-b \pm \sqrt{b^2 - ac}}{2a}$ C. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{a}$
 D. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ E. $y = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$

- Problems 2–5: Identify the values for a , b , and c in each of the quadratic equations below.

	Equation	a -term	b -term	c -term
2.	$-2x^2 - 7x + 5 = 0$	-2	-7	5
3.	$8x^2 = 2x + 4$	8	-2	-4
4.	$4x^2 = 12$	4	0	-12
5.	$2x(x - 5) = 0$	2	-10	0

6. Use the quadratic formula to show the equation $x^2 + 6x + 10 = 0$ has no solutions. Show or explain how you know.

$$\begin{aligned} x^2 + 6x + 10 &= 0 \\ a &= 1 \quad b = 6 \quad c = 10 \\ x &= \frac{-6 \pm \sqrt{6^2 - 4(1)(10)}}{2(1)} \\ x &= \frac{-6 \pm \sqrt{36 - 40}}{2} \\ x &= \frac{-6 \pm \sqrt{-4}}{2} \end{aligned}$$

There are no solutions for x because the $\sqrt{-4}$ is not a real number.

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- Problems 7–10: Here is the equation $x^2 - 8x = -7$.

7. Write the equation in standard form.

$$x^2 - 8x + 7 = 0$$

8. Identify the values for a , b , and c .

$$a = 1, \quad b = -8, \quad c = 7$$

9. Substitute the values for a , b , and c into the quadratic formula. (You do not need to perform any operations.)

$$x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(7)}}{2(1)}$$

10. Show or explain how the expression you wrote is related to solving $x^2 - 8x = -7$ by completing the square.

Explanations vary. The expression takes all the steps for solving by completing the square and bundles them up into one expression. Instead of having to evaluate every step, you can do the calculation all at once.

$$\begin{aligned} x^2 - 8x + 7 &= 0 \\ x^2 - 8x + 16 - 16 + 7 &= 0 \\ (x - 4)^2 - 9 &= 0 \\ (x - 4)^2 &= 9 \\ \sqrt{(x - 4)^2} &= \sqrt{9} \\ x - 4 &= \pm \sqrt{9} \\ x &= 4 \pm \sqrt{9} \quad \leftarrow \text{This is a simpler form of the values in the quadratic formula.} \\ x &= 4 \pm 3 \\ x &= 7 \text{ and } x = 1 \end{aligned}$$

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Unit 8 Lesson 14

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.B.4.A
2	1	HSA.REI.B.4.B
3	1	HSA.REI.B.4.B
4	1	HSA.REI.B.4.B
5	1	HSA.REI.B.4.B
6	2	HSA.REI.B.4.B
7	1	HSA.REI.B.4.B
8	1	HSA.REI.B.4.B
9	1	HSA.REI.B.4.B
10	2	HSA.REI.B.4.A

Notes:

Additional Practice

A1.8.15

1. Each equation is in the form $ax^2 + bx + c = 0$. Determine the values of a , b , and c .

a $2x^2 + 5x + 1 = 0$

$a = 2, b = 5, c = 1$

b $-4x^2 + 3x + 6 = 0$

$a = -4, b = 3, c = 6$

c $x^2 - 7x + 4 = 0$

$a = 1, b = -7, c = 4$

d $-x^2 + 10x - 8 = 0$

$a = -1, b = 10, c = -8$

e $24 + 2x - x^2 = 0$

$a = -1, b = 2, c = 24$

f $\frac{3}{4}x^2 - 8x = \frac{1}{4}$

$a = \frac{3}{4}$ (or equivalent), $b = -8, c = -\frac{1}{4}$
(or equivalent)

g $x^2 = 100$

$a = 1, b = 0, c = -100$

h $5x - 2x^2 = -19$

$a = -2, b = 5, c = 19$

2. What are the values of x if $x = \frac{7 \pm \sqrt{49 - (-72)}}{4}$?

A. 0 and $-\frac{9}{2}$

B. 0 and 1

C. -1 and $\frac{9}{2}$

D. $-\frac{9}{2}$ and 1

3. Determine whether the solutions for each quadratic equation are correct.

a Equation: $x^2 - x - 56 = 0$

Solutions: $x = -7$ and $x = 8$

Yes

b Equation: $x^2 + 4x - 45 = 0$

Solutions: $x = -5$ and $x = 9$

No

c Equation: $2x^2 - 8x + 3 = 0$

Solutions: $x = \frac{4 + \sqrt{10}}{2}$ and $x = \frac{4 - \sqrt{10}}{2}$

Yes

4. Solve each quadratic equation using the quadratic formula. Show your thinking.

a $x^2 + 5x - 36 = 0$

$x = -9$ and $x = 4$. *Work varies.*

$$x = \frac{-5 \pm \sqrt{(5)^2 - 4(1)(-36)}}{2(1)}$$

$$x = \frac{-5 \pm \sqrt{169}}{2}$$

$$x = \frac{-5 + 13}{2} \text{ and } x = \frac{-5 - 13}{2}$$

$$x = \frac{8}{2} = 4 \text{ and } x = \frac{-18}{2} = -9$$

b $2x^2 + 9x - 35 = 0$

$x = \frac{5}{2}$ and $x = -7$. *Work varies.*

$$x = \frac{-9 \pm \sqrt{(9)^2 - 4(2)(-35)}}{2(2)}$$

$$x = \frac{-9 \pm \sqrt{361}}{4}$$

$$x = \frac{-9 + 19}{4} \text{ and } x = \frac{-9 - 19}{4}$$

$$x = \frac{10}{4} = \frac{5}{2} \text{ and } x = \frac{-28}{4} = -7$$

c $2x^2 - 16x + 30 = 0$

$x = 3$ and $x = 5$. *Work varies.*

$$x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4(2)(30)}}{2(2)}$$

$$x = \frac{16 \pm \sqrt{16}}{4}$$

$$x = \frac{16 + 4}{4} \text{ and } x = \frac{16 - 4}{4}$$

$$x = \frac{20}{4} = 5 \text{ and } x = \frac{12}{4} = 3$$

d $x^2 + 8x + 4 = 0$

$x = -4 \pm \frac{\sqrt{48}}{2}$ (or equivalent). *Work varies.*

$$x = \frac{-8 \pm \sqrt{(8)^2 - 4(1)(4)}}{2(1)}$$

$$x = \frac{-8 \pm \sqrt{48}}{2}$$

$$x = \frac{-8}{4} \pm \frac{\sqrt{48}}{2} = -4 \pm \frac{\sqrt{48}}{2}$$

5. Solve each quadratic equation using any strategy.

a $x^2 - 2x - 63 = 0$

$x = -7$ and $x = 9$

b $2x^2 + 18x + 40 = 0$

$x = -4$ and $x = -5$

c $4x^2 - 45x + 11 = 0$

$x = \frac{1}{4}$ and $x = 11$

d $3x^2 + 15x - 2 = 0$

$x = \frac{-15 \pm \sqrt{249}}{6}$ (or equivalent)

6. Diego is solving $5x^2 - x - 8 = 0$ using the quadratic formula. His work is shown. Are his solutions correct? Explain your thinking.

No. Explanations vary. In the second line, Diego wrote -1 for $(-1)^2$ rather than 1 . The correct solutions are $x = \frac{1 + \sqrt{161}}{10}$ and $x = \frac{1 - \sqrt{161}}{10}$.

Diego

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(5)(-8)}}{2(5)}$$

$$x = \frac{1 \pm \sqrt{-1 + 160}}{10}$$

$$x = \frac{1 + \sqrt{159}}{10} \text{ and } x = \frac{1 - \sqrt{159}}{10}$$

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Additional Practice

A1.8.15

1. Each equation is in the form $ax^2 + bx + c = 0$. Determine the values of a , b , and c .

a $2x^2 + 5x + 1 = 0$

$a = 2, b = 5, c = 1$

b $-4x^2 + 3x + 6 = 0$

$a = -4, b = 3, c = 6$

c $x^2 - 7x + 4 = 0$

$a = 1, b = -7, c = 4$

d $-x^2 + 10x - 8 = 0$

$a = -1, b = 10, c = -8$

e $24 + 2x - x^2 = 0$

$a = -1, b = 2, c = 24$

f $\frac{3}{4}x^2 - 8x = \frac{1}{4}$

$a = \frac{3}{4}$ (or equivalent), $b = -8$, $c = -\frac{1}{4}$ (or equivalent)

g $x^2 = 100$

$a = 1, b = 0, c = -100$

h $5x - 2x^2 = -19$

$a = -2, b = 5, c = 19$

2. What are the values of x if $x = \frac{7 \pm \sqrt{49 - (-72)}}{4}$?

A. 0 and $-\frac{9}{2}$

B. 0 and 1

C. -1 and $\frac{9}{2}$

D. $-\frac{9}{2}$ and 1

3. Determine whether the solutions for each quadratic equation are correct.

a Equation: $x^2 - x - 56 = 0$

Solutions: $x = -7$ and $x = 8$

Yes

b Equation: $x^2 + 4x - 45 = 0$

Solutions: $x = -5$ and $x = 9$

No

c Equation: $2x^2 - 8x + 3 = 0$

Solutions: $x = \frac{4 + \sqrt{10}}{2}$ and $x = \frac{4 - \sqrt{10}}{2}$

Yes

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4. Solve each quadratic equation using the quadratic formula. Show your thinking.

a $x^2 + 5x - 36 = 0$

$x = -9$ and $x = 4$. Work varies.

$x = \frac{-5 \pm \sqrt{(5)^2 - 4(1)(-36)}}{2(1)}$

$x = \frac{-5 \pm \sqrt{169}}{2}$

$x = \frac{-5 + 13}{2}$ and $x = \frac{-5 - 13}{2}$

$x = \frac{8}{2} = 4$ and $x = \frac{-18}{2} = -9$

b $2x^2 + 9x - 35 = 0$

$x = \frac{5}{2}$ and $x = -7$. Work varies.

$x = \frac{-9 \pm \sqrt{(9)^2 - 4(2)(-35)}}{2(2)}$

$x = \frac{-9 \pm \sqrt{361}}{4}$

$x = \frac{-9 + 19}{4}$ and $x = \frac{-9 - 19}{4}$

$x = \frac{10}{4} = \frac{5}{2}$ and $x = \frac{-28}{4} = -7$

c $2x^2 - 16x + 30 = 0$

$x = 3$ and $x = 5$. Work varies.

$x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4(2)(30)}}{2(2)}$

$x = \frac{16 \pm \sqrt{16}}{4}$

$x = \frac{16 + 4}{4}$ and $x = \frac{16 - 4}{4}$

$x = \frac{20}{4} = 5$ and $x = \frac{12}{4} = 3$

d $x^2 + 8x + 4 = 0$

$x = -4 \pm \sqrt{48}$ (or equivalent). Work varies.

$x = \frac{-8 \pm \sqrt{(8)^2 - 4(1)(4)}}{2(1)}$

$x = \frac{-8 \pm \sqrt{48}}{2}$

$x = \frac{-8 \pm \sqrt{48}}{2} = -4 \pm \frac{\sqrt{48}}{2}$

5. Solve each quadratic equation using any strategy.

a $x^2 - 2x - 63 = 0$

$x = -7$ and $x = 9$

b $2x^2 + 18x + 40 = 0$

$x = -4$ and $x = -5$

c $4x^2 - 45x + 11 = 0$

$x = \frac{1}{4}$ and $x = 11$

d $3x^2 + 15x - 2 = 0$

$x = \frac{-15 \pm \sqrt{249}}{6}$ (or equivalent)

6. Diego is solving $5x^2 - x - 8 = 0$ using the quadratic formula. His work is shown. Are his solutions correct? Explain your thinking.

No. Explanations vary. In the second line, Diego wrote -1 for $(-1)^2$ rather than 1 . The correct solutions are $x = \frac{1 + \sqrt{161}}{10}$ and $x = \frac{1 - \sqrt{161}}{10}$.

Diego
 $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(5)(-8)}}{2(5)}$
 $x = \frac{1 \pm \sqrt{-1 + 160}}{10}$
 $x = \frac{1 + \sqrt{159}}{10}$ and $x = \frac{1 - \sqrt{159}}{10}$

Unit 8 Lesson 15

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.B.4.B
2	1	HSA.REI.B.4.B
3	2	HSA.REI.B.4.B
4	2	HSA.REI.B.4.B
5	2	HSA.REI.B.4.B
6	2	HSA.REI.B.4.B

Notes:

Additional Practice

A1.8.16

1. The function $g(t) = -4.9t^2 + 40t + 6$ models the height, in meters, of an apple t seconds after it has been launched into the air from a toy cannon.
 - a Write an equation to determine when the apple will hit the ground.
 $-4.9t^2 + 40t + 6 = 0$ (or equivalent)
 - b Determine when the apple hits the ground. Round to the nearest thousandth.
8.311 seconds

2. Two objects are launched into the air. The function $h(t) = 45 + 130t - 16t^2$ models the height, in feet, of Object A t seconds after it has been launched. The function $f(t) = 18 + 70t - 16t^2$ models the height, in feet, of Object B t seconds after it has been launched.
 - a After 5 seconds have passed, which object is still in the air?
Object A
 - b Write an equation for each object to determine when it will hit the ground.
Object A: $45 + 130t - 16t^2 = 0$ (or equivalent); Object B: $18 + 70t - 16t^2 = 0$ (or equivalent)
 - c Determine when each object hits the ground. Round to the nearest thousandth.
Object A: 8.458 seconds; Object B: 4.619 seconds

3. A picture is 10 inches wide, 15 inches long, and has a frame x inches thick surrounding it. The equation $(10 + 2x)(15 + 2x) = 336$ represents the total area, in square inches, of the picture in its frame.
 - a Rewrite the equation $(10 + 2x)(15 + 2x) = 336$ in standard form.
 $4x^2 + 50x - 186 = 0$
 - b Determine the thickness of the frame.
3 inches

4. A picture is 8 inches wide and 10 inches long. It has a frame x inches thick surrounding it. The equation $(8 + 2x)(10 + 2x) = 195$ represents the total area, in square inches, of the picture and the frame.
- a Rewrite the equation $(8 + 2x)(10 + 2x) = 195$ in standard form.
 $4x^2 + 36x - 115 = 0$
- b Solve your equation. What do the solutions represent?
 $x = -11.5$ and $x = 2.5$. The solutions represent the thickness of the frame. Only the positive solution makes sense in this context. So, the thickness of the frame is 2.5 inches.
5. A ball is thrown up into the air. Its height in inches is modeled by the function $h(t) = -16t^2 + 28t + 3$, where t is the time after the ball is thrown, measured in seconds. Solve the equation $-16t^2 + 28t + 3 = 0$. What do the solutions tell you about the ball?
 $t \approx -0.101$ and $t \approx 1.851$. The ball landed on the ground about 1.851 seconds after it was thrown up in the air. The negative solution does not make sense in this context.
6. A projectile is launched into the air and its height above the ground, in feet, is modeled by the function $f(t) = 6 + 32t - 16t^2$, where t is the number of seconds since the projectile was launched.
- a What are the solutions to the equation $0 = 6 + 32t - 16t^2$?
 $t \approx -0.173$ and $t \approx 2.173$
- b Do both of the solutions have meaning in this context? Explain your thinking.
No. Explanations vary. The negative value for time does not make sense in this context. The positive value means the projectile hits the ground about 2.173 seconds after it was launched.

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Additional Practice

A1.8.16

1. The function $g(t) = -4.9t^2 + 40t + 6$ models the height, in meters, of an apple t seconds after it has been launched into the air from a toy cannon.

a Write an equation to determine when the apple will hit the ground.
 $-4.9t^2 + 40t + 6 = 0$ (or equivalent)

b Determine when the apple hits the ground. Round to the nearest thousandth.
8.311 seconds

2. Two objects are launched into the air. The function $h(t) = 45 + 130t - 16t^2$ models the height, in feet, of Object A t seconds after it has been launched. The function $f(t) = 18 + 70t - 16t^2$ models the height, in feet, of Object B t seconds after it has been launched.

a After 5 seconds have passed, which object is still in the air?
Object A

b Write an equation for each object to determine when it will hit the ground.
Object A: $45 + 130t - 16t^2 = 0$ (or equivalent); Object B: $18 + 70t - 16t^2 = 0$ (or equivalent)

c Determine when each object hits the ground. Round to the nearest thousandth.
Object A: 8.458 seconds; Object B: 4.619 seconds

3. A picture is 10 inches wide, 15 inches long, and has a frame x inches thick surrounding it. The equation $(10 + 2x)(15 + 2x) = 336$ represents the total area, in square inches, of the picture in its frame.

a Rewrite the equation $(10 + 2x)(15 + 2x) = 336$ in standard form.
 $4x^2 + 50x - 186 = 0$

b Determine the thickness of the frame.
3 inches

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4. A picture is 8 inches wide and 10 inches long. It has a frame x inches thick surrounding it. The equation $(8 + 2x)(10 + 2x) = 195$ represents the total area, in square inches, of the picture and the frame.

a Rewrite the equation $(8 + 2x)(10 + 2x) = 195$ in standard form.
 $4x^2 + 36x - 115 = 0$

b Solve your equation. What do the solutions represent?
 $x = -11.5$ and $x = 2.5$. The solutions represent the thickness of the frame. Only the positive solution makes sense in this context. So, the thickness of the frame is 2.5 inches.

5. A ball is thrown up into the air. Its height in inches is modeled by the function $h(t) = -16t^2 + 28t + 3$, where t is the time after the ball is thrown, measured in seconds. Solve the equation $-16t^2 + 28t + 3 = 0$. What do the solutions tell you about the ball?
 $t \approx -0.101$ and $t \approx 1.851$. The ball landed on the ground about 1.851 seconds after it was thrown up in the air. The negative solution does not make sense in this context.

6. A projectile is launched into the air and its height above the ground, in feet, is modeled by the function $f(t) = 6 + 32t - 16t^2$, where t is the number of seconds since the projectile was launched.

a What are the solutions to the equation $0 = 6 + 32t - 16t^2$?
 $t \approx -0.173$ and $t \approx 2.173$

b Do both of the solutions have meaning in this context? Explain your thinking.
No. Explanations vary. The negative value for time does not make sense in this context. The positive value means the projectile hits the ground about 2.173 seconds after it was launched.

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.B.4
2	1	HSF.IF.B.4
3	2	HSF.IF.B.4
4	2	HSF.IF.B.4
5	2	HSF.IF.B.4
6	2	HSF.IF.B.4

Notes:

Additional Practice

A1.8.17

1. Select *all* the expressions that are rational.

☒ A. $\frac{1}{4} \cdot \sqrt{9}$

☐ B. $\sqrt{5} + \sqrt{9}$

☒ C. $2.6 + 5.3$

☒ D. $-\sqrt{7} + \sqrt{7}$

☒ E. $\sqrt{16} \cdot \sqrt{4}$

☐ F. $6 + \sqrt{6}$

2. Select *all* the following numbers that will result in a rational number when multiplied by $\sqrt{2}$.

☐ A. 2

☐ B. 8

☐ C. $\frac{1}{2}$

☒ D. $\sqrt{2}$

☐ E. $\sqrt{4}$

☒ F. $\sqrt{8}$

3. Determine whether each of the following statements are *always*, *sometimes*, or *never* true. Explain your thinking.

a Two rational numbers added together will be rational.

Always true. Explanations vary. There is no way to get an irrational number from adding two rational numbers.

b Multiplying an irrational number by another irrational number will result in an irrational number.

Sometimes true. Explanations vary. When an irrational number is multiplied by itself, it results in a rational number. For example, $\sqrt{7} \cdot \sqrt{7} = \sqrt{49} = 7$.

c The sum of a rational number that and an irrational number is rational.

Never true. Explanations vary. There is no way to get a rational number from adding a rational and an irrational number.

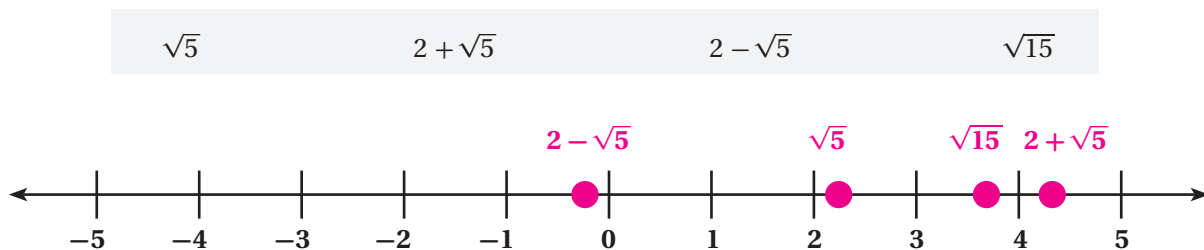
4. Consider the statement: *The product of a rational number and an irrational number is irrational.* Which example shows this statement is false?

- A. $\sqrt{7} \cdot \sqrt{7} = 7$
 B. $2 \cdot \sqrt{2} = 2\sqrt{2}$
 C. $\sqrt{4} \cdot \sqrt{16} = 8$
 D. $\sqrt{7} \cdot 0 = 0$

5. Here are the solutions to some quadratic equations. Decide if the solutions are rational or irrational.

a	$5 \pm \sqrt{25}$	Rational	Irrational
b	$-8 \pm \sqrt{6}$	Rational	Irrational
c	$\frac{3}{4} \pm \sqrt{\frac{70}{7}}$	Rational	Irrational
d	2 ± 9	Rational	Irrational
e	$3 \pm \sqrt{2}$	Rational	Irrational
f	8 ± 0.5	Rational	Irrational

6. Here is a number line with four points on it. Estimate each expression and plot it on the number line.



7. Consider the expressions $\sqrt{2} + a$ and $b\sqrt{2}$. Determine a value of a and b so that both expressions will result in a rational number.

Responses vary. $a = -\sqrt{2}$, $b = \sqrt{2}$

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Additional Practice

A1.8.17

1. Select all the expressions that are rational.

- ☒ A. $\frac{1}{4} \cdot \sqrt{9}$ ☐ B. $\sqrt{5} + \sqrt{6}$
☒ C. $2.6 + 5.3$ ☒ D. $-\sqrt{7} + \sqrt{7}$
☒ E. $\sqrt{16} \cdot \sqrt{4}$ ☐ F. $6 + \sqrt{6}$

2. Select all the following numbers that will result in a rational number when multiplied by $\sqrt{2}$.

- ☐ A. 2 ☐ B. 8
☐ C. $\frac{1}{2}$ ☒ D. $\sqrt{2}$
☐ E. $\sqrt{4}$ ☒ F. $\sqrt{8}$

3. Determine whether each of the following statements are always, sometimes, or never true. Explain your thinking.

- a. Two rational numbers added together will be rational.
Always true. Explanations vary. There is no way to get an irrational number from adding two rational numbers.
- b. Multiplying an irrational number by another irrational number will result in an irrational number.
Sometimes true. Explanations vary. When an irrational number is multiplied by itself, it results in a rational number. For example, $\sqrt{7} \cdot \sqrt{7} = 49 = 7$.
- c. The sum of a rational number and an irrational number is rational.
Never true. Explanations vary. There is no way to get a rational number from adding a rational and an irrational number.

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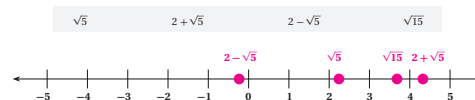
4. Consider the statement: *The product of a rational number and an irrational number is irrational.* Which example shows this statement is false?

- A. $\sqrt{7} \cdot \sqrt{7} = 7$
B. $2 \cdot \sqrt{2} = 2\sqrt{2}$
C. $\sqrt{4} \cdot \sqrt{16} = 8$
☒ D. $\sqrt{7} \cdot 0 = 0$

5. Here are the solutions to some quadratic equations. Decide if the solutions are rational or irrational.

- | | | |
|--|---|---|
| a. $5 \pm \sqrt{25}$ | Rational | Irrational |
| b. $-8 \pm \sqrt{6}$ | Rational | ☒ Irrational |
| c. $\frac{3}{4} \pm \sqrt{\frac{76}{7}}$ | Rational | ☒ Irrational |
| d. 2 ± 9 | ☒ Rational | Irrational |
| e. $3 \pm \sqrt{2}$ | Rational | ☒ Irrational |
| f. 8 ± 0.5 | ☒ Rational | Irrational |

6. Here is a number line with four points on it. Estimate each expression and plot it on the number line.



7. Consider the expressions $\sqrt{2} + a$ and $b\sqrt{2}$. Determine a value of a and b so that both expressions will result in a rational number.

Responses vary. $a = -\sqrt{2}$, $b = \sqrt{2}$

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSN.RN.B.3
2	2	HSN.RN.B.3
3	1	HSN.RN.B.3
4	2	HSN.RN.B.3
5	2	HSN.RN.B.3
6	3	HSN.RN.B.3
7	3	HSN.RN.B.3, HSA.REI.B.4

Notes: